RB4107 ES1 rescue carrier, 1:5 prototype

Track segments: complete pack (Nazim, 16 Oct 2026)

The track unit on Nhien's Rev B frame: his side plates P1 / P2 and the parts in his BOM. The track itself has two options, A my printed track or B his bought belt with bonded cleats; frame, wheels, shafts, bearings and every joint are identical, so Part A applies to both and only the option sections differ.

Interactive 3D assembly (slide it apart and back together in 11 steps, bend the hinge, switch the track, click any part):

https://claude.ai/artifact/6gnfWidMkD9kDk2BQc1tBw

FEASIBILITY: can this be built as drawn, and what changed after the 16 Oct design review

0 Feasibility check: 25 items measured on the shared model, verdicts, fixes; the build-day sequence with go / no-go checks; drive, hinge and power check

PART A: the build, common to both track options

1 How it works: assembled with the 15 numbered joints and the drive train (16, 17); bent down the ES1 stairs; where the hinge stops

2 Piece it together: five of the 11 build steps, connection by connection

3 Parts catalogue, one page per part, highlighted in the assembly, with clickable vendor links (track: options A and B)

4 Procurement list: track options A and B, wheels, and Nhien's parts, with links, prices and delivery (clickable)

5 Layout with Nhien's plates: where everything sits, what changed

6 Connection map: one side pulled apart, every joint with its fastener

7 Drawings TS-06 Toothed wheel (reamed, pinned), TS-09 Idler tensioner, TS-10 Rail carrier angle, TS-11 Stack-ups and fits, each with its explainer

OPTION A: my printed track

8 Option page (what it is, deck clearance, what to order), drawing TS-01 Track and its explainer

OPTION B: Nhien's bought belt with bonded cleats (B10)

9 Option page (what it is, the deck notch, what to order), drawing TS-01B Track belt and its explainer

PART B: Nhien's Rev B sheets for the parts the track runs on, unchanged from his fabrication pack

Files on the branch: track_print/stl (print files), track_print/step (CAD, STEP), track_print/viewer (this interactive page), track_print/assembly (this pack and its sources).

0. Feasibility check: measured on the shared model, 16 and 18 Oct 2026

Every item below was measured on the model that the viewer and this pack are rendered from (Nhien's Rev B STEP plus the track), or comes from the design review feedback. 'FIX' items are drawn on TS-06, TS-09, TS-10 and TS-11 and are in the procurement list.

Item	As drawn / measured	Consequence	Fix or action	Verdict
Geometry, from the model				
Lane width, plate face to plate face	20.0 (bars 160 and 110, plates 5)	pulley 19.0 over flanges, 0.5 each side	OK if bars are cut to +-0.1 and square	**OK**
Pulley float along the hinge shaft	0.5 + 0.5 mm gaps, no shims at the plates	pulley faces can rub a bearing outer ring	one 8x11x0.5 shim per pulley + 0.3-0.5 feeler gap, pinned (TS-11)	**FIX**
Idler flange vs rail end	rail ends x 207, flange edge at x 204.6 when the idler is pushed fully in	jams 2.4 mm	rails trimmed to 186; idler never in past x 223 (TS-10, TS-11)	**FIX**
Idler screw stack vs shoulder	25.5 stack on a 25 shoulder	nut clamps the inner-ring stack: correct	keep the 0.5 shim; slider 3 instead of washer 1.6 still leaves 1.5 mm of thread past the nut	**OK**
Plate slot for the shoulder screw	8.1 wide, 12.2 long	Ø8 f9 shoulder: 0.1-0.15 clearance	laser +0.1/0; usable travel 9.6 mm = 19 mm of loop	**OK**
Hinge bearing seats	Ø16.00 drawn, no fit	laser alone gives press or slip at random	laser Ø15.8, ream Ø16 H7, Loctite 641 (TS-11)	**FIX**
Deck over the front track	underside 25.0 above the axis	printed track 19.7: 5 mm clear; belt 25.0: rubs	option B: notch the deck 13 x 135 or 5 mm cleats	**OK / FIX B**
Hinge range	up-stops B18 at 0; P14 corner hits deck slot H at 18.8	stairs need 35	slot H 74 long (+3) or chamfer P14	**FIX**
Hinge sector travel	sector spans 31.4 deg behind to 42.3 deg ahead of vertical; pinion 10.4 deg ahead (x 11, y 60)	41.8 deg of teeth behind the pinion, less 5-7 deg to stay in mesh: 35-37 deg; the stairs need 35	Nhien: two more teeth at the rear end (+7 deg); slot H 3 mm longer rearward	**FIX**
Sector gear to pinion centre distance	61.0 = exact pitch sum (R51 + 10)	zero backlash: binds	61.1 to 61.2 (Nhien, P13 / motor bracket)	**FIX**
Track tooth tips on the rails	tips r 13.34, rail faces at 13.3	slides on UHMW as intended	no change	**OK**
Flanges vs 2020 bar and rails	nose flange reaches x 207 at y 0; bar ends 210.9; rail leg of TS-10 ends at 204	3.7 and 3 mm clear	no change	**OK**
Drive and fastening, from the design review				
Hinge shaft drive	D-flat, M4 grub on the front pair	reviewer: no D-shaft	round Ø8 h7 shaft, Ø2.5 spring pin per pulley, drilled at assembly (TS-06)	**FIX**
Cross pin drilling	'drilled together at assembly' in a V-block	the shaft cannot reach the drill press with its pulleys on: they sit between the plates	hub holes drilled on the press when boring; shaft drilled through the hub hole in place	**FIX**
Shaft axial location	E-clips in Ø6 grooves at both ends	reviewer: no E-clips; groove weakens the shaft	pinned front pulley between the plate bearings locates the shaft (TS-11)	**FIX**
Track tension	shoulder screw clamped in the slot by friction	reviewer: no tension, will slip	jack-screw tensioner, 2-3 mm deflection preload (TS-09)	**FIX**
Rail bracket contact area	4 clips of 10 x 14.6 per rail	reviewer: too small	full-length 10x10x1.5 angle, 186 x 10 both sides (TS-10)	**FIX**
Plate to bar brackets P18 / P19	13 and 18 long angle offcuts	reviewer: more contact	standard cast 2020 / 3030 corner brackets (Nhien)	**FIX**
Drive, hinge and power (page 4)				
Drive motors, Pololu 2828	150:1, 67 rpm, 49 kg.cm stall, 5.5 A	stairs need about 2.5 kg.cm per side at 64 rpm	no change; ratio has 20x margin	**OK**
Hinge motor 5840-31ZY	speed variant not in the BOM	needs about 5 kg.cm at the motor; 12-40 rpm versions give 15-70 kg.cm	Nhien to order the 12 to 40 rpm version, not 160 or 470	**FIX**
Power	3S 3000 mAh, 15 A fuse, 2 x MDD10A, 5 V 5 A	17.5 A if every motor stalls at once; 2-3 A when climbing	fuse 20 A or a software current limit; otherwise fine	**OK**
Process and supply				
Plates	5 mm Al 6061 laser + tap, 2 + 2	vendor laser, workshop tap and ream	1 week; add the M3 tab hole and the 15.8 seats to the DXF	**OK**
Pulleys	8 plain-bore MISUMI, 2 stage-2	ream 8 H7 x 6, bore 16 H7 x 4, hub cross holes on the press, shaft drilled through them at assembly	lathe or drill press + V-block; 1-2 weeks queue	**OK**
Track, option A	4 TPU rings, 3 h each, Fab Lab UP300	fit-test slice first; ring fits the bed flat	print week 1	**OK**
Track, option B	4 belts RS + 88 cleats bonded	belt shop 1-2 weeks; deck notch	order week 1	**OK**
Bearings, shoulder screws, shafts, pins, angles	all stock items, RS SG / MISUMI SG	days	order with the pulleys	**OK**

Verdict: buildable. Nothing needs a part that cannot be bought or made at the Professional Workshop and the Fab Lab; the fixes are small parts, reamed holes and two notes to Nhien's DXF. Longest lead items: the pulley bore work and, for option B, the belt-shop cleats.

0. Build-day sequence with go / no-go checks

Same order as the 11 steps of the interactive page. A check that fails is fixed before the next step, not after.

Before the build day

Plates back from the laser with the 15.8 seats, the M3 tab holes at (+-223, +11) and the six carrier holes and slot H at 74. Ream the six Ø16 H7 seats; tap the plates (M3 / M4 / M5 per Rev B sheets 4, 5).

Pulleys back from the workshop: 6 x Ø8 H7 reamed with their Ø2.5 hub cross holes, 4 x Ø16 H7 bored. Check each on the shaft and on a 688: the 8 H7 ones slide without shake, the 16 H7 ones take a bearing with a light press.

Tracks printed (option A: fit-test slice first, then 4 rings) or belts back from the belt shop (option B). Rails trimmed to 186, carrier angles made (TS-10), tabs and sliders made (TS-09).

Bring: 688-2RS x14, shoulder screws x4 with 0.5 shims, sleeves and nylocs, M3 / M4 / M5 / M6 hardware and T-nuts, Loctite 641 and 243, feeler gauges, Ø2.5 drill and spring pins, V-block, digital calipers, hex keys, a square.

1. Bearings (step 1 of the viewer)

Loctite 641 on the outer ring, press the 688s into P1 x2, P2 x2, P17 x2 and into the four idler pulleys (two each, sleeve between). Check: a pulley on its bearings spins free with no rock.

2. Rear frame (steps 2-3)

P2 plates onto the 3030 bars with the corner brackets, P17 supports on the middle bar. Check with a square and calipers: inner faces of P2 at 160.0 +-0.2 apart, both plates parallel, hinge holes in line (a Ø8 shaft passes both freely).

Rear rails on their carrier angles (TS-10), angle to the plate with M3 x 6, rail to the angle with the thread-forming screws.

3. Front frame (steps 4-5)

P1 plates onto the 2020 bars with the brackets and the sector gear on P14. Check: inner faces 110.0 +-0.2 apart; front rails on their carriers. Lower the front frame between the rear plates: lane each side 20.0 +-0.2 (plate face to plate face), hinge holes in line with the rear ones (shaft passes all four).

4. Hinge wheels and shafts (steps 6-7)

Per side: front hinge pulley into the front lane, rear hinge pulley into the rear lane, stage-2 pulley with its 1 mm shim inboard. Push the plain shaft in from outside through rear pulley, P2 bearing, front pulley, P1 bearing, P17 bearing.

Set the front pulley: 0.5 shim against the P1 bearing, 0.3-0.5 feeler gap to the P2 bearing. Turn the hub's pre-drilled hole up, drill Ø2.5 through the shaft with the hub hole as the guide, drive the spring pin. Check: shaft turns freely, no axial play beyond the feeler gap.

Same for the rear pulley (0.3-0.5 gap to its bearing) and the stage-2 pulley (shim against P17). Pinned: no grub screws, no E-clips.

5. Deck, posts, motors, tray (steps 8-9)

Nhien's sheet 3. Check before the tray goes on: sector-gear backlash at the pinion 0.1-0.2 mm (centre distance 61.1-61.2); hinge motor swings the front frame nose-down to 35 with nothing touching slot H; it rests on the foam up-stops at 0.

6. Idlers and tracks (steps 10-11)

Per idler: pulley into the track loop, shoulder screw through plate and bearings with the head washer outside and the 0.5 shim inside, slider and nyloc loose inboard, then the tab with its M3. Wind the jack screw until the lower run, hooked at a cleat at mid-span and pulled off its rail with 4 N (machine on blocks), gives 2 to 3 mm; pushing a run inward moves nothing because the rails sit under both runs; tighten the nyloc; back the jack screw off a quarter turn.

Check each track: turns by hand a full loop without a tooth skipping or the cleats touching the deck; flanges clear the rail ends and the bar; idler not inside x 223.

7. First run

Drive each side on blocks: no tooth jump under a hand-braked load. Hinge: 0 to 35 nose-down and back, worm holds at any angle with the motor off. Then the stairs.

0. Drive, hinge and power check

Hand calculation from the model's geometry, Nhien's BOM and the vendors' published figures. Masses are estimates from the part volumes; everything else is measured or quoted. Full-size numbers are not scaled from these.

Mass budget (1:5)

Plates P1 / P2 4 x 0.10, deck and tray 2 x 0.71, bars and posts 0.5, drive motors 2 x 0.21, hinge motor 0.35, battery 0.25, electronics 0.15, pulleys 0.24, tracks 0.13, bearings, shafts, rails and screws 0.3: about 4.2 kg. With a 1:5 dummy load of 0.6 kg: 5 kg, used below.

Drive: can it climb the ES1 slope?

ES1 at 1:5 is 35 rise on 50 going: a 35 degree slope. Tractive force = m g (sin 35 + 0.1 cos 35) = 5 x 9.81 x 0.656 = 32 N, 16 N per side. Wheel pitch radius 15.9 mm: 0.26 N.m per side at the hinge shaft.

The two belt stages are close to 1:1 (stage-2 pulleys 12T to 12T, stage-1 about 15T to 16T in the model), so the motor sees about 0.25 N.m = 2.5 kg.cm. Pololu 2828 (BOM B1): 150:1, 67 rpm no load, 49 kg.cm stall at 5.5 A. Margin on stall 20x, on a 25 per cent continuous rating 5x.

Speed: about 64 rpm under this load = 0.107 m/s at the wheel, one step every half second, a 14-step flight in 7 s. If the team wants it faster, the 70:1 or 100:1 of the same motor still has 2x to 3x margin.

Hinge: can the worm motor swing and hold the front segment?

Front segment about 0.9 kg with its centre of mass 110 mm ahead of the hinge: 0.97 N.m to lift it in the air. Worst case on the stairs, when the front segment carries half the vehicle on a step edge 100 mm ahead of the hinge: 2.5 N.m at the hinge.

Sector gear R51 driven by the m1 20T pinion (pitch radius 10): ratio 5.1, so the motor needs about 0.5 N.m = 5 kg.cm. The 5840-31ZY is sold in many speeds under one name: 12 rpm at 70 kg.cm, 40 rpm at 15 kg.cm, 160 and 470 rpm at a few kg.cm. The BOM does not say which.

Verdict: order the 12 to 40 rpm version (3x to 14x margin; 35 degrees in 0.8 to 2.5 s). The worm is self-locking, so the angle holds with the motor off. Tangential load on the sector teeth is about 50 N: trivial for 5 mm steel.

Power and electronics

Stall currents: 2 x 5.5 A drive + up to 6.5 A hinge = 17.5 A if everything stalls at once; climbing draws about 0.5 A per drive motor and 1 A on the hinge while it moves, 2 to 3 A average. The 3S 3000 mAh battery gives about an hour of climbing and must deliver 20 A peaks (any 5C pack does).

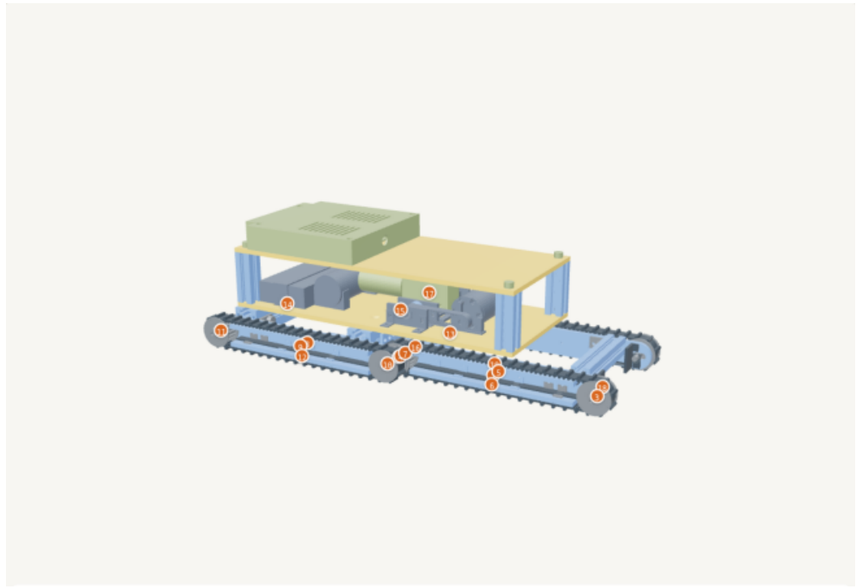
MDD10A drivers: 10 A continuous per channel, 30 A peak: fine for the 37D motors and the worm motor. The 15 A fuse (E6) would blow only in a double stall; a 20 A fuse or a current limit in the ESP32 code avoids a nuisance trip. The 5 V 5 A regulator runs the ESP32, IMU and encoders with margin. Encoders: 64 CPR x 150 = 9600 counts per wheel turn, plenty for left-right sync.

Sources

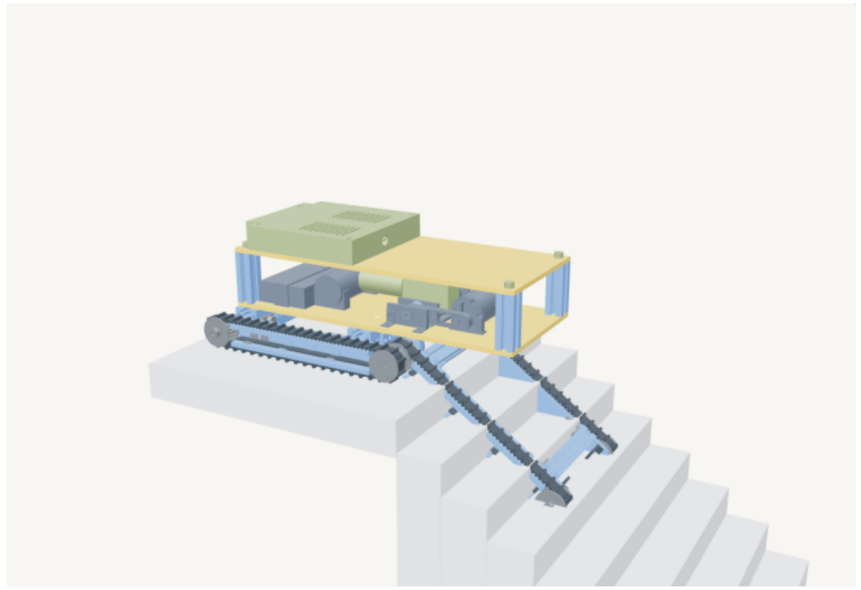
Pololu 2828 product page (150:1 37D with 64 CPR encoder); 5840-31ZY listings at Sigmanortec, RobotShop and Probots (12 rpm / 70 kg.cm; 40 rpm / 15 kg.cm variants). Figures as published on 16 Oct 2026; vendor stall figures are extrapolated, hence the margins above.

1. How it works

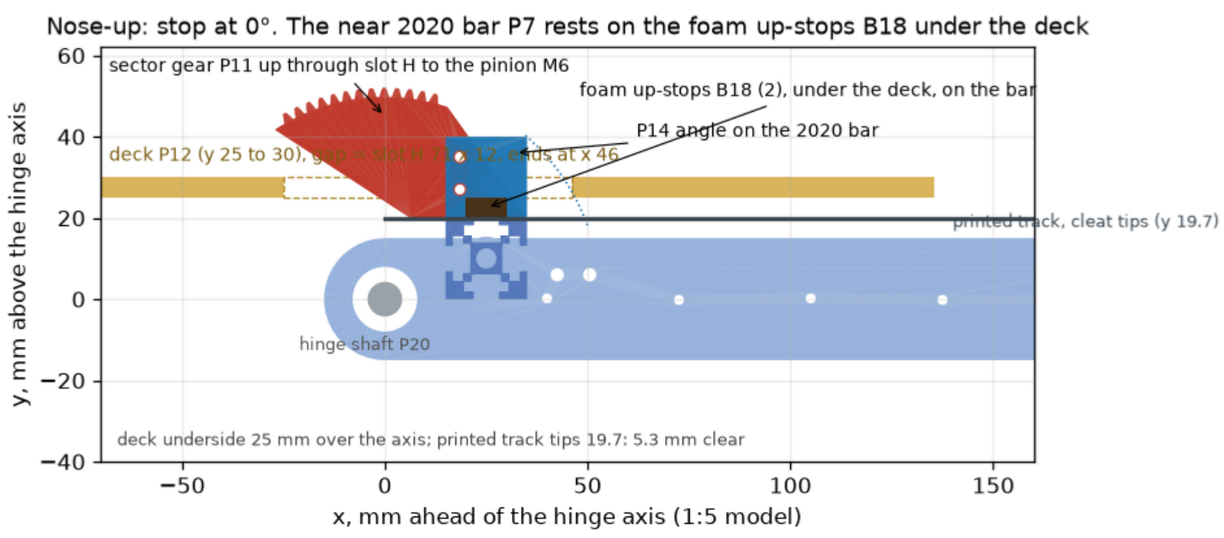
Nhien's Rev B frame (aluminium plates P1 / P2, 2020 and 3030 bars, deck and tray) carrying the tracks on bought HTD 5M wheels. Shown with option A, the printed track; option B looks the same with taller cleats. Same model as the interactive

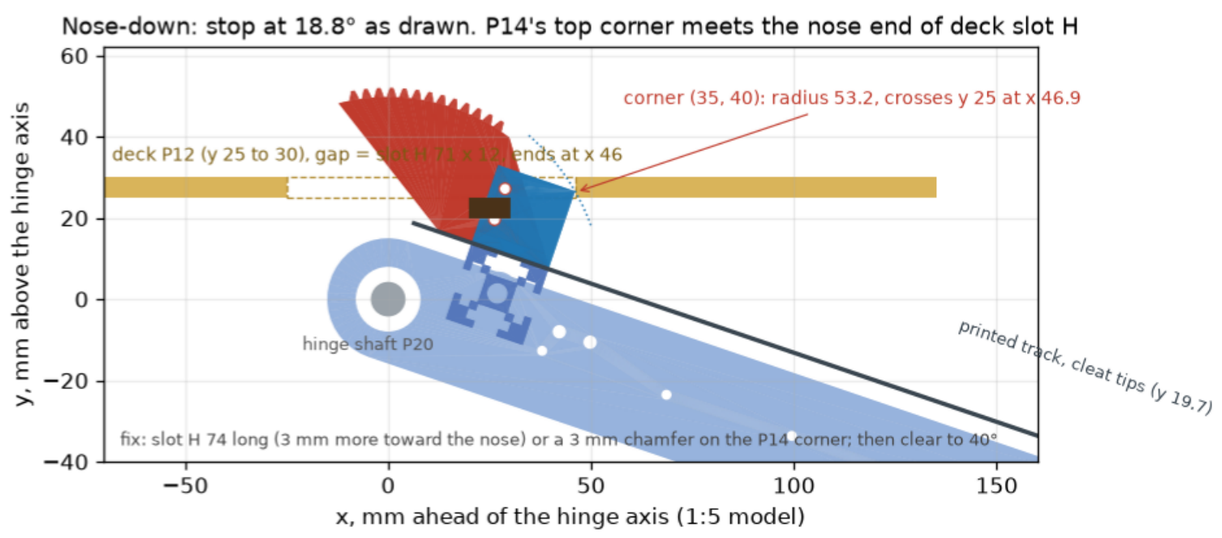


Assembled. Orange 1 to 15 are the joints of the connection map (page 6), each one connection with its own fastener. 16 is the drive pulley on the hinge shaft (the middle sprocket, between the plates), 17 the two drive motors on the deck that belt down to it.



Going down the ES1 stairs: the front segment bent 35° nose-down about the hinge shaft, held by the worm-gear hinge motor; the rear segment flat on the landing. The whole front frame swings: plates P1, both 2020 bars, sector gear, rails, nose idlers and the track.





Where the hinge stops (section at the sector gear, from the shared model, checked part against part at 0.1° steps). Left: it cannot lift. At 0° the near 2020 bar rests on the two foam up-stops B18 that Nhien glues under the deck, and the deck itself runs 135 mm ahead of the hinge, 5 mm over the printed track (touching the bought belt's cleats). So Rev B bends nose-down only: down the stairs front-first, up the stairs rear-first. Right: nose-down, as drawn, it jams at 18.8°, where the top corner of the sector mount angle P14 (swinging on a 53 mm radius) meets the nose end of the deck's sector slot H, 46 mm ahead of the hinge. With slot H 3 mm longer toward the nose (74 instead of 71), or a 3 mm chamfer on that corner, the swing is clear to 40°. The interactive page stops at both and names them; its 'Deck slot H +3 mm' box shows the fixed version.

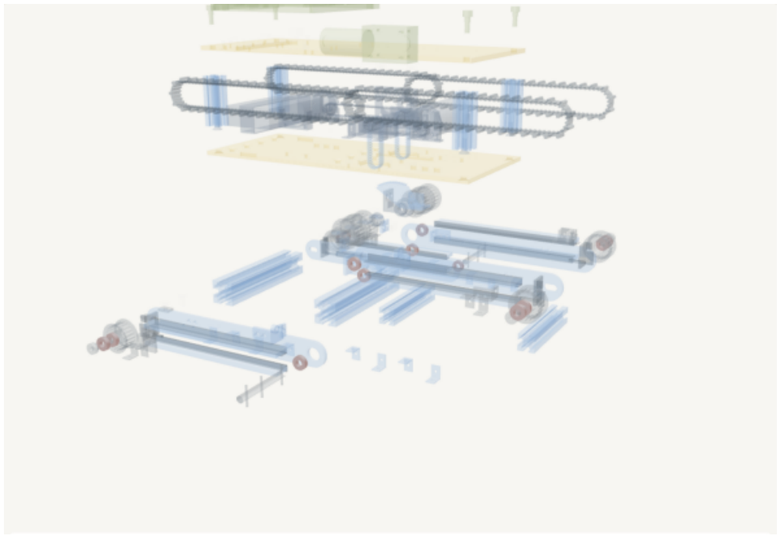
Checks for Nhien

Deck slot H: make it 74 x 12 (3 mm more toward the nose), or chamfer the P14 corner 3 mm. Otherwise the hinge stops at 19° instead of the 35° the stairs need. Needed for both track options.

Deck clearance over the front track: the deck underside is 25 mm above the axis. Option A's 3 mm cleats top out at 19.7 mm, 5 mm clear. Option B's 7.9 mm cleats reach exactly 25 mm and rub along 135 mm: notch P12 13 mm deep over each front lane from 25 mm behind the hinge to 110 mm ahead of it, or cut the cleats to 5 mm (see the option B page).

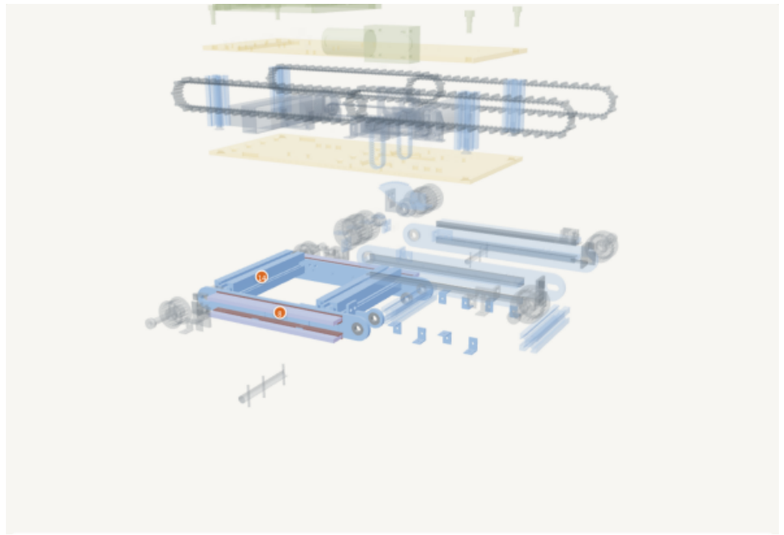
Colours: dark grey = the track, light grey = bought wheels, bearings, shafts, blue = Nhien's plates, rails and bars, yellow = deck and tray, slate = motors and belt drive, olive = electronics. Fasteners faded.

2. Piece it together: five of the 11 build steps (slide the interactive page for every step in between)



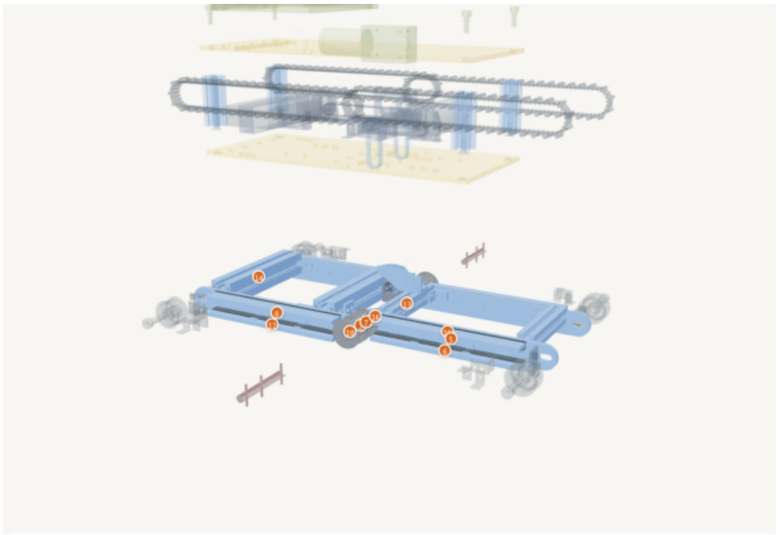
Taken apart

Everything waits in the order it goes on: rear plates and hinge shafts out to the sides, front frame, hinge wheels, deck, tray and tracks stacked above. Step 1: press the 688 bearings into the plates, the P17 supports and the idler wheels.



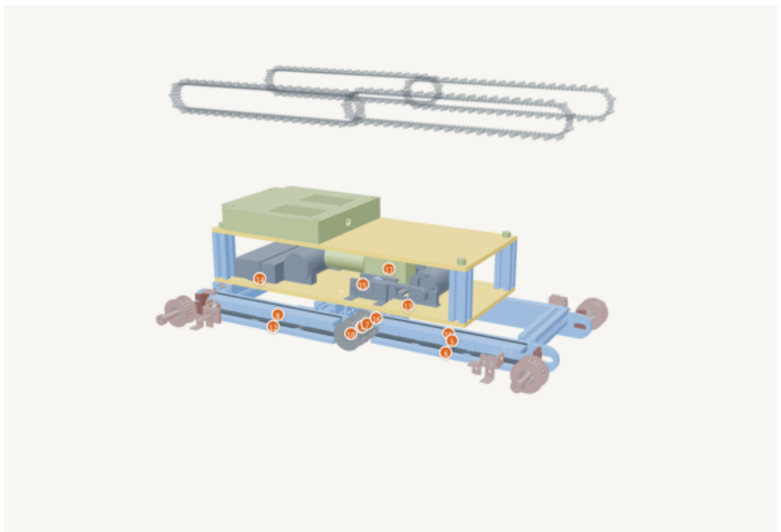
Steps 2 to 3: rear frame

Rear plates P2 onto the 3030 bars with P18 corner brackets (M6x10 + T-nut, 2x M5x8 into the plate) and the P17 hinge supports on the middle bar; then the rear slide rails P5 / P6 onto the outboard faces.



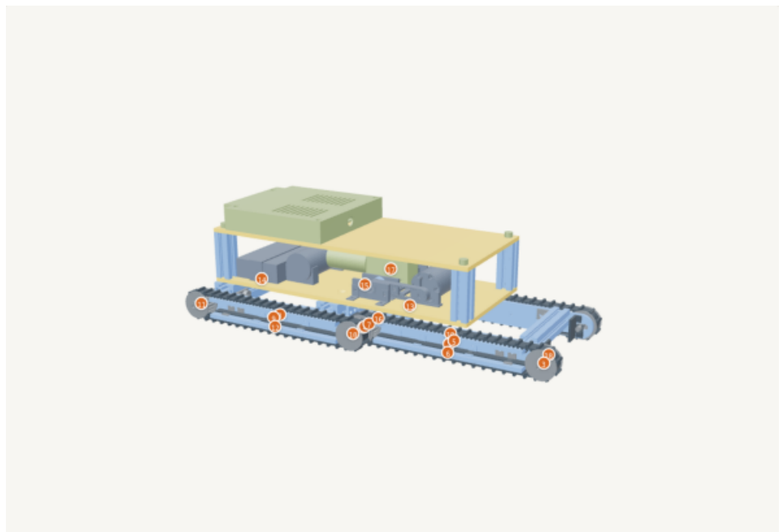
Steps 4 to 7: front frame and hinge

Front plates P1 with the 2020 bars, P19 brackets (M5x8 + T-nut, 2x M4x8) and the sector gear lowered in between the rear plates, rails on. Hinge wheels and drive pulleys dropped in; the plain round hinge shafts push in from outside through rear wheel, rear plate, front wheel, front plate and P17. Shims and feeler gaps set, then one Ø2.5 spring pin drilled through each pulley and the shaft (TS-11): no D-flat, no E-clips.



Steps 8 to 10: deck and idlers

Deck onto the 3030 bars (8x M5x12 csk + T-nuts), posts, drive motors, belts and hinge motor; tray, controller, battery and cover. Then the idler wheels onto their shoulder screws through the plate slots, with the sliders, nuts and tensioner tabs (TS-09).



Step 11: tracks on

Each track (printed, or the cleated belt) over its two wheels from above; wind the jack screws until the lower run, pulled off its rail with 4 N at mid-span, gives 2 to 3 mm; tighten the nyloc nuts. Complete.

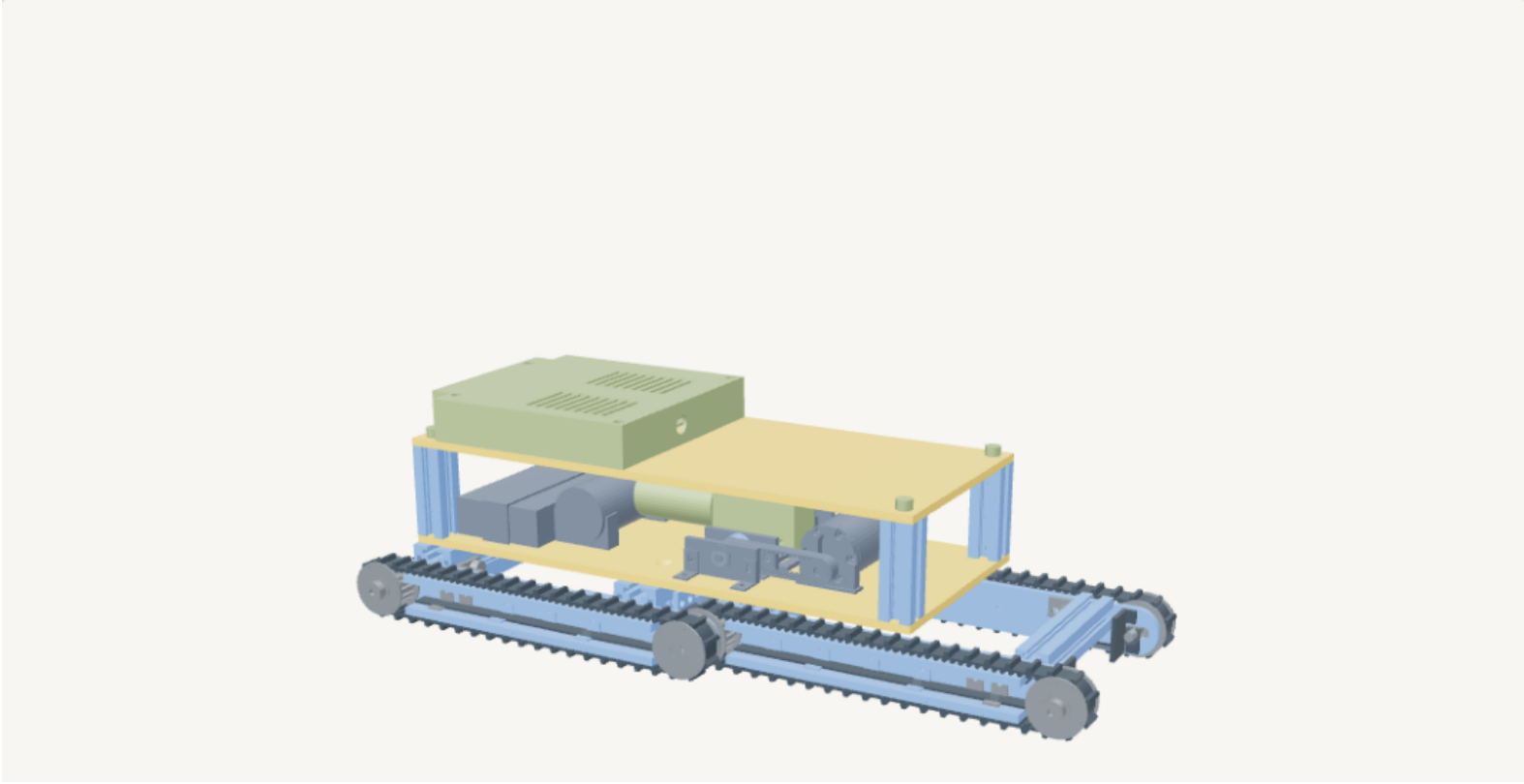
Interactive version

Open the page, drag the Assemble slider left to take the robot apart, then right to rebuild it one connection at a time. Attached parts ride with the part they are fixed to, so nothing passes through anything else. The panel names the joint being made and its fasteners; click any part or number for its details. The Track menu switches between the printed track and the bought belt; Show: drive train only, or frame and tracks only. The Hinge slider stops where the real parts would touch; the stairs toggle puts it on the ES1 flight.

https://claude.ai/artifact/6gnfWidMkD9kDk2BQc1tBw

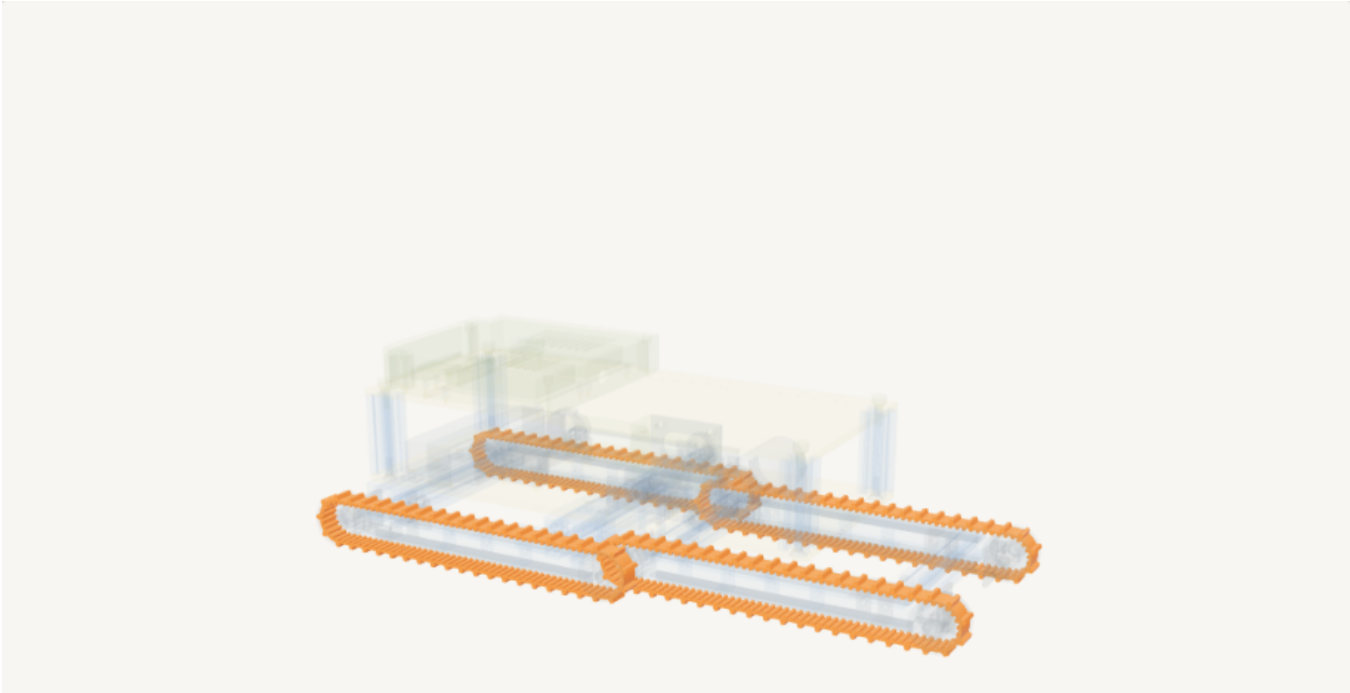
Parts catalogue: the track unit on Nhien's Rev B frame

One page per part. Orange = that part in the shared model of Nhien's Rev B assembly. Blue lines are links: click the vendor line to open the product page. Rev B sheet numbers refer to 'RB4107 Fabrication drawings (Rev B).pdf'.



Track, two options (next two pages):
A my printed TPU track x4, or
B Nhien's belt x4 + 88 bonded cleats.
Pulleys x8 bought plain bore, either way.
Nhien's: plates, rails, shafts, bearings, shoulder screws, bars, deck, motors, electronics (his BOM).

Option A: Track, printed



Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Quantity: 4

MAKE: 3D print (mine)

What it does

The printed TPU belt: 110 teeth inside on the toothed wheels, 45 cleats 3 mm high outside, spaced so five span one ES1 step. Fits Rev B's pulleys and 225 mm spacing; its cleat tips sit 5 mm under the deck. The other version of this pack uses Nhien's stock belt with bonded cleats instead.

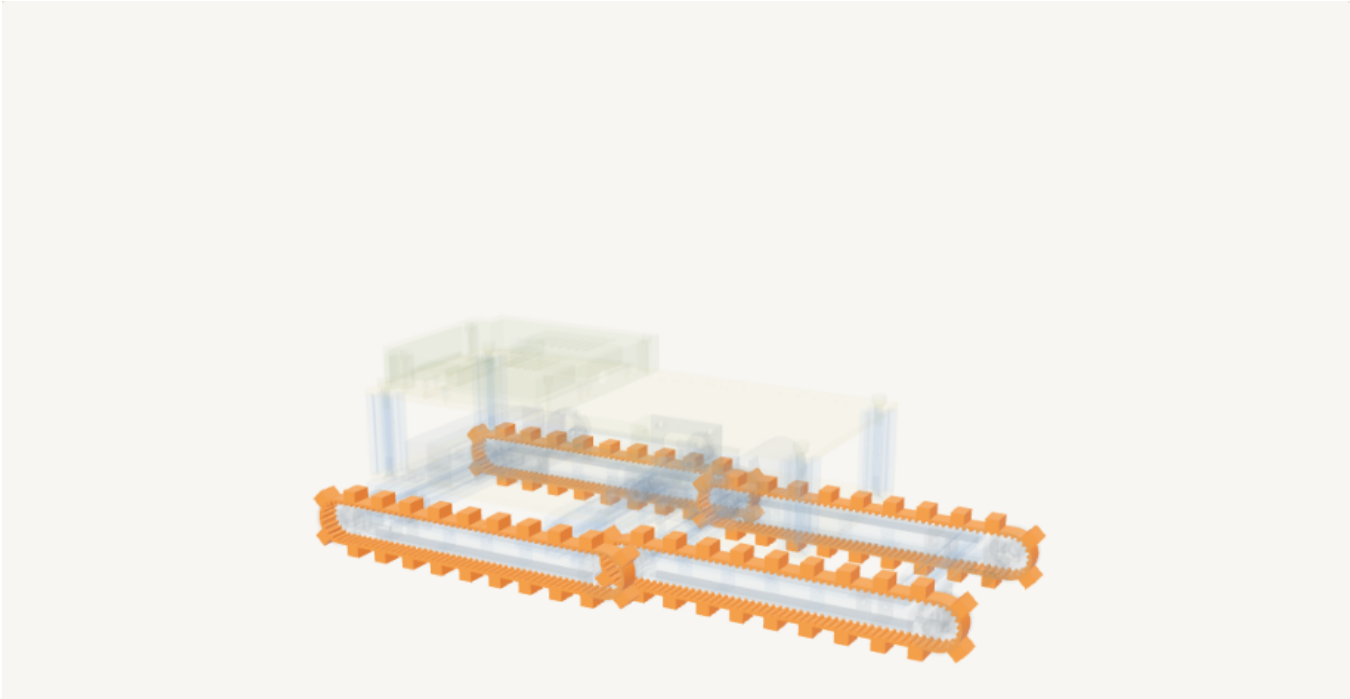
Where to get it

Fab Lab E4-02-06, Tiertime UP300 with the TPU head. File stl/track_TPU_x4.stl, standing as a ring, no supports, 0.2 mm, 100% infill. Filament: Fab Lab stock or Element14 SG TPU 95A (link). Drawing TS-01.

Link: Element14 SG: TPU 95A filament

https://sg.element14.com/multicomp-pro/mc011460/3d-printer-filament-tpu-1-75

Option B: Track belt HTD 550-5M-15 + 22 cleats (Rev B B10)



Quantity: 4 belts, 88 cleats

BUY belt; belt shop bonds the cleats

What it does

Nhien's stock-belt option on the same 20T wheels and 225 mm centres. The teeth inside drive; the bonded cleats grip the step edges. The cleat tips sit 25 mm above the hinge axis, level with the deck underside, so the deck needs a 13 x 135 notch over each front lane (or 5 mm cleats). The other version of this pack uses my printed track instead.

Where to get it

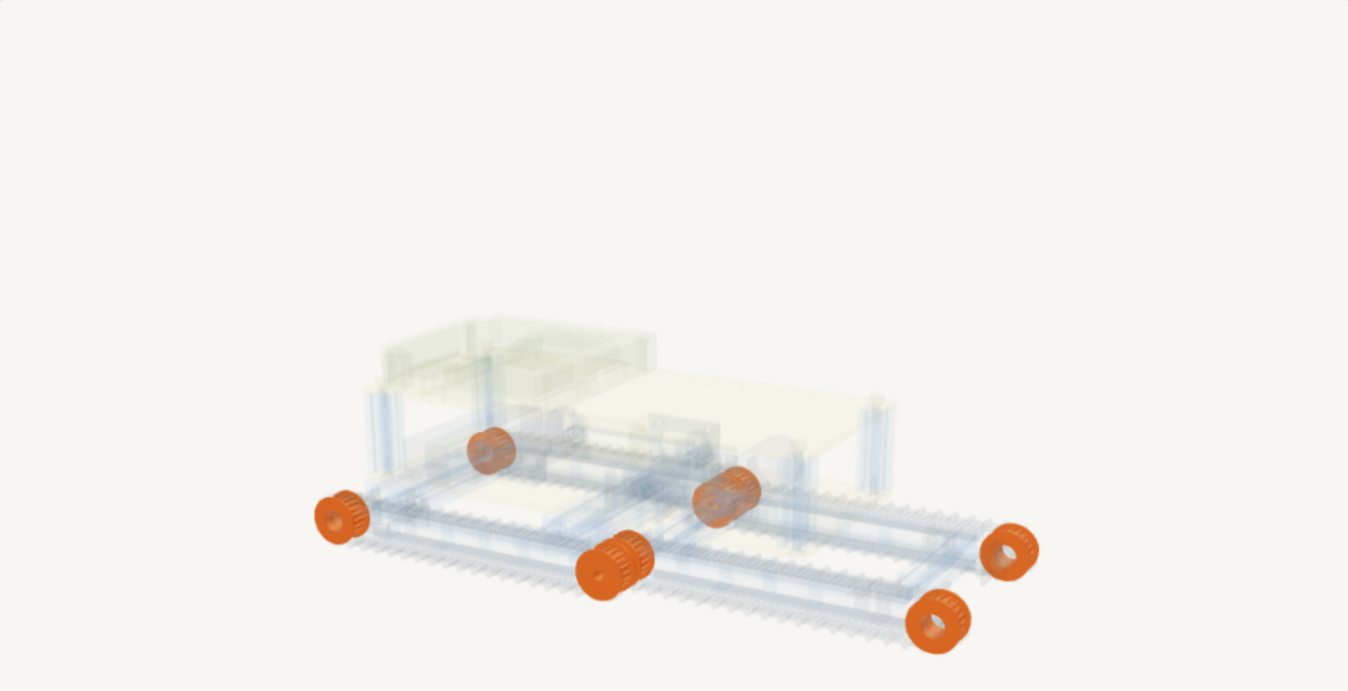
RS SG stock no. 058-6535 (SKF PHG 550-5M-15, link): 110 teeth, 15 wide. Cleats: rubber 10 x 7.9 x 15, one every 5th tooth, 22 per belt, cut and bonded by a belt shop (Fenner, Megadyne or Gates distributor). Drawing TS-01B.

Link: RS SG: search 058-6535

https://sg.rs-online.com/web/c/?searchTerm=058-6535

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Toothed wheel (timing pulley) HTD 5M 20T



Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Quantity: 8

BUY plain bore + MODIFY

What it does

Rev B's lanes are only 19-20 mm wide, so there is no room for a set-screw hub. The hinge ones are pinned to the plain round shaft (one spring pin each, design review); the end ones spin on bearings on a shoulder screw.

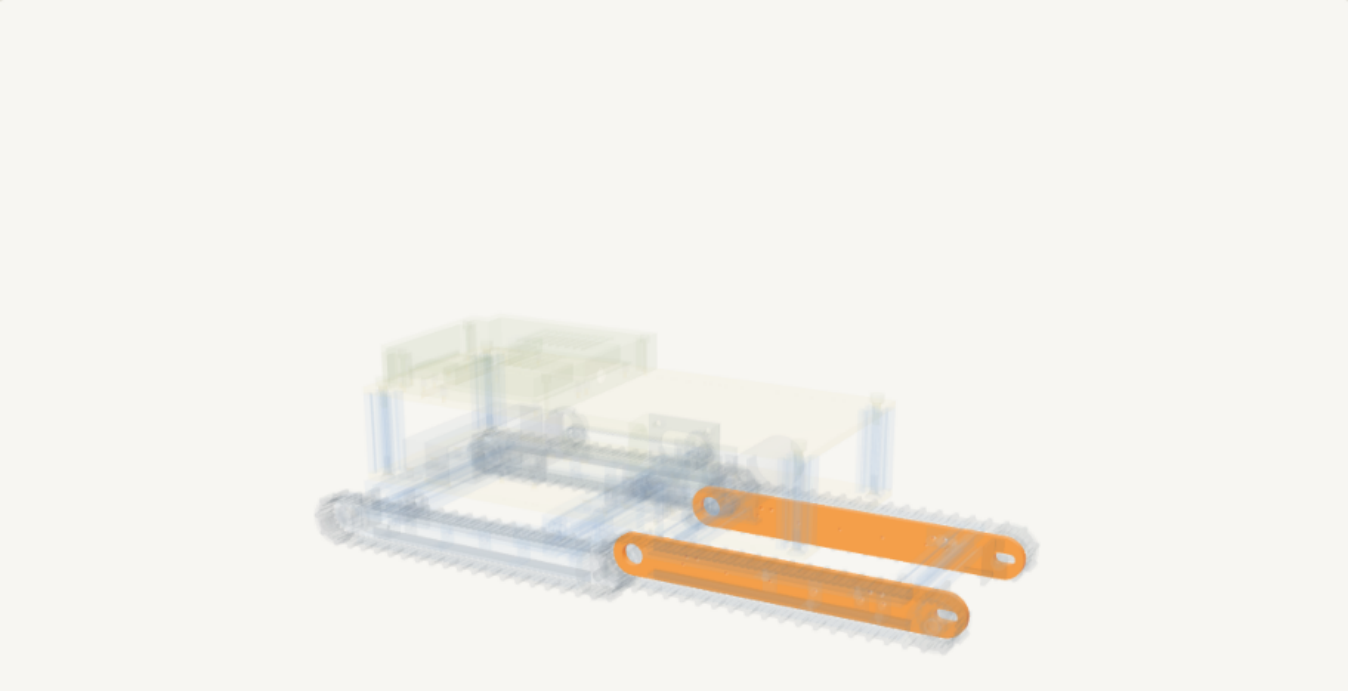
Where to get it

MISUMI SG, Timing Pulleys HTD 5M (link): 20 T, 15 mm belt, flanges both sides, PILOT (plain) bore. Then Professional Workshop: 4 reamed Ø8 H7 (hinge shaft, M1), 4 bored Ø16 H7 for two 688 bearings each (end idlers, M2). Cross holes Ø2.5 drilled with the shaft at assembly. TS-06, TS-11.

Link: MISUMI SG: HTD 5M timing pulleys

https://sg.misumi-ec.com/vona2/detail/110310532889/

Side plate, front segment (Rev B P1)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

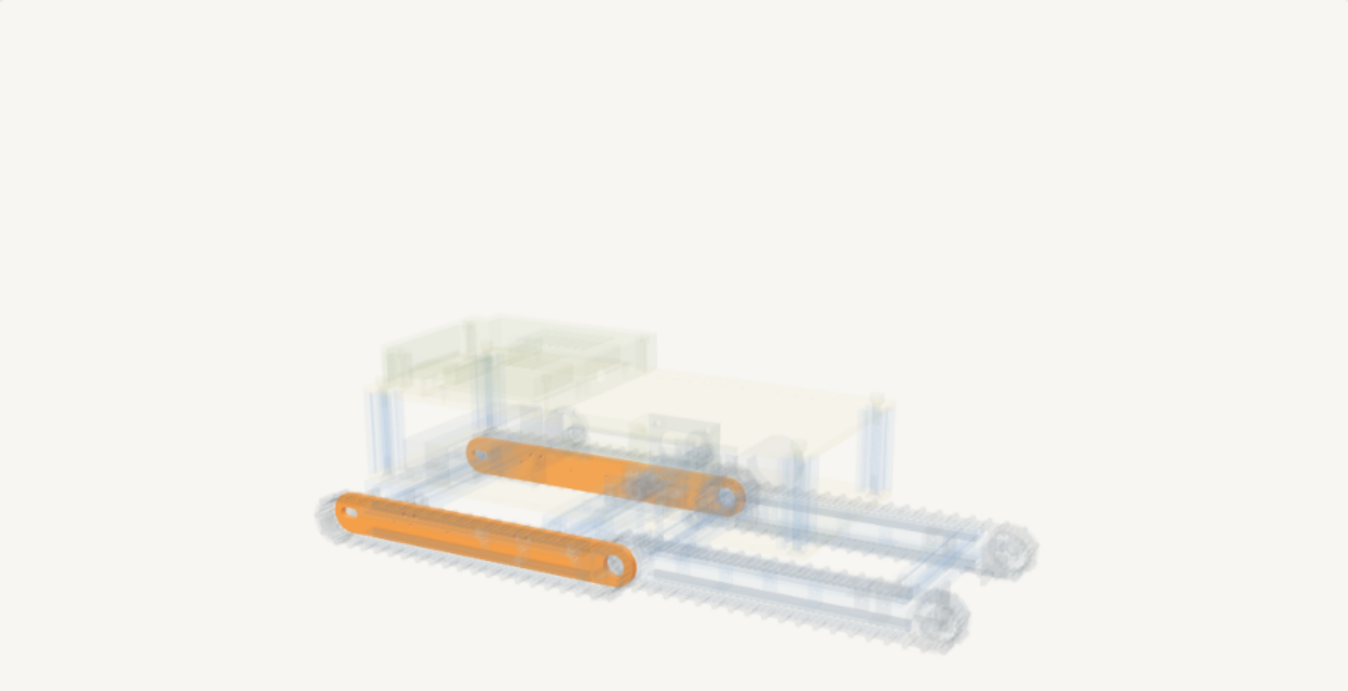
255 x 30.4 plate, inboard (z 55-60). Ø16 hole for the hinge bearing, a slot at the front end for the idler's shoulder screw (slide it to tension the track), tapped holes for the rail brackets and the 2020 bar brackets.

Where to get it

5 mm Al 6061, laser cut from DXF/P1 Side plate - FRONT segment.dxf, then drill and tap per Rev B sheet 4. Fab Lab laser does not cut aluminium: Professional Workshop or an outside laser shop on the vendor list.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Side plate, rear segment (Rev B P2)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

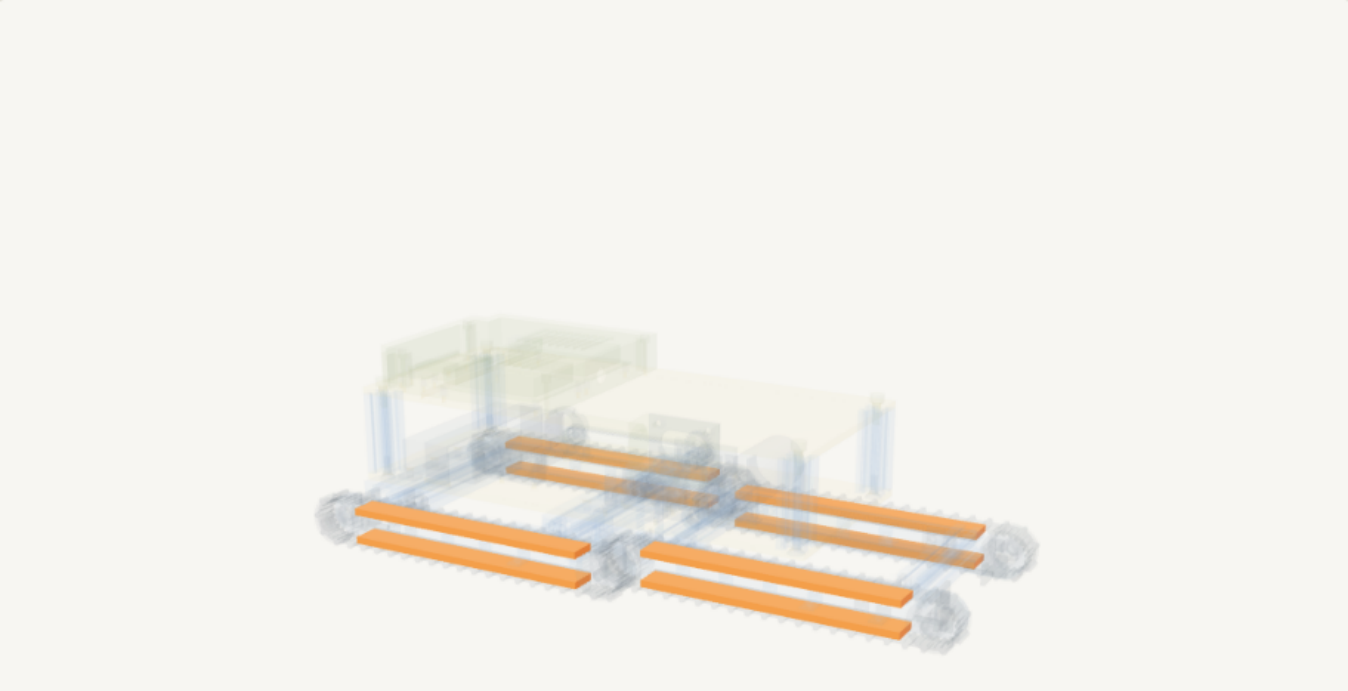
Outboard plate (z 80-85). The rear track runs outside it on a shoulder screw. Holes for the 3030 bar brackets and the rails.

Where to get it

As P1, from DXF/P2 Side plate - REAR segment.dxf, Rev B sheet 5.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Slide rails (Rev B P3-P6)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Quantity: 8

BUY strip + drill, trim to 186

What it does

The strips the track's upper and lower runs slide on so the track cannot sag between the wheels. The track's tooth tips ride on the lower one.

Where to get it

MISUMI SG UHMW-PE strip UPA-189-20-5 (link), 189 x 20 x 5, 3 pilot holes each, Rev B sheet 18. Trimmed to 186 at the idler end and carried on a full-length angle (TS-10) instead of the four RS clips.

Correction 19 Oct: do not drill the Rev B sheet 18 pilots (they overlap the new ones). Pilots 2.2 x 4 deep, blind, at 20, 93, 166 from the rail's hinge end, 7.5 from the plate face (TS-10).

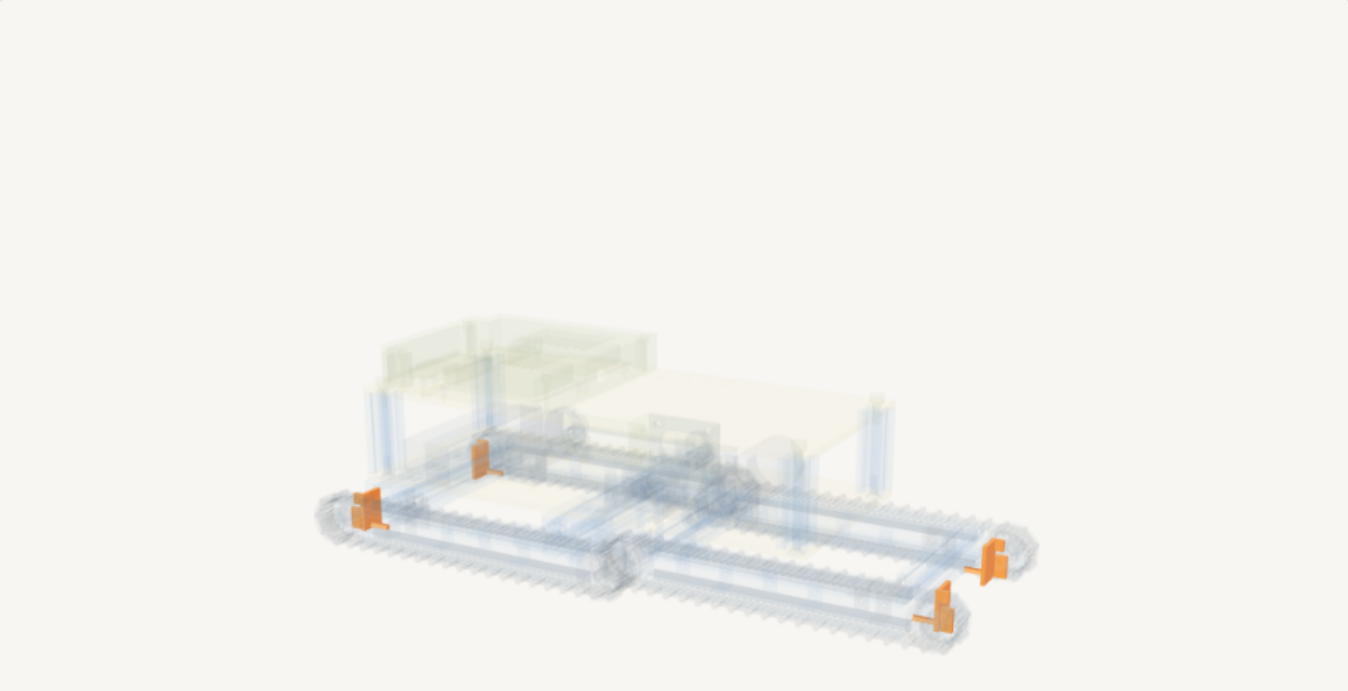
Link: MISUMI SG: plastic strips (UHMW)

https://sg.misumi-ec.com/vona2/mech/M1500000000/M1501000000/

Idler tensioner: tab, slider, jack screw (TS-09)

Quantity: 4 sets

MAKE on site



What it does

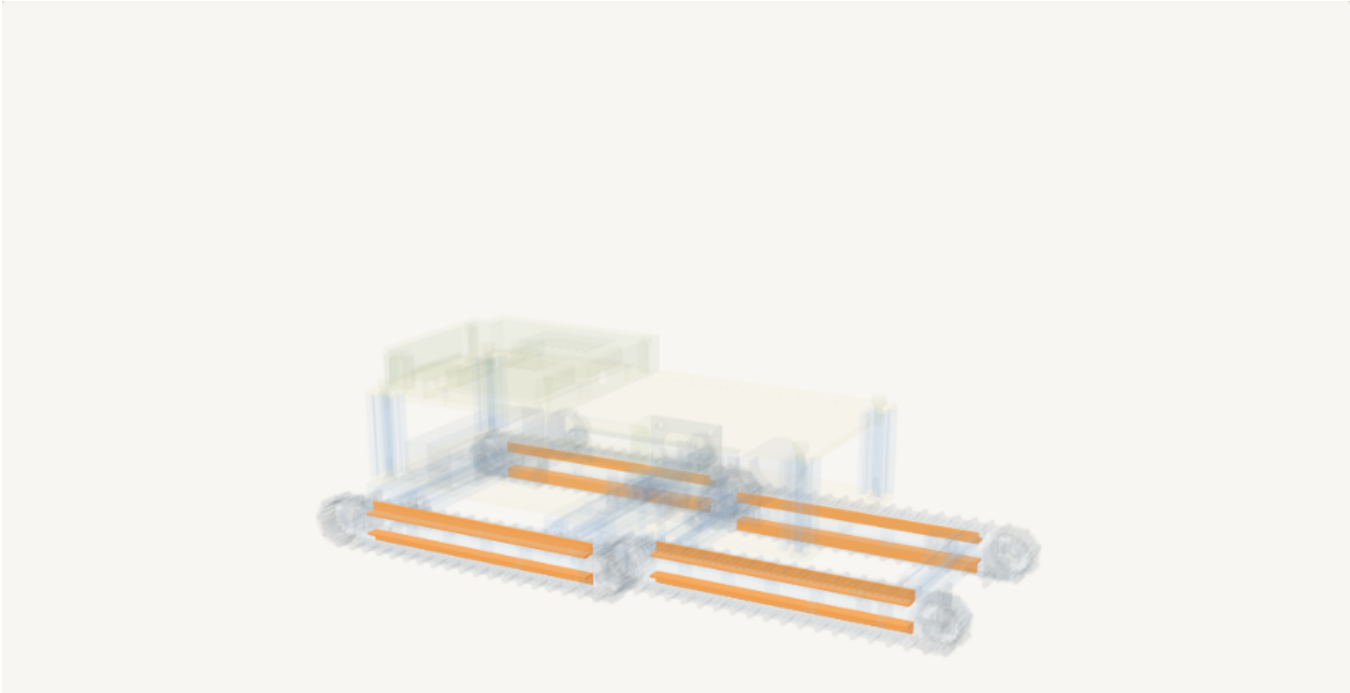
Gives the track a set tension. The jack screw pushes the idler outward along the plate slot until the upper run deflects 2 to 3 mm under a firm push, then the nyloc is tightened. Before this the shoulder screw was held by friction only.

Where to get it

Angle 20 x 20 x 3 cut 28 with an 8 x 6 ear and an M4 tapped hole; a 3 mm aluminium slider 8 x 16 with a Ø6.4 hole; an M4 x 20 jack screw; one M3 x 6 into a new tapped hole in the plate at (223, +10).

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Rail carrier angle (TS-10)



Quantity: 8

MAKE from angle stock

What it does

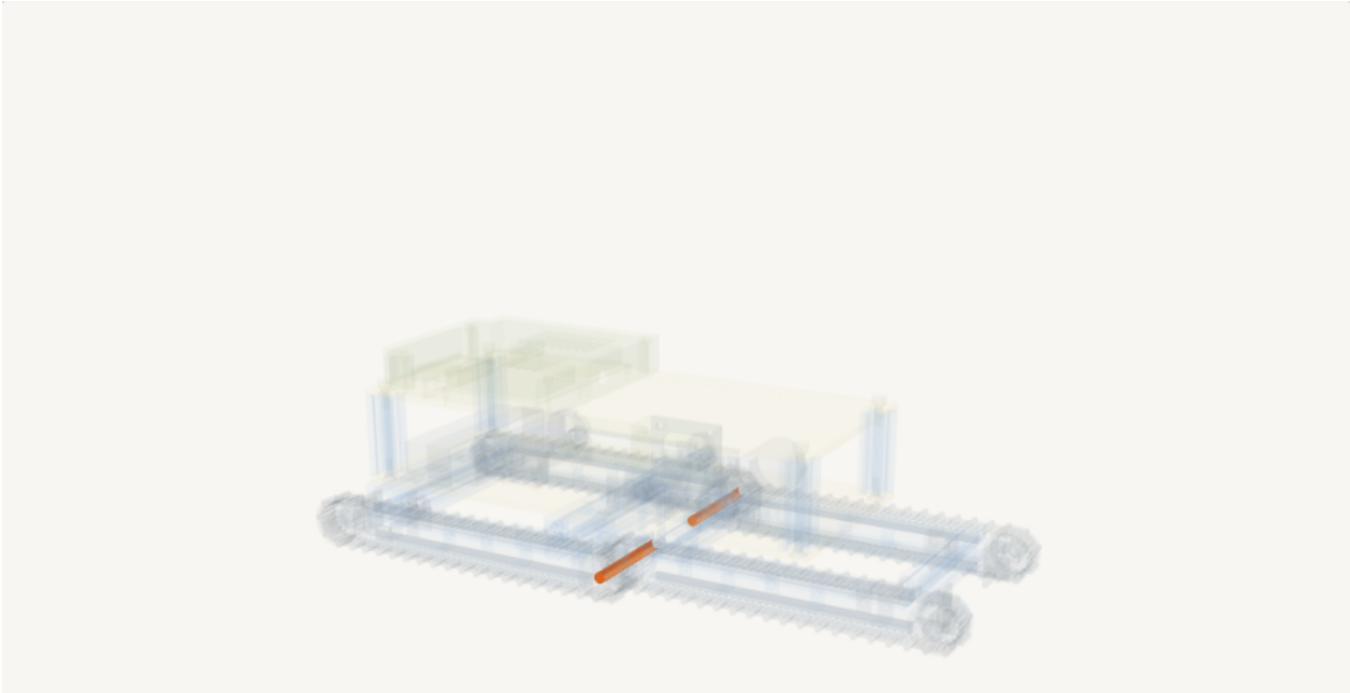
Replaces the four RS 427-991 clips per rail with 186 mm of contact on the plate and on the rail (design review: more bracket contact). Sits on the rail face away from the track.

Where to get it

Aluminium angle 10 x 10 x 1.5, 186 long, one leg cut to 8; 3 x Ø3.4 for the plate screws at the existing clip holes, 3 x Ø2.6 for the thread-forming rail screws. About 1.6 m of angle.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Hinge shaft (Rev B P20 A + B, now plain round)



Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Quantity: 2 (or 1 if one motor)

BUY round shaft, cut

What it does

The hinge axis and the drive. Runs in 688 bearings (reamed Ø16 H7 seats, Loctite 641) in both plates and in the P17 support. Located along its axis by the pinned front hinge pulley between the two plate bearings; three Ø2.5 spring pins per shaft.

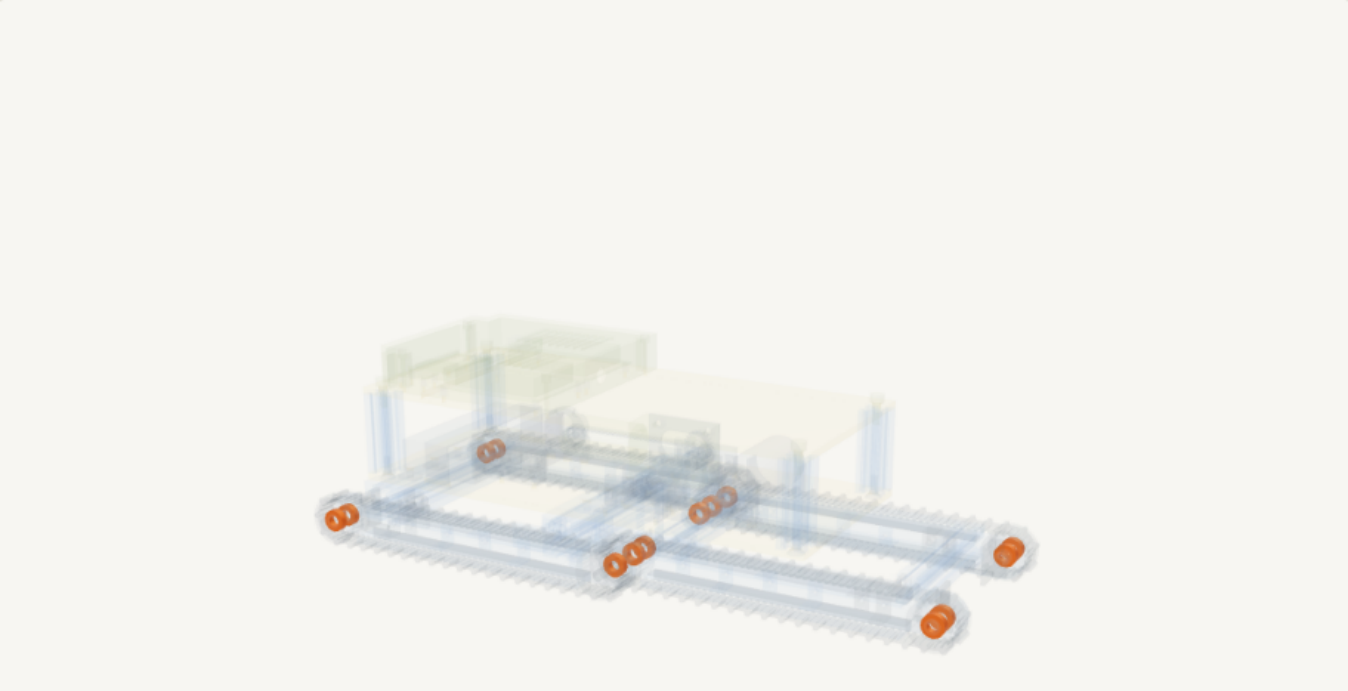
Where to get it

Ø8 h7 ground round shaft, 200 mm stock (MISUMI SG PSFJ8 or RS SG linear shafts, link). Cut to 82.3 (A) and 77.3 (B), chamfer. No D-flat and no E-clip grooves after the design review: the pulleys are pinned (TS-11). One motor per side -> two half shafts.

Link: RS SG: linear shafts

https://sg.rs-online.com/web/c/bearings-seals/linear-motion/linear-shafts/

Ball bearing 688-2RS 8 x 16 x 5



Quantity: 14

BUY: RS Singapore

What it does

Small OD (16) because the plate is only 30 mm tall. Pressed in first, before anything else goes together (build step 1).

Where to get it

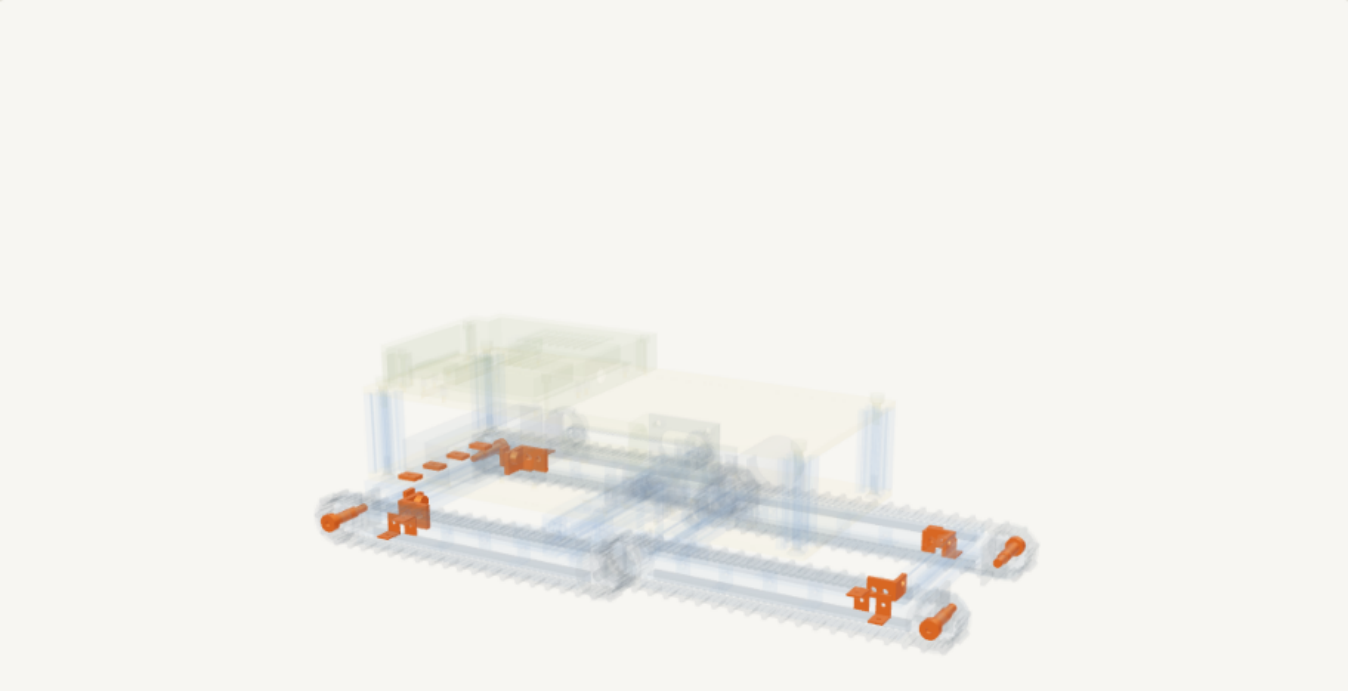
RS SG ball bearings (link): search 688-2RS. 4 in the plate hinge holes, 2 in the P17 supports, 8 inside the four idler pulleys.

Link: RS SG: ball bearings

https://sg.rs-online.com/web/c/bearings-seals/bearings/ball-bearings/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Idler stub axle: shoulder screw Ø8 x 25, M6



Quantity: 4

BUY: MISUMI SG or RS SG

What it does

One per plate end, through the slot; the end wheel spins on two 688s on its shoulder. Slide outwards in the slot and tighten the nut to tension the track.

Where to get it

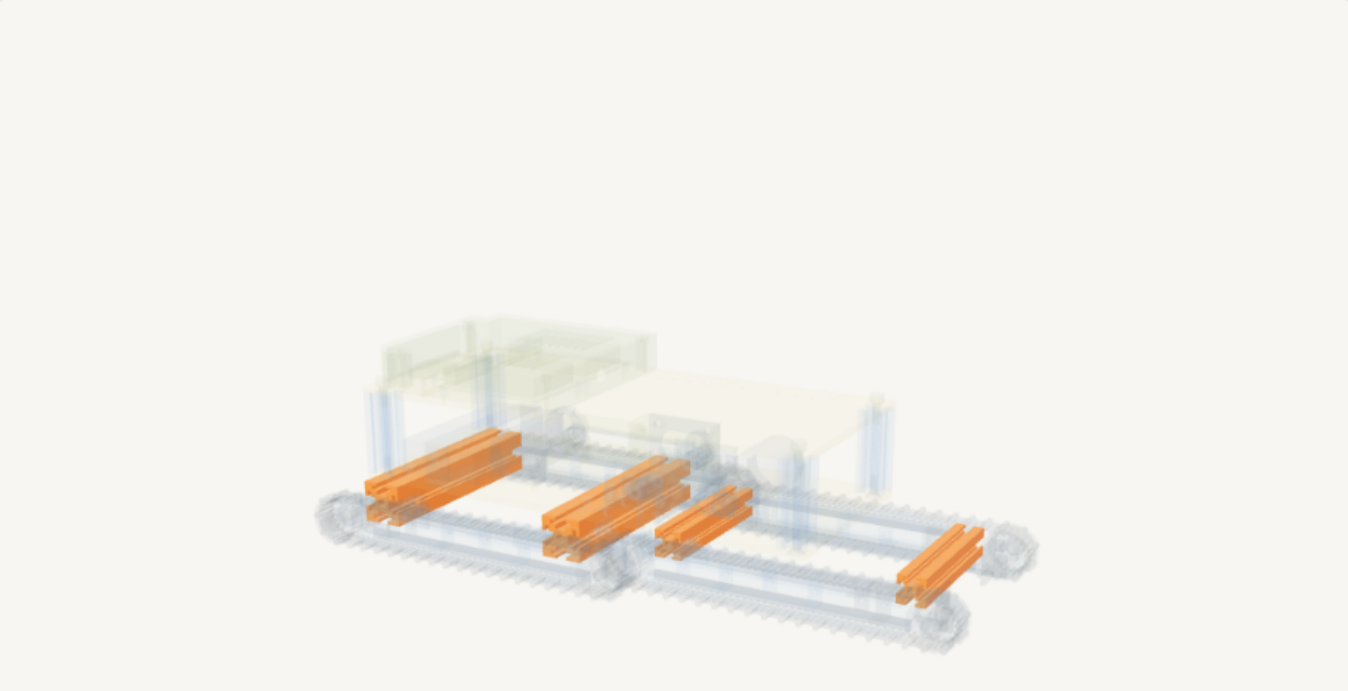
ISO 7379 8 x 25 shoulder screw (link), with 8 x 11 x 1 washer, 8 x 11 x 0.5 shim, 8 x 10 x 9 spacer, M6 washer and nyloc. Rev B sheet 3, idlers.

Link: RS SG: shoulder screws

https://sg.rs-online.com/web/c/fasteners-fixings/screws-bolts/shoulder-screws/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Cross bars: 2020 x 110 and 3030 x 160 (Rev B P7, P8)



Quantity: 2 + 2

BUY profile, cut

What it does

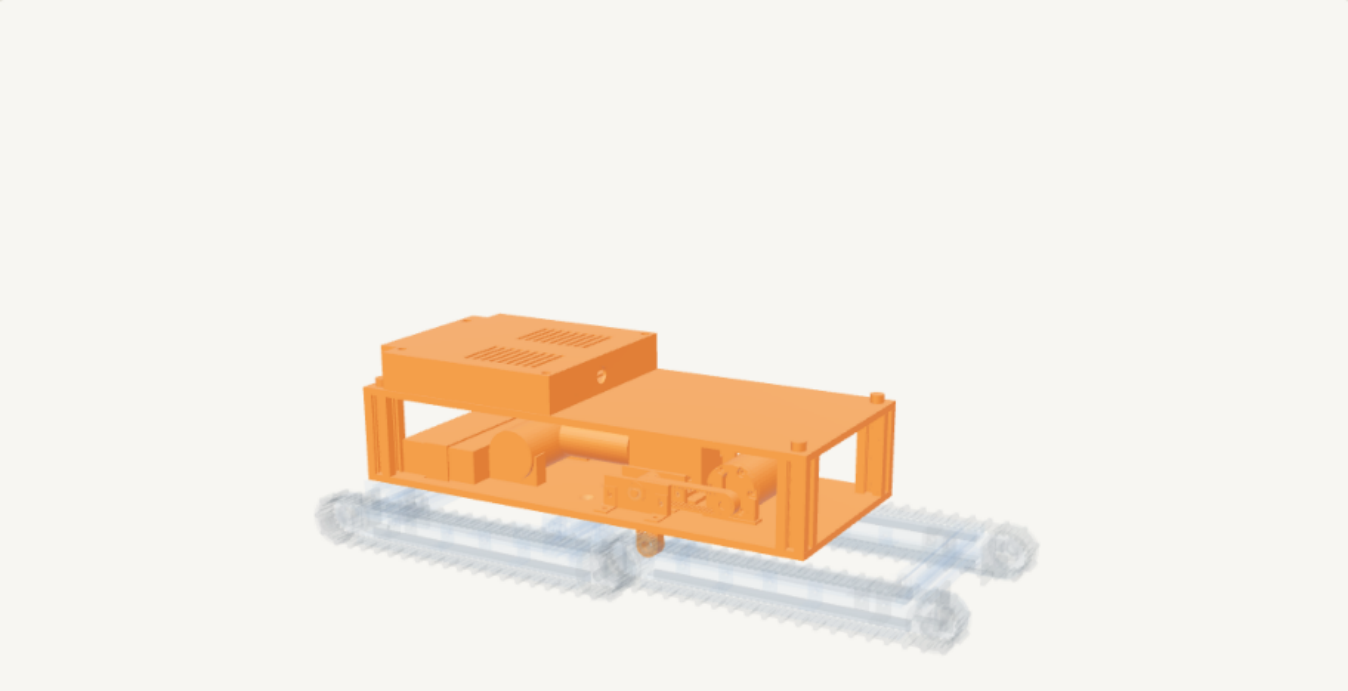
They hold the plates apart and carry the deck. The 2020 bars belong to the front segment and swing with it; the 3030 bars are what the deck bolts to. The near 2020 bar also carries the sector gear angle P14 and rests on the foam up-stops at 0°.

Where to get it

2020 B-type slot 6 (1 m) and 3030 slot 8 (0.5 m) from MISUMI SG or a local supplier; cut per Rev B sheet 19. Corner brackets P18/P19 from aluminium angle, T-nuts M5/M6.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Deck and tray (Rev B P12, P16) and everything on them



Quantity: 1 + 1

Nhien's parts

What it does

The 'box': 160 mm wide between the rear plates. Motors, drivers, battery and the platform mount sit here. Its edges overhang the front track lanes by 12 mm at 25 mm above the axis; slot H lets the sector gear through.

Where to get it

Al 5 mm laser cut, Rev B sheets 6-8; posts P15, controller cover P22, motors, electronics per the Rev B BOM.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Procurement: the track unit on Nhien's Rev B frame

Date 16 Oct 2026, updated 19 Oct (3 x 5 rail screws, stage-2 pins 2.5 x 12). Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien's pack (`RB4107 Rev B`), whose BOM covers everything in section 2. The track comes in two options, A and B; everything else is the same whichever is chosen, so section 2 and the pulleys can be ordered first.

1A. Track, option A: my printed track

Part	Qty	Make / buy	Where	Price	Arrives
Track, TPU 95A, `stl/track_TPU_x4.stl` (drawing TS-01)	4	**Make**: Fab Lab UP300 (TPU head)	Element14 SG TPU 95A filament if the lab has none	~S$45/kg, uses 130 g	filament 1-2 days; 3 h print each
Fit-test slice, `stl/fit_test_track_5mm_TPU.stl`	1	Make	same	-	10 min
Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl`	1	Make, PLA	any printer	-	1 h

Deck: no change needed; the 3 mm cleats sit 5 mm under the deck.

1B. Track, option B: Nhien's bought belt with cleats (Rev B B10)

Part	Qty	Make / buy	Where	Price	Arrives
Track belt HTD 550-5M-15, 110 teeth, 15 wide (e.g. SKF PHG 550-5M-15), drawing TS-01B	4	**Buy**	RS SG, stock no. 058-6535 or MISUMI SG HTD 5M belts	see listing (about S$25-35 each)	RS SG stock: 1-3 days
Cleats, rubber 10 x **5** x 15 (cut down from 7.9, Nhien 19 Oct), bonded on the belt back every 5th tooth (25 mm), 22 per belt	88	**Buy + bond**: belt shop	Local belt / conveyor shop (Fenner, Megadyne or Gates distributor), with the belts	quote	allow 1-2 weeks
Deck change for the belt: **not needed**, the 5 mm cleats pass 2.9 under the deck (Nhien 19 Oct)	0	-	-	-	-

1C. Wheels, both options

Part	Qty	Make / buy	Where	Price	Arrives
Timing pulley HTD 5M 20T, 15 mm belt, flanged (19 Oct: MISUMI economy HTD 5M, code **H5M**, not S5M). **Hinge wheels: 4 + 1 with a finished Ø8 bore**: shaft-hole type H (plain round, no tapped hole) if the configurator offers it for 20T, else P (round with tapped hole, ignore the screw; pin hole 90° from it). **Idlers: 4 with the smallest bore offered**, bored to Ø16 at the workshop (20T stops at about Ø15, so no 16 bore). Before ordering, check on the drawing: width over the flanges 19 or less (the lane is 20) and no hub (shape B has a Ø19 hub)	8 + 1	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x bore Ø16 H7 (idlers, M2) only; the hinge wheels come bored Ø8. Stage-2 pulleys M3 (Nhien's): ream Ø8 H7. Ø2.5 cross holes drilled on our drill press in a V-block, shaft drilled through them at assembly (TS-06, TS-11)	4 + 2	**Make**: Professional Workshop request	E3-01-06	free	workshop queue, allow 1-2 weeks
Spring pins ISO 8752 2.5 x 24 (hinge pulleys, replaces E-clips F22 and the D-flats)	4 + 4 spare	**Buy**	RS SG spring pins	~S$5 / pack	RS SG stock
Spring pins ISO 8752 2.5 x 12 (stage-2 pulleys M3: **pinned, Nhien 19 Oct** (Ø17.5 OD, tooth roots about Ø13.4: a longer pin would reach into the belt grooves))	2 + 2 spare	**Buy**	RS SG spring pins	~S$5 / pack	RS SG stock
~~Set screws M4 x 5 cone point~~: **not needed**, the stage-2 pulleys are pinned (Nhien 19 Oct)	0	-	-	-	-

Part	Qty	Make / buy	Where	Price	Arrives
Hinge shaft: plain round Ø8 h7 ground steel, 300 mm (100 mm to practise on) (replaces the D-shaft B13; cut to 82.3 and 77.3, no grooves)	1	**Buy**, cut	RS SG linear shafts or MISUMI SG PSFJ8	~S$10	RS SG stock

1D. Tension and rail support (from the design review), both options

Part	Qty	Make / buy	Where	Price	Arrives
Tensioner tab, Al angle 20 x 20 x 3 cut 30 with a 7.5 x 8 ear, M4 tapped (TS-09)	4	**Make**: hacksaw, drill, M4 tap	from BOM B16 angle (order 300 mm instead of 100)	-	1 h
Tensioner slider, an 8 mm slice of the same angle, leg 19 with Ø6.4, foot 8 (TS-09; replaces the M6 washer under the nut)	4	**Make**: hacksaw, drill	from the same angle (order 400 mm)	-	30 min
Thread-forming screws 3 x 5 pan head for plastics (rail to carrier angle, 3 per rail; 3 x 5 not 3 x 10: the rail is 5 thick)	24 + 6 spare	**Buy**	RS SG thread-forming screws	~S$6 / pack	RS SG stock
M3 x 6 SHCS ISO 4762 (carrier angles to plates 24, tensioner tab ears 4)	28 + 4 spare	**Buy**	RS SG	~S$5 / pack	RS SG stock
Jack screw M4 x 20 socket head + M3 x 6 for the tab	4 + 4	**Buy**	RS / local	-	stock
New M3 tapped holes in P1 and P2: (x +-223, y +11) for the tab (TS-09) and six at (x +-40, +-105, +-170; y +-3.5) for the carriers (TS-10)	4	**Make**: drill 2.5 + tap M3 on site (Nhien 19 Oct: no DXF re-issue)	with the plates	-	with the plates
Rail carrier angle, Al 10 x 10 x 1.5 x 186, plate leg cut to 6.5, 3 + 3 holes (TS-10; replaces the 32 RS 427-991 clips)	8	**Make**: from 1.6 m of angle	local metal supplier / MISUMI SG	~S$10	1-3 days
Rails P3-P6 trimmed from 189 to 186 at the idler end (TS-10, TS-11)	8	Make: hacksaw	-	-	10 min

2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
~~Rail brackets RS PRO 427-991~~: **do not order**, replaced by the TS-10 carrier angles (1D)	0	-	-

Part	Qty	Make / buy	Where
Hinge shafts P20 A 82.3 + B 77.3, now plain round Ø8 h7 (section 1C), no D-flat, no E-clip grooves	2	Buy stock, cut	see 1C
Sector gear P11: proposed KHK SSAY1-100 (module 1, 100 teeth, 20°, face 6, bore 10, S45C, flat plate) cut to a 23-tooth sector (plan 5.14); **Nhien OK 19 Oct**: centre distance 60.1 to 60.2, base already changed for the new sector	1	Buy, cut	KHK catalog (thin-face hubless SSAY series, about US$43) or MISUMI SG (KHK brand), search SSAY1-100; plan B any 1M 100T 45# steel gear from Shopee / AliExpress, faced to 6 mm at the Professional Workshop
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets: standard cast 2020 and 3030 inside corner brackets instead of the P18 / P19 angle offcuts (bigger contact area, design review); T-nuts, shims 8x11x0.5 and 8x11x1; E-clips F22 no longer needed	-	Buy	MISUMI SG / local
Deck P12 (slot H 3 mm longer, see pack page 2), tray P16, cover P22, motors, drivers, ESP32, battery	-	Theirs	Rev B BOM
Bearing seats in P1, P2, P17: laser-cut Ø15.8, reamed Ø16 H7 on site, 688-2RS retained with Loctite 641 (TS-11)	6	Make	Professional Workshop, 16 H7 reamer

3. Choosing between A and B

19 Oct: option B, Nhien's belts. He has cut the cleats to 5 mm (no deck notch) and will have the belt maker bond them, since they need a chemical bond to the belt. Allow 1 to 2 weeks for that, so the belts go out first. Option A (printed track) stays as the fallback if the bonded belts are late.

4. Order of events (19 Oct)

1. Mon 19 to Tue 20 Oct: belts and cleats to the belt maker (Nhien, 1 to 2 weeks: now on the critical path). Check the MISUMI H5M 20T 150 drawing (19 or less over the flanges, no hub), then order the pulleys (4 + 1 bored 8, 4 idlers smallest bore), the RS order (shaft, pins, screws, jack screws) and the aluminium angle. Print the fit-test slice and stand-in pulley (option A).
2. 22 to 29 Oct: plates from Nhien's existing DXF (no re-issue); drill and tap the new M3 holes on site. As parts arrive, make the tabs, sliders and carrier angles, trim and drill the rails, drill the hinge wheels' Ø2.5 holes.
3. When the idlers arrive (about 23 to 26 Oct): one workshop request for the idler bores (16 H7), the stage-2 pulley reams (8 H7) and, once the plates are back, the bearing seats (16 H7). Allow 1 to 2 weeks.
4. 9 to 10 Nov: assemble; 11 to 13 Nov: tension and run tests; 16 Nov: advisor demo.

Layout with Nhien's side plates (Rev B P1 / P2): what changes on the track side

Read from `RB4107 Rev B/STEP` (plates P1, P2 and the full assembly, 15 Oct 2026). Rev B's frame origin is the hinge axis with the A side at z < 60; below everything is converted to **z from the vehicle centre line**, one side (the other mirrors), mm at 1:5.

Their plates

	Rev B plate	My printed plate (for comparison)
Material	5 mm aluminium 6061, laser cut, holes tapped	7 mm PETG, printed
Size	255 x 30.4 (R15 ends)	265 x 45 (R20 ends)
Axle spacing	225 (hinge hole to the middle of the idler slot)	225
Hinge hole	Ø16 for a 688-2RS bearing (8 x 16 x 5)	Ø22.2 for a 608ZZ
End hole	slot 220.4-232.6 for an Ø8 shoulder-screw stub axle: slide to tension the track	round pocket
Rail	separate UHMW strips P3-P6, screwed on through the C holes with small brackets	printed as part of the plate
Cross bars	2020 profile x 110 (front), 3030 profile x 160 (rear), corner brackets + T-nuts	M4 rod + printed spacer

Where everything sits across the width (per side, from the centre line)

z	What
0 to 24 (A side) / 29 (B side)	free: hinge shaft ends here, hinge pinion / sector gear region
36 to 41	P17 hinge-shaft bearing support (688 bearing), bolted to the middle 3030 bar
55 to 60	**front plate P1** (688 bearing in it at the hinge)
60.5 to 79.5	front segment track lane: pulley 19 over flanges, track 62.5 to 77.5
80 to 85	**rear plate P2** (688 bearing at the hinge)
85.5 to 104.5	rear segment track lane, **outboard of the rear plate**; track 87.5 to 102.5
106	end of the hinge shaft

So: 205 over the track faces, 209 over the pulley flanges, and the frame between the rear plates is 160 wide inside (not 80 as in my layout): the box can be up to 160 wide.

Consequences for the parts I was making

Part	With their plates

Track x4	unchanged: 550 loop, 110 teeth, 15 wide, 45 cleats. Still printed. The only printed part left.
Side plates	theirs (P1, P2). Not printed.
Spacers	**none.** The 2020 and 3030 bars hold the plates apart.
Cross bar rods, nuts, washers	**none.** Replaced by the bars, P18/P19 corner brackets, T-nuts.
Middle axle 230	replaced by Rev B's two hinge half-shafts P20, now Ø8 h7 plain round (design review: no D-flat, no E-clip grooves; the pulleys are pinned, TS-11), A = 82.3 long (z 14 to 106 on the A side), B = 77.3 (z 31 to 106 on the B side). Two halves because Rev B has one motor per side. If the team goes to **one motor**, use one Ø8 D-shaft 212 long (z -106 to +106) instead, same stock.
Front and rear axles	**none.** The end pulleys ride on Ø8 x 25 M6 shoulder screws (ISO 7379) through the plate slots, one per plate, 4 in all, each with two 688-2RS inside a Ø16-bored pulley.
Bearings	608ZZ x10 -> **688-2RS x14** (8 x 16 x 5): 4 in the plate hinge holes, 2 in the P17 supports, 8 in the idler pulleys.
Pulleys	same HTD 5M 20T 15 mm belt, but the lanes are only 19-20 wide, so there is **no room for a set-screw hub**. They are modified: the 4 on the hinge shaft are reamed Ø8 H7 and cross-pinned to the plain shaft (TS-06, TS-11); the 4 idlers are bored Ø16 H7 for two 688 bearings each.
Set screws	none on the pulleys: one Ø2.5 spring pin per hinge pulley drives it and locates it (design review). No E-clips; shims 8x11x0.5 and feeler gaps per TS-11.

What I need from Nhien to finish the hybrid

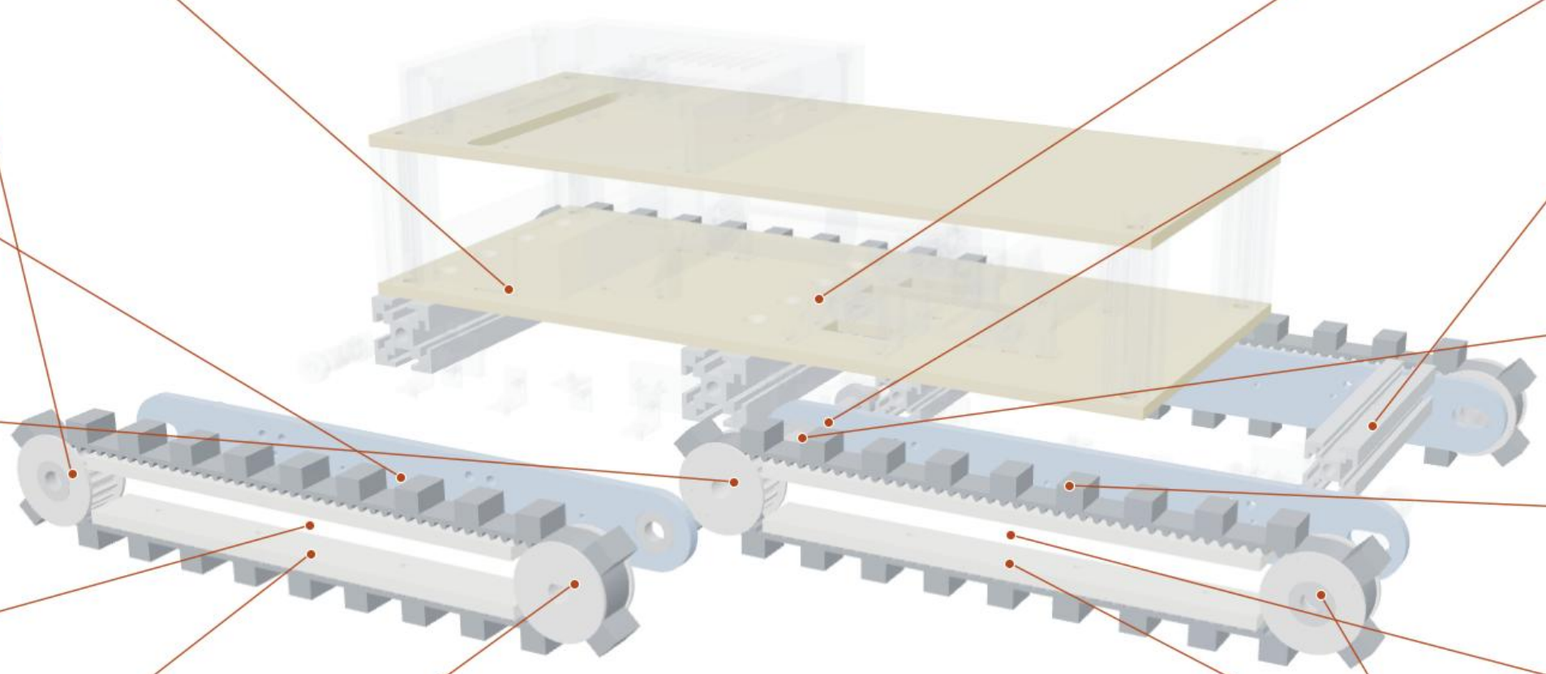
1. One motor or two? It decides whether the hinge shaft is one piece (212) or two (82.3 + 77.3).
2. Confirm the rail strips P3-P6 are coming (MISUMI UPA-189-20-5 UHMW, 8 off) or whether the lower run just rides on nothing; my track's tooth tips sit 13.34 below the axle and need the rail.
3. The track is still mine to print; the step-matched cleat pitch (12.22) replaces Rev B's 22 cleats at 25 mm on a stock belt. Both fit the same pulleys, so either can be tested on their frame.

Design review, 16 Oct 2026

The reviewer asked for tolerances and part spacing, no E-clips on the axle, no D-shaft for the pulleys, positive track tension and more bracket contact. Answered by TS-06 (reamed, pinned pulleys), TS-09 (idler tensioner), TS-10 (rail carrier angles, rails trimmed to 186), TS-11 (stack-ups and fits) and the feasibility pages of the pack. Nhien's P20 and M1 / M3 rows are superseded by these sheets.

How it all connects: one side pulled apart, every joint and its fastener

Nhien's Rev B frame with the track. Near side exploded outwards from the hinge shaft (plate, then track and wheels, then rear plate, then rear track); far side left assembled. Bought parts grey, Nhien's blue, deck yellow, electronics ghosted. Fasteners updated to the 16 Oct des



14 CROSS BARS 3030 x 160 (P8)
Hold the rear plates 160 apart and carry the deck: 8x M5x12 csk + slot-8 T-nuts. P17 hinge supports on the middle bar: 2x M5x10 + T-nuts.

11 REAR IDLER WHEEL (M2)
Same shoulder-screw set and tensioner as the nose idler, through the rear plate's slot.

8 REAR PLATE P2 (5 mm Al)
688 in the reamed hinge seat; rear idler on a shoulder screw through its slot with its own TS-09 tensioner. Joins the 3030 bars with cast corner brackets or P18: M6x10 + slot-8 T-nut into the bar, 2x M5x8 into the plate.

2 HINGE WHEEL, front (M1)
HTD5M 20T, bore reamed Ø8 H7 on the plain round shaft. One Ø2.5 x 24 spring pin through wheel and shaft (drilled at assembly) drives it and holds it: shim 8x11x0.5 to the plate bearing, 0.3-0.5 feeler gap the other side.

9 REAR TRACK
Same track, outboard of the rear plate, on the rear hinge wheel and the rear idler; tensioned at the rear idler.

12 SLIDE RAILS P5 / P6 (trimmed to 186)
Same carrier angle and screws as the front rails, on the outboard face of the rear plate.

10 HINGE WHEEL, rear (M1)
Outermost on the hinge shaft, reamed Ø8 H7, pinned with 0.3-0.5 gap to the P2 bearing. The shaft ends 1.5 past it.

15 DECK P12, posts, tray, motors
Posts P15: M6x16 csk from below; tray P16: M6x16 from top. Drive motors on P9 shim + Pololu bracket (M3x12). Hinge motor 5840 on its L-bracket: 4x M4x8 csk, 2x M5x8 into the deck. Slot H 74 long; option B needs the lane notches.

4 HINGE SHAFT P20 (Ø8 h7 plain round)
No D-flat, no E-clip grooves (design review). Runs in 688-2RS in reamed Ø16 H7 seats (Loctite 641) in the front plate, the rear plate and P17. Located by the pinned front wheel; three pins per shaft. The stage-2 pulley on it drives this side.

13 CROSS BARS 2020 x 110 (P7)
Hold the two front plates 110 apart and swing with them. Sector gear P11 on angle P14 on the hinge-side bar: 2x M3x8 csk into the gear, 2x M4x8 csk + T-nuts into the bar. Rests on the foam up-stops at 0 degrees.

7 BEARING 688-2RS
8 x 16 x 5 in a seat laser-cut Ø15.8 and reamed Ø16 H7, retained with Loctite 641. 14 in all: 4 in the plates, 2 in P17, 8 in the idler wheels.

5 FRONT PLATE P1 (5 mm Al)
688 in the reamed Ø16 H7 hinge seat. Joins the 2020 bars with cast corner brackets (review) or P19: M5x8 + slot-6 T-nut into the bar, 2x M4x8 into the tapped plate. New M3 tapped hole at (223, +10) for the tensioner tab.

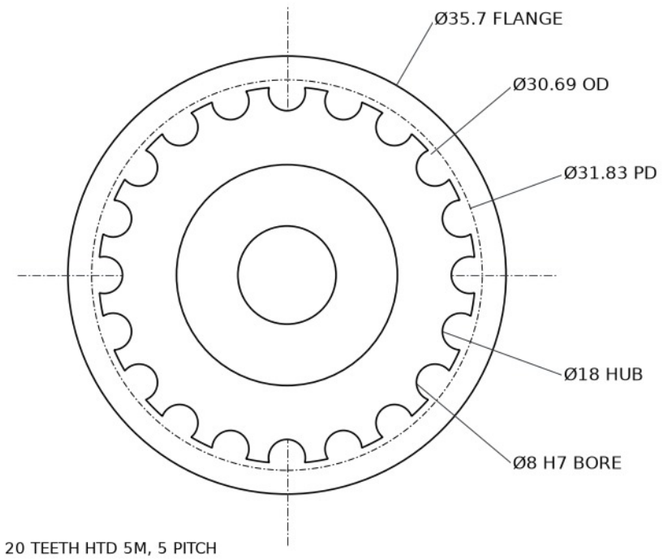
1 FRONT TRACK (option A printed TPU, or option B belt)
Teeth mesh in the wheel grooves, no fastener. Tension: the TS-09 jack screw pushes the nose idler out along the plate slot to 2-3 mm of deflection; nut tightened.

6 SLIDE RAILS P3 / P4 (UHMW, trimmed to 186)
On a full-length TS-10 carrier angle: 3x M3x6 into the plate's clip holes, 3x 3x10 thread-forming into the rail. The track's lower run slides on P4.

3 NOSE IDLER WHEEL (M2)
Bore 16 H7, 2x 688-2RS inside with the 8x10x9 sleeve. Spins on an ISO 7379 Ø8x25 M6 shoulder screw through the plate slot: head washer 8x11x1, shim 8x11x0.5, TS-09 slider + nyloc inboard, tensioner tab with M3x6.

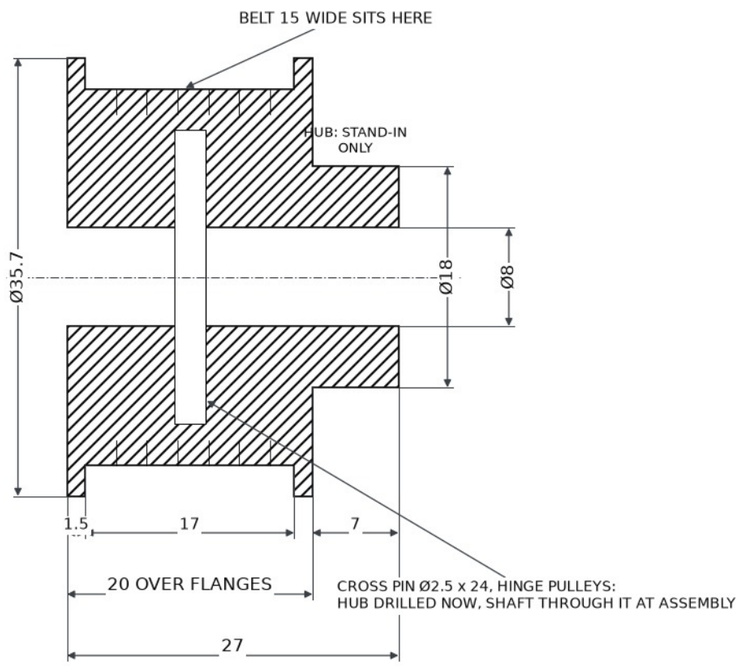
Assembly order (Rev B sheet 3, reviewed): 1 bearings into reamed seats, P17 and idlers 2 rear frame: P2 + 3030 bars + brackets, rails on carriers 3 front frame: P1 + 2020 bars + brackets + sector gear, rails on carriers 4 hinge wheels, shafts in, shims and feeler gaps set, pins drilled and driven 5 deck, posts, tray 6 motors, belts 7 idlers on shoulder screws with sliders and tensioner tabs, wind the jack screws, tighten the nuts 8 electronics, cover. Fasteners from Rev B sheet 3 and TS-09 / TS-10 / TS-11.

FRONT VIEW



20 TEETH HTD 5M, 5 PITCH

SECTION (ALONG THE AXLE)



NOTES:

1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, FLANGES BOTH SIDES, PLAIN PILOT BORE.
   MISUMI SINGAPORE (APPROVED VENDOR). 8 OFF, PLUS 1 TO PRACTISE THE BORING AND PINNING ON.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83) AND THE 15 MM BELT. ON NHIEN'S PLATES THE LANE IS 19-20 WIDE, SO NO HUB FITS:
   THE HUB AND SET SCREW DRAWN ARE THE PLA STAND-IN'S ONLY.
3. WORKSHOP: 4 x REAM Ø8 H7 (HINGE SHAFT, PLAIN ROUND, NO D-FLAT). Ø2.5 CROSS HOLE AT MID-FACE: DRILL THE HUB NOW,
   ON THE PRESS IN A V-BLOCK. AT ASSEMBLY THE SHAFT IS DRILLED THROUGH THE HUB HOLE IN PLACE. SPRING PIN ISO 8752 2.5 x 24.
4. THE PIN DRIVES AND LOCATES THE PULLEY: NO GRUB SCREW, NO E-CLIPS (TS-11).
   4 x BORE Ø16 H7 THROUGH FOR 2 x 688-2RS EACH (IDLERS ON Ø8 SHOULDER SCREWS, SLEEVE 8 x 10 x 9 BETWEEN).
5. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

TIMING PULLEY HTD 5M 20T (BOUGHT, PLAIN BORE)

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: ALUMINIUM (VENDOR'S)

FINISH: VENDOR'S

DWG NO. TS-06

QTY. 8

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

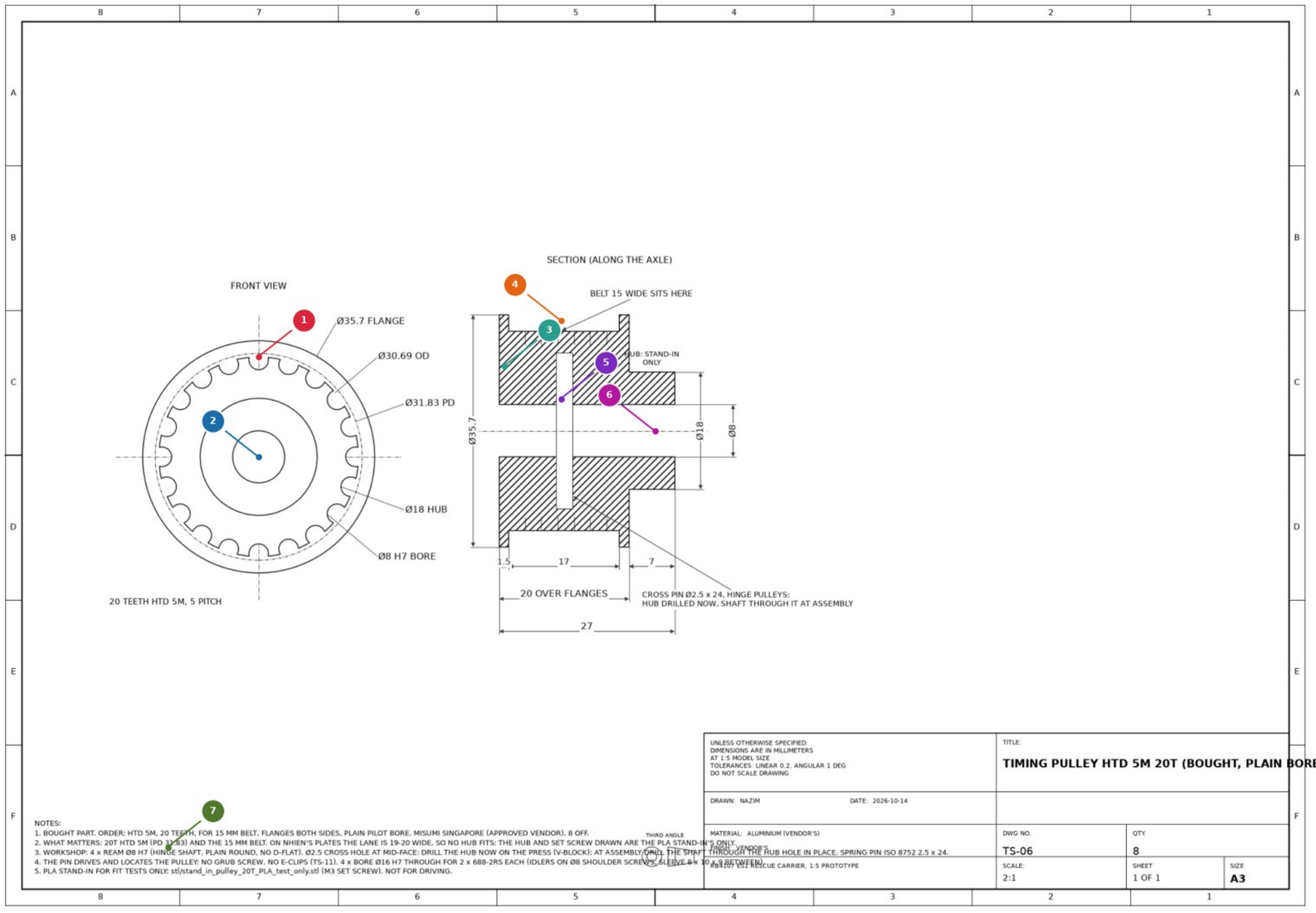
SCALE: 2:1

SHEET 1 OF 1

SIZE **A3**

How to read TS-06: the toothed wheel (bought, reamed, pinned)

Markers below sit on a reduced copy of sheet TS-06 (previous page, full size). Numbers match the key on the right.



1 Teeth

20 HTD 5M teeth on a 31.83 pitch circle: the same tooth form as the track's inside, so they mesh without slip.

2 Bore

Bought with a small pilot bore. The 4 hinge wheels and the 2 stage-2 pulleys are reamed Ø8 H7 to run on the plain round shaft; the 4 idlers are bored Ø16 H7 for two 688 bearings each.

3 Flange

Ø35.7 flanges both sides, 1.5 thick, keep the track on the wheel. 19 over the flanges, 0.5 clear of each plate in the 20 lane.

4 Belt face

16 between the flanges; the 15 wide track sits here.

5 Cross pin

Ø2.5 x 24 spring pin through wheel and shaft at mid-face: the hub is drilled on the press when the pulley is bored, the shaft is drilled through that hole at assembly, in place. It drives the wheel and holds it along the shaft: no D-flat, no grub screw, no E-clips (design review, TS-11).

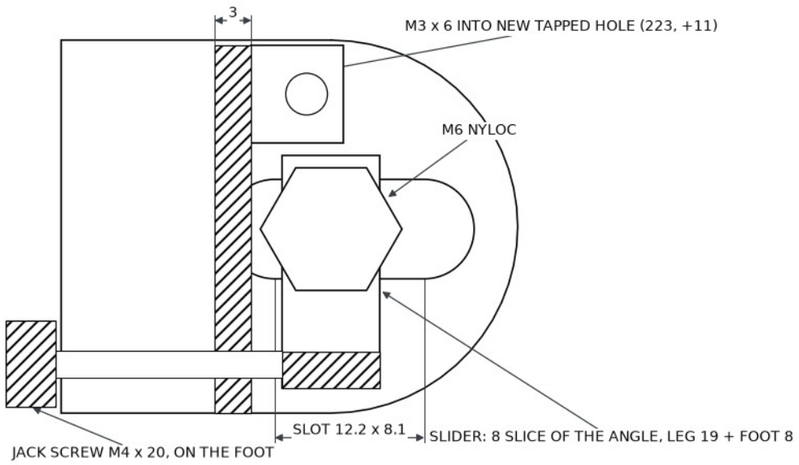
6 Hub

Only the printed stand-in has a hub and set screw. The bought wheel is a plain flanged disc, because the lane has no room for a hub.

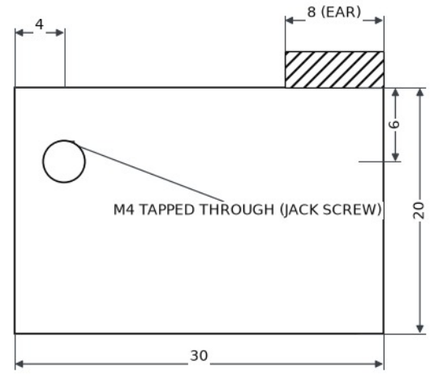
7 Notes

Order 8 from MISUMI SG plain bore; the Professional Workshop reams and bores them; pins go in on the build day.

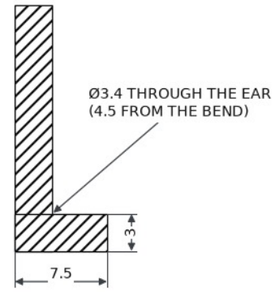
NOSE END OF P1, SEEN FROM INBOARD (THE z 55 FACE)



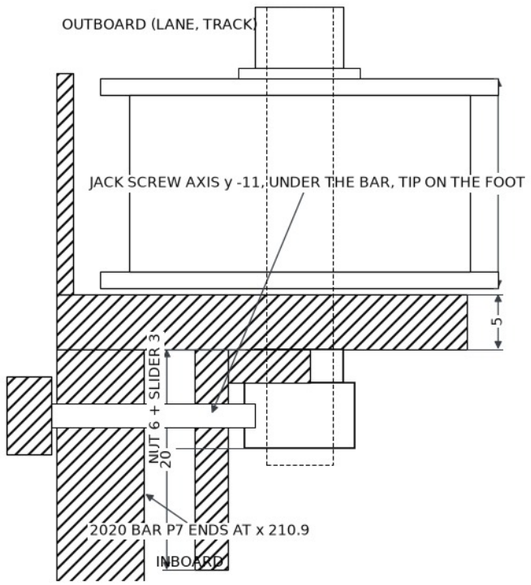
TAB: AL ANGLE 20x20x3, CUT 30, LEG A CUT TO A 7.5 x 8 EAR



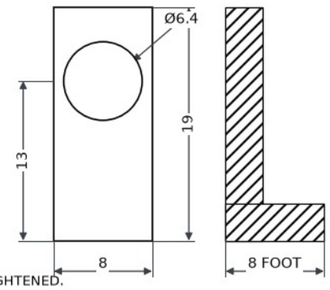
TAB, END VIEW



PLAN FROM ABOVE: PLATE, LANE, IDLER, TAB



SLIDER: 8 SLICE OF THE 20x20x3 ANGLE, FACE AND END VIEW



NOTES:
1. THE SHOULDER SCREW IS CLAMPED IN THE SLOT BY FRICTION ONLY. THE JACK SCREW SETS THE IDLER, THEN THE NUT IS TIGHTENED.
2. SLIDER REPLACES THE M6 WASHER UNDER THE NUT (3 THICK). TAB: NEW M3 TAPPED HOLE IN P1 / P2 AT x = +-223, y = +11 (ADD TO THE DXF, OR DRILL 2.5 AND TAP ON SITE).
3. FIT: IDLER INTO THE LOOP, SHOULDER SCREW THROUGH, SLIDER, NUT LOOSE, THEN THE TAB. TENSION (MACHINE ON BLOCKS): HOOK A CLEAT OF THE LOWER RUN AT MID-SPAN, PULL IT OFF ITS RAIL WITH 4 N, WIND THE JACK SCREW UNTIL IT GIVES 2-3 MM, TIGHTEN THE NUT, BACK OFF 1/4 TURN. PUSHING A RUN INWARD MOVES NOTHING: THE RAILS SIT UNDER BOTH RUNS.
4. DO NOT PUSH THE IDLER IN PAST x = 223 (RAIL END): TRAVEL 223 TO 232.6 = 9.6 MM = 19 MM OF LOOP. JACK SCREW M4 x 20 SOCKET HEAD, KEY FROM THE HINGE SIDE, UNDER THE BAR; IT PUSHES THE SLIDER'S FOOT. LEG B 1.5 OFF THE SLOT END.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

IDLER TENSIONER: TAB, SLIDER AND JACK SCREW

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: AL ANGLE 20x20x3; AL PLATE 3

FINISH: DEBURRED

DWG NO. TS-09

QTY. 4 SETS

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

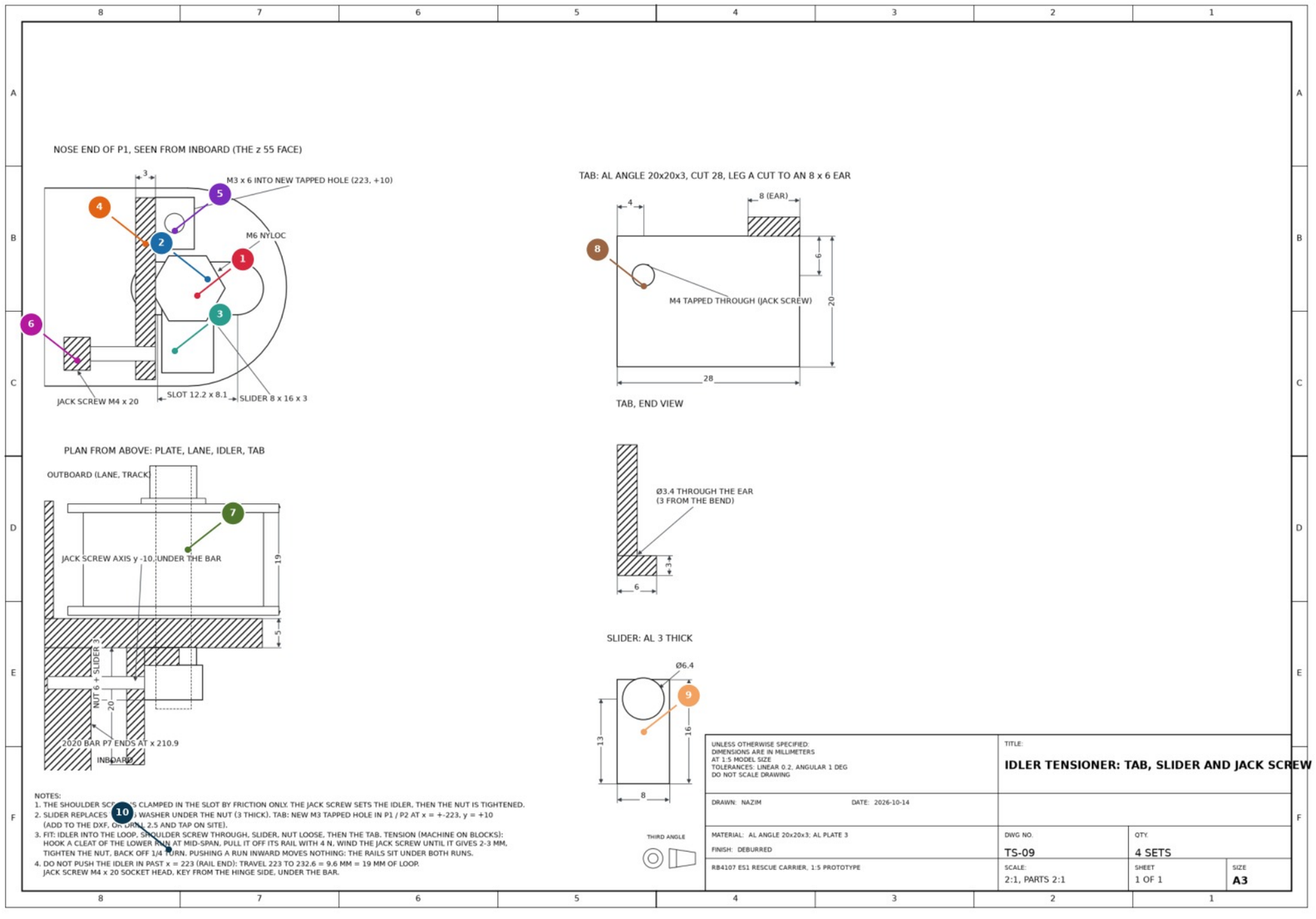
SCALE: 2:1, PARTS 2:1

SHEET 1 OF 1

SIZE **A3**

How to read TS-09: the idler tensioner

Markers below sit on a reduced copy of sheet TS-09 (previous page, full size). Numbers match the key on the right.



1 Slot and shoulder screw

The idler's Ø8 shoulder screw sits in the plate's 12.2 x 8.1 slot. Sliding it outward tightens the track; before this sheet only the nut's friction held it.

2 Nyloc nut

M6 nyloc on the inboard side. Tightened last, after the jack screw has set the tension.

3 Slider

An 8 mm slice of the same 20 x 20 x 3 angle in place of the M6 washer: a 19 mm leg on the plate with the Ø6.4 hole, and an 8 mm foot standing off the plate. The jack screw pushes on the foot, so nothing pushes on the nut.

4 Tab, leg B

The angle's long leg stands off the plate, inboard. The M4 jack screw threads through it along the slot direction.

5 Ear and M3

The short leg is cut down to a 7.5 x 8 ear above the slot and bolted to the plate with one M3 x 6 into a new tapped hole at (223, +11). Leg B stands 1.5 mm clear of the slot end, so the nut's corners never touch it.

6 Jack screw

M4 x 20 socket head. Wind it in until the lower run, hooked at a cleat at mid-span and pulled off its rail with 4 N (machine on blocks), gives 2 to 3 mm; pushing a run inward moves nothing because the rails sit under both runs, tighten the nut, then back it off a quarter turn.

7 Plan: the lane

Seen from above: plate P1 at z 55 to 60, the idler wheel in the lane with the screw head outside, the nut, slider and tab inboard, the 2020 bar ending at x 210.9 just behind the tab.

8 Tab, flat

Angle 20 x 20 x 3 cut 30 long; the M4 hole is 4 from the bottom end and 6 from the bend; the ear is the top 8 of the other leg, trimmed to 7.5 wide.

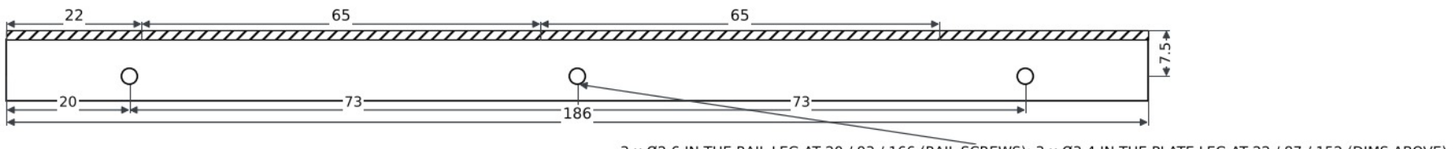
9 Slider

8 mm slice of the angle: leg 19 with the Ø6.4 hole 13 from the bottom, foot 8 bent inboard.

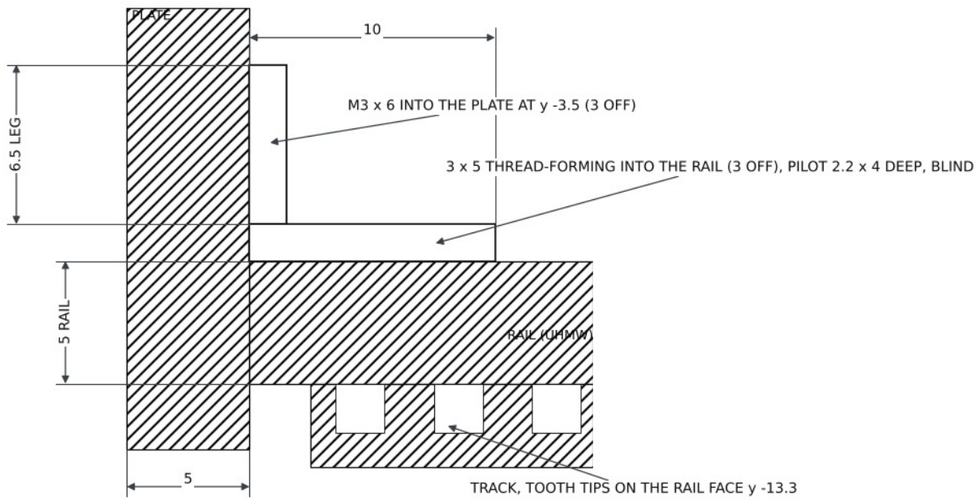
10 Notes

Four sets: nose and rear idler on each side. The rear tab mirrors the front; the jack screw always pushes outward, never in past x 223.

PLAN OF ONE ANGLE, 1:1 (RAIL LEG FLAT)



SECTION ACROSS THE LANE AT THE LOWER RAIL (4:1): PLATE, ANGLE, RAIL, TRACK



NOTES:

1. WHY: EACH UHMW RAIL HUNG ON 4 SMALL CLIPS (10 x 14.6 FEET). THE ANGLE RUNS THE FULL RAIL LENGTH: 186 x 10 OF CONTACT ON THE PLATE AND ON THE RAIL INSTEAD OF 4 SMALL PATCHES.
2. TRIM EACH RAIL FROM 189 TO 186 (3 OFF THE IDLER END): THE IDLER PULLEY FLANGE REACHES x 207 AT ITS INNERMOST SLOT POSITION AND THE RAIL ENDED AT 207 (TS-11).
3. PLATE LEG CUT TO 6.5 SO THE UPPER AND LOWER ANGLES DO NOT MEET ON THE PLATE'S CENTRELINE. 3 x Ø3.4 IN THE PLATE LEG AT x 40, 105, 170 FROM THE HINGE, y +-3.5: SIX NEW M3 HOLES IN P1 / P2 (THE OLD CLIP HOLES ON THE CENTRELINE ARE NOT USED). 3 x Ø2.6 IN THE RAIL LEG AT 20, 93, 166 FROM THE RAIL'S HINGE END (x 38, 111, 184), 7.5 FROM THE PLATE FACE: 3 x 5 THREAD-FORMING SCREWS. DO NOT DRILL THE OLD CLIP PILOTS IN THE RAIL: THEY OVERLAP THE NEW ONES AT x 38 / 40 AND 184 / 185.
4. THE ANGLE SITS ON THE RAIL FACE AWAY FROM THE TRACK: UNDER THE UPPER RAIL, ON TOP OF THE LOWER RAIL. THE TRACK'S TOOTH TIPS RIDE ON THE OTHER FACE (y 13.3).

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

RAIL CARRIER ANGLE (REPLACES RS 427-991 CLIPS)

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: AL ANGLE 10x10x1.5

FINISH: DEBURRED

DWG NO. TS-10

QTY. 8

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

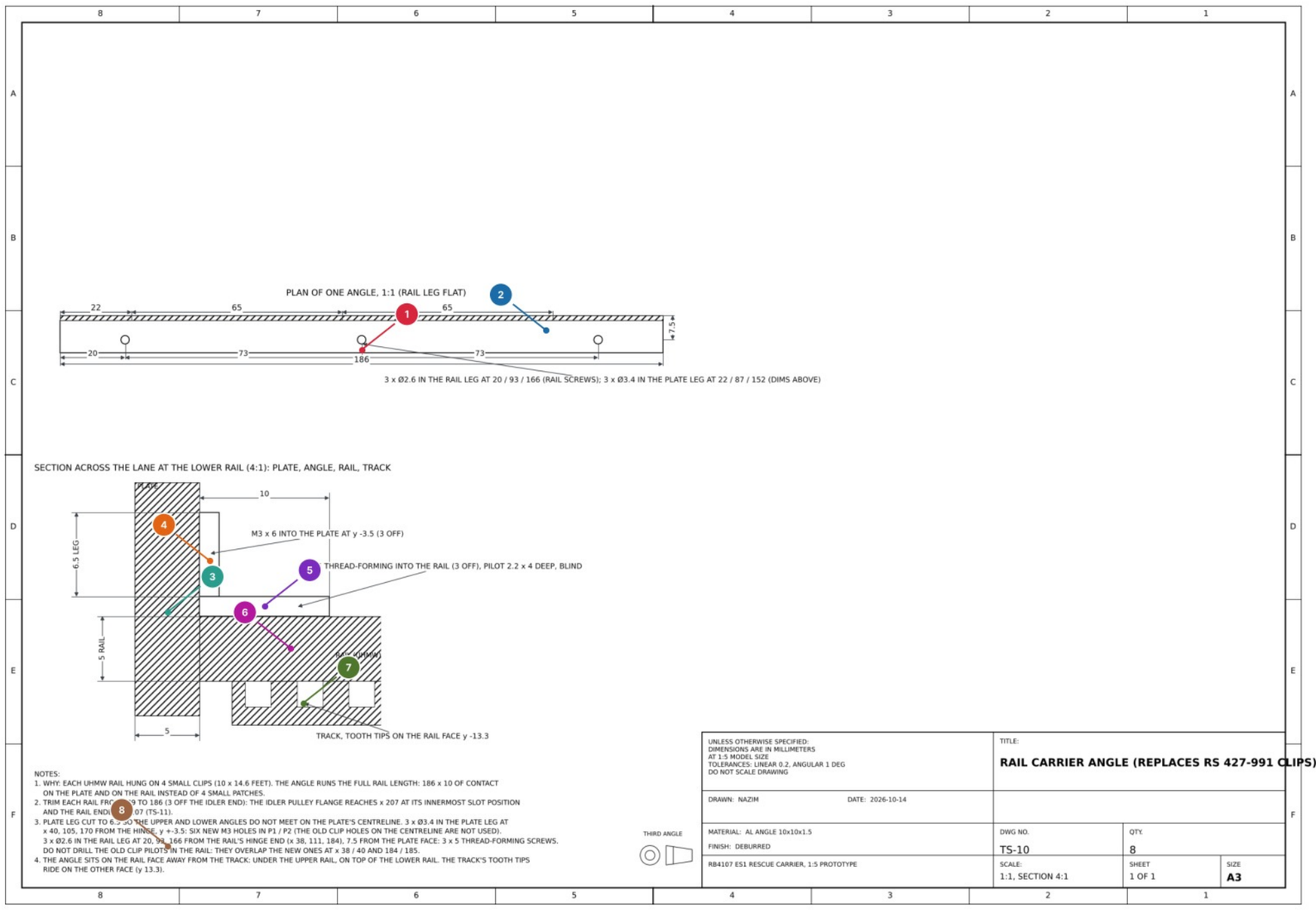
SCALE: 1:1, SECTION 4:1

SHEET 1 OF 1

SIZE **A3**

How to read TS-10: the rail carrier angle

Markers below sit on a reduced copy of sheet TS-10 (previous page, full size). Numbers match the key on the right.



1 Angle, plan
One 10 × 10 × 1.5 aluminium angle per rail, 186 long: three Ø2.6 holes in the rail leg for the thread-forming screws at 20, 93 and 166 from the rail's hinge end, 7.5 from the plate face, three Ø3.4 in the plate leg at 22, 87 and 152 from the hinge end (x 40, 105, 170), 3.5 from the leg's edge, for M3 x 6 into six new tapped holes in the plate.

2 Plate leg
Seen edge on above the rail leg. It is cut down to 8 so it stays inside the 15.4 plate height.

3 Plate
P1 or P2, 5 thick. The three M3 × 6 go through the angle into the plate's tapped holes, the same ones the four RS clips used.

4 Angle on the plate
The 8 leg lies flat on the plate's lane-side face, full length: 186 × 8 of contact instead of four 10 × 14.6 clip feet.

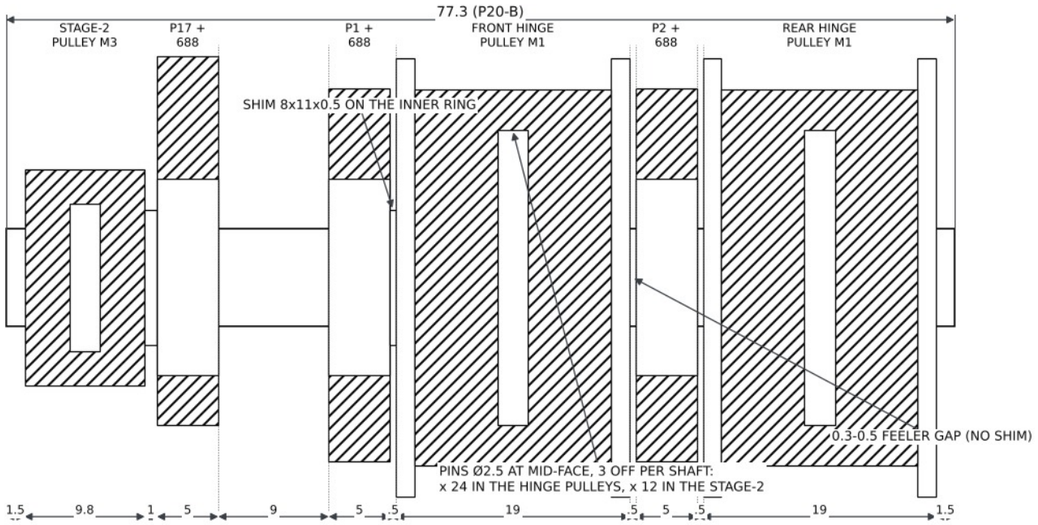
5 Angle on the rail
The 10 leg lies on the rail's face away from the track: on top of the lower rail, under the upper one. Three 3 × 5 thread-forming screws into blind Ø2.2 pilots 4 deep: the rail is only 5 thick and its far face carries the track, so a longer screw would come out of the sliding face.

6 Rail
UHMW strip 20 wide, 5 thick, trimmed from 189 to 186 at the idler end so the idler wheel's flange can never reach it.

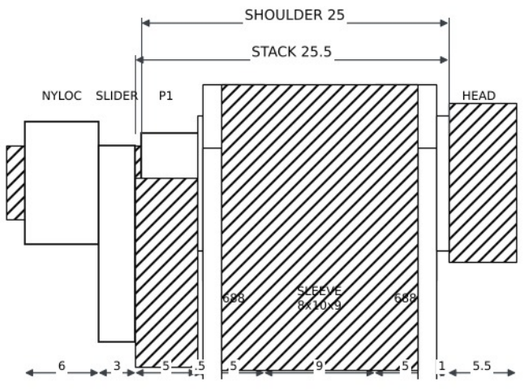
7 Track
The track's lower run rides tooth-tips-down on the rail's other face at y -13.3; the angle never touches the track.

8 Notes
Eight angles from 1.6 m of stock; the RS 427-991 clips and their 24 M3 × 6 are no longer needed (keep the thread-forming screws).

SECTION ALONG THE HINGE SHAFT P20-B, FROM THE CENTRE (z 28.7) OUTWARD TO 106. PINS REPLACE THE E-CLIPS.



SECTION ALONG THE NOSE IDLER SCREW (ISO 7379 8 x 25 M6), INBOARD TO OUTBOARD



FITS AND CLEARANCES

P1 / P2 / P17 bearing seat	laser Ø15.8, ream Ø16 H7	688-2RS, Loctite 641
hinge pulley bore M1, stage-2 M3	ream Ø8 H7	shaft Ø8 h7 round, pinned
idler pulley bore M2	bore Ø16 H7	2 x 688-2RS, light press
plate slot	8.1 +0.1 / 0 wide	shoulder Ø8 f9
lane, plate face to plate face	20.0 +-0.2	pulley 19.0: 0.5 each side
pulley to bearing gaps	0.5 shim + 0.3-0.5 feeler	never on the outer ring
rail end to idler flange	rail 186, idler >= x 223	2 mm minimum
deck over the front track	25 above axis	printed 19.7 / belt 25.0

NOTES:
1. FITS: 688-2RS SEATS IN P1, P2, P17 LASER-CUT Ø15.8 THEN REAMED Ø16 H7, RETAINED WITH LOCTITE 641. HINGE PULLEYS Ø8 H7 ON A PLAIN Ø8 h7 ROUND SHAFT (NO D-FLAT, NO GROOVES).
2. IDLER PULLEYS Ø16 H7, 688-2RS LIGHT PRESS. SLOTS 8.1 (+0.1 / 0) FOR THE Ø8 f9 SHOULDER. LANE 20.0 +-0.2 = (160 - 110) / 2 - 5 FROM THE BAR CUT LENGTHS: CUT BARS TO +-0.1 AND SQUARE.
3. AXIAL LOCATION ON THE HINGE SHAFT: PINS, NOT E-CLIPS. PIN THE FRONT HINGE PULLEY FIRST: ONE 8x11x0.5 SHIM AGAINST THE P1 BEARING, 0.3-0.5 FEELER GAP TO THE P2 BEARING. THAT FIXES THE SHAFT.
4. THEN PIN THE REAR PULLEY (0.3-0.5 GAP TO ITS BEARING) AND THE STAGE-2 PULLEY (1 MM SHIM AGAINST THE P17 BEARING). SHIMS BEAR ON INNER RINGS ONLY: NO PULLEY FACE ON AN OUTER RING OR A PLATE.
5. IDLER SCREW: SHOULDER 25 vs STACK 25.5 (HEAD WASHER 1 + BEARINGS 10 + SLEEVE 9 + SHIM 0.5 + PLATE 5): THE NUT CLAMPS THE INNER-RING STACK AGAINST THE PLATE, AS INTENDED. KEEP THE 0.5 SHIM.
6. IDLER FLANGE MEETS THE RAIL END IF THE IDLER IS PUSHED IN PAST x 223: RAILS TRIMMED TO 186 (TS-10), TENSIONER TS-09 ONLY EVER PUSHES OUTWARD.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
STACK-UPS AND FITS: HINGE SHAFT AND IDLER SCREW

DRAWN: NAZIM DATE: 2026-10-14

THIRD ANGLE

MATERIAL: SEE PARTS
FINISH: -

DWG NO. TS-11

QTY. PER SIDE

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

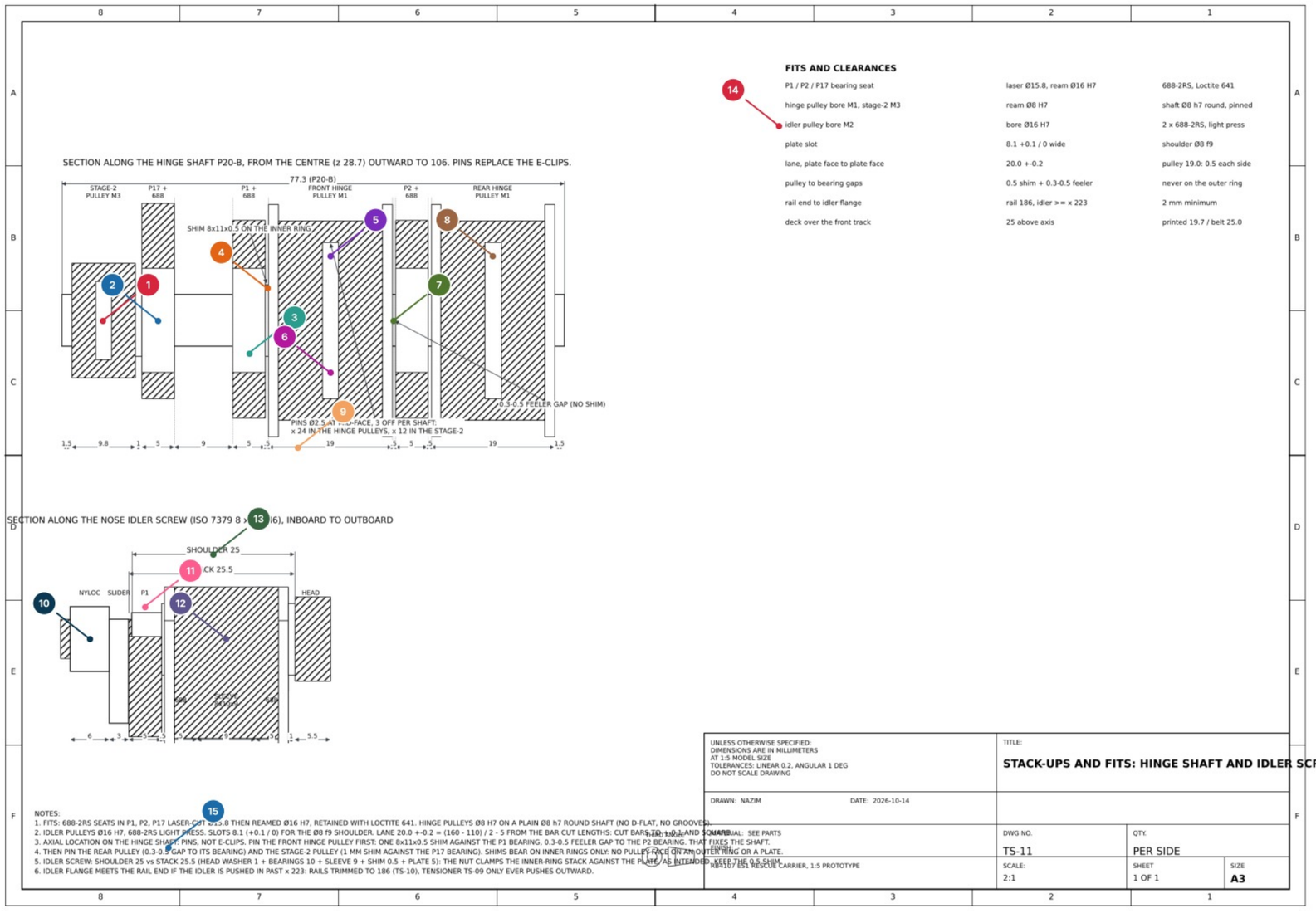
SCALE: 2:1

SHEET 1 OF 1

SIZE **A3**

How to read TS-11: the spacing between parts

Markers below sit on a reduced copy of sheet TS-11 (previous page, full size). Numbers match the key on the right.



1 Stage-2 pulley
The HTD 5M drive pulley on the hinge shaft, 9.8 wide, z 30.2 to 40, pinned, with a 1 mm shim between it and the P17 bearing.

2 P17 and its bearing
The hinge-shaft support on the middle bar, bearing at z 41 to 46. 9 mm of bare shaft follow before the front plate.

3 P1 and its bearing
Front plate at z 55 to 60; the 688 sits flush in a reamed Ø16 H7 seat with Loctite 641.

4 Shim 0.5
An 8 × 11 × 0.5 shim on the bearing's inner ring, so the pulley face never touches the outer ring. The pulley is pinned hard against it.

5 Front hinge pulley
19 over the flanges in the 20 lane, z 60.5 to 79.5. Pinned to the shaft first: it is what locates the whole shaft between the two plate bearings.

6 Cross pin
Ø2.5 spring pin at mid-face, drilled through pulley and shaft together. Three per shaft: 2.5 × 24 in the front and rear pulleys, 2.5 × 12 in the stage-2 pulley (Ø17.5).

7 Feeler gap
0.3 to 0.5 mm between the front pulley and the P2 bearing, set with a feeler gauge before pinning. This is the shaft's total end float.

8 Rear hinge pulley
Outboard of P2 at z 85.5 to 104.5, pinned with its own 0.3 to 0.5 gap to the bearing. The shaft ends 1.5 past it; no E-clip.

9 Station chain
Every width along the shaft from the centre outward. Sum from 28.7: the 77.3 mm half shaft P20-B.

10 Idler: nut
M6 nyloc inboard, 6 tall, then the 3 mm slider of TS-09.

11 Idler: plate
P1, 5 thick; the shoulder's end sits 0.5 inside it, so the M6 washer face never touches the slot.

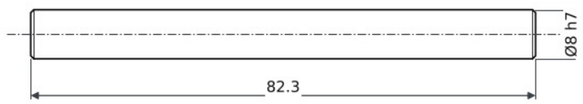
12 Idler: bearings and sleeve
Two 688s with the 8 × 10 × 9 sleeve between their inner rings, inside the Ø16 H7 bore of the idler wheel.

13 Shoulder vs stack
Shoulder 25 against a 25.5 stack: the nut clamps washer, inner rings, sleeve, shim and plate together. That is correct; do not leave the shim out.

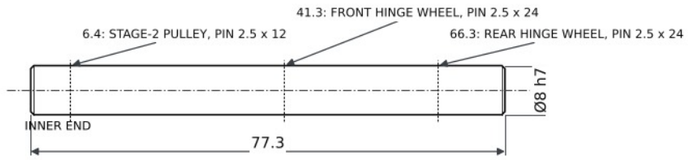
14 Fits table
The fit or clearance for every mating pair that matters, and what the laser, reamer or saw has to hold.

15 Notes
The order of pinning and the shim rule in words.

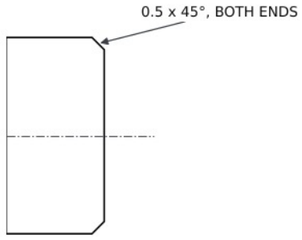
SHAFT P20-A, 82.3, QTY 1 (1:1)



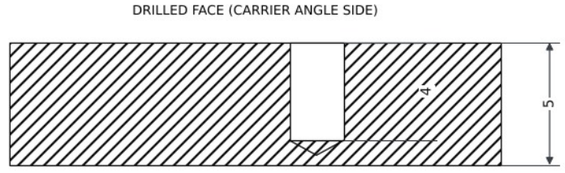
SHAFT P20-B, 77.3, QTY 1 (1:1). PIN POSITIONS NOMINAL, DRILLED AT ASSEMBLY



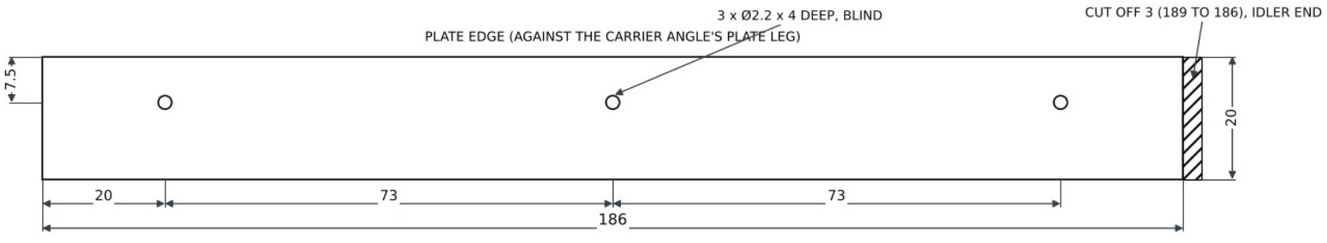
DETAIL: SHAFT END (4:1)



SECTION THROUGH A PILOT (4:1)



RAIL P3-P6, DRILLING FACE UP (1:1). PLATE EDGE AT THE TOP, HINGE END ON THE LEFT

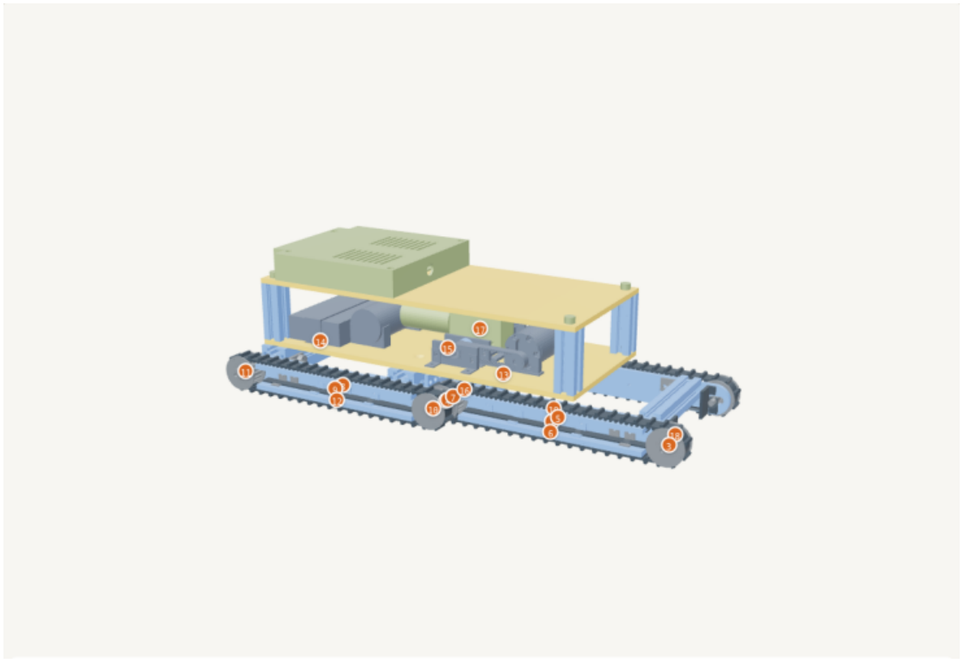


NOTES:
1. SHAFTS: CUT BOTH FROM ONE 300 MM Ø8 h7 GROUND SHAFT: A = 82.3, B = 77.3 (+-0.1). FACE SQUARE, CHAMFER 0.5 x 45° BOTH ENDS. PLAIN ROUND: NO D-FLAT, NO E-CLIP GROOVES. KEEP THE GROUND SURFACE CLEAN: IT RUNS IN THE 688 BEARINGS.
2. DO NOT DRILL THE Ø2.5 PIN HOLES NOW. AT ASSEMBLY EACH ONE IS DRILLED THROUGH THE SHAFT USING THE PULLEY'S HUB HOLE AS THE GUIDE, AFTER THE SHIM AND FEELER GAP ARE SET (TS-11). THE PIN POSITIONS SHOWN ON B ARE NOMINAL, FOR REFERENCE ONLY; A IS THE SAME STACK.
3. RAILS: MISUMI UPA-189-20-5. CUT 3 OFF THE IDLER END: 186 LONG. THE SAME PART SERVES AS UPPER AND LOWER RAIL: DRILL THE FACE AWAY FROM THE TRACK, WHERE THE CARRIER ANGLE SITS (TS-10). MARK THE HINGE END AND THE PLATE EDGE ON EACH RAIL BEFORE DRILLING.
4. PILOTS Ø2.2 x 4 DEEP, BLIND, WITH A DEPTH STOP: THE OTHER FACE IS WHAT THE TRACK SLIDES ON AND MUST STAY UNBROKEN. DO NOT DRILL THE REV B SHEET 18 PILOTS: THEY OVERLAP THESE AT 20 / 22 AND 166 / 167.

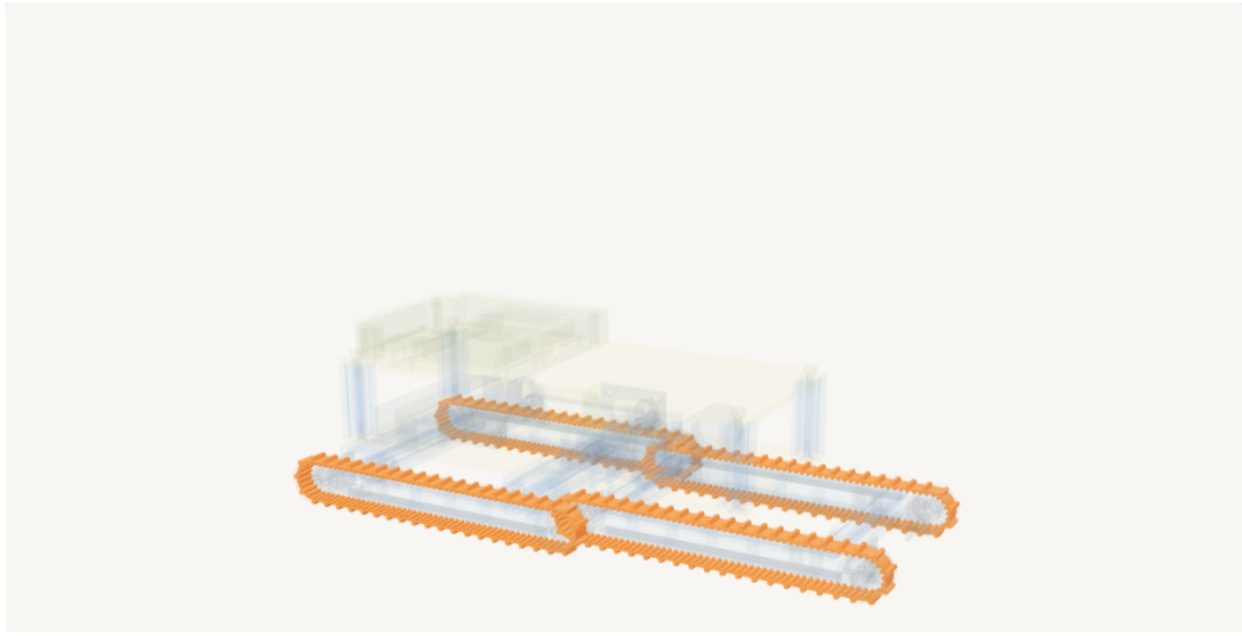
THIRD ANGLE

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS AT 1:5 MODEL SIZE TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG DO NOT SCALE DRAWING	TITLE: **HINGE SHAFTS A + B, RAIL TRIM AND PILOTS**		
DRAWN: NAZIM DATE: 2026-10-14			
MATERIAL: Ø8 h7 GROUND STEEL; UHMW-PE 20 x 5 FINISH: DEBURRED	DWG NO. TS-12	QTY. 2 SHAFTS, 8 RAILS	
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE	SCALE: 1:1, DETAILS 4:1	SHEET 1 OF 1	SIZE **A3**

OPTION A: my printed track



Assembled with this track (same model as the interactive page; Track menu: my printed track).



The four tracks of this option highlighted; everything else is common to both options.

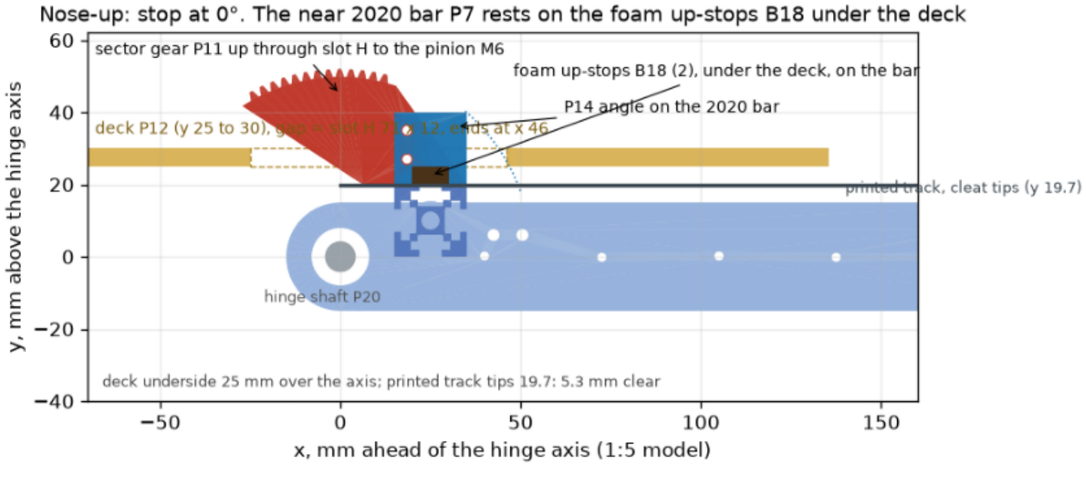
What it is TPU 95A, printed standing as a ring on the Fab Lab UP300: 110 HTD 5M teeth inside, 45 cleats 3 mm high outside, spaced so five span one ES1 step. Four tracks, about 32 g and 3 h each. Drawing TS-01 and its explainer follow.

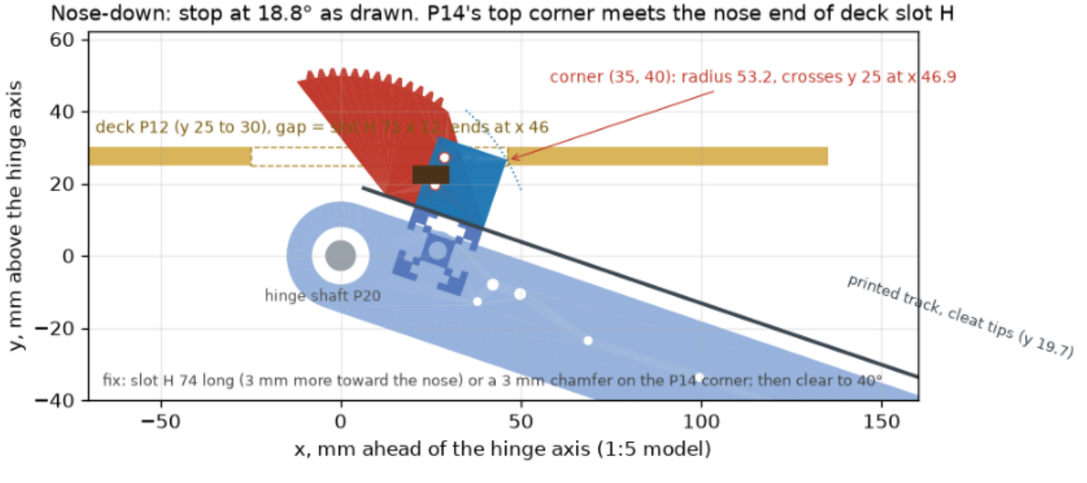
Deck clearance Deck: the cleat tips are 19.7 mm above the hinge axis, 5 mm under the deck. No deck change.

What to order To order: TPU 95A filament if the Fab Lab has none (Element14 SG); nothing else is specific to this option. Print the 5 mm fit-test slice first, then the four tracks as UP300 slots come free. Procurement list section 1A.

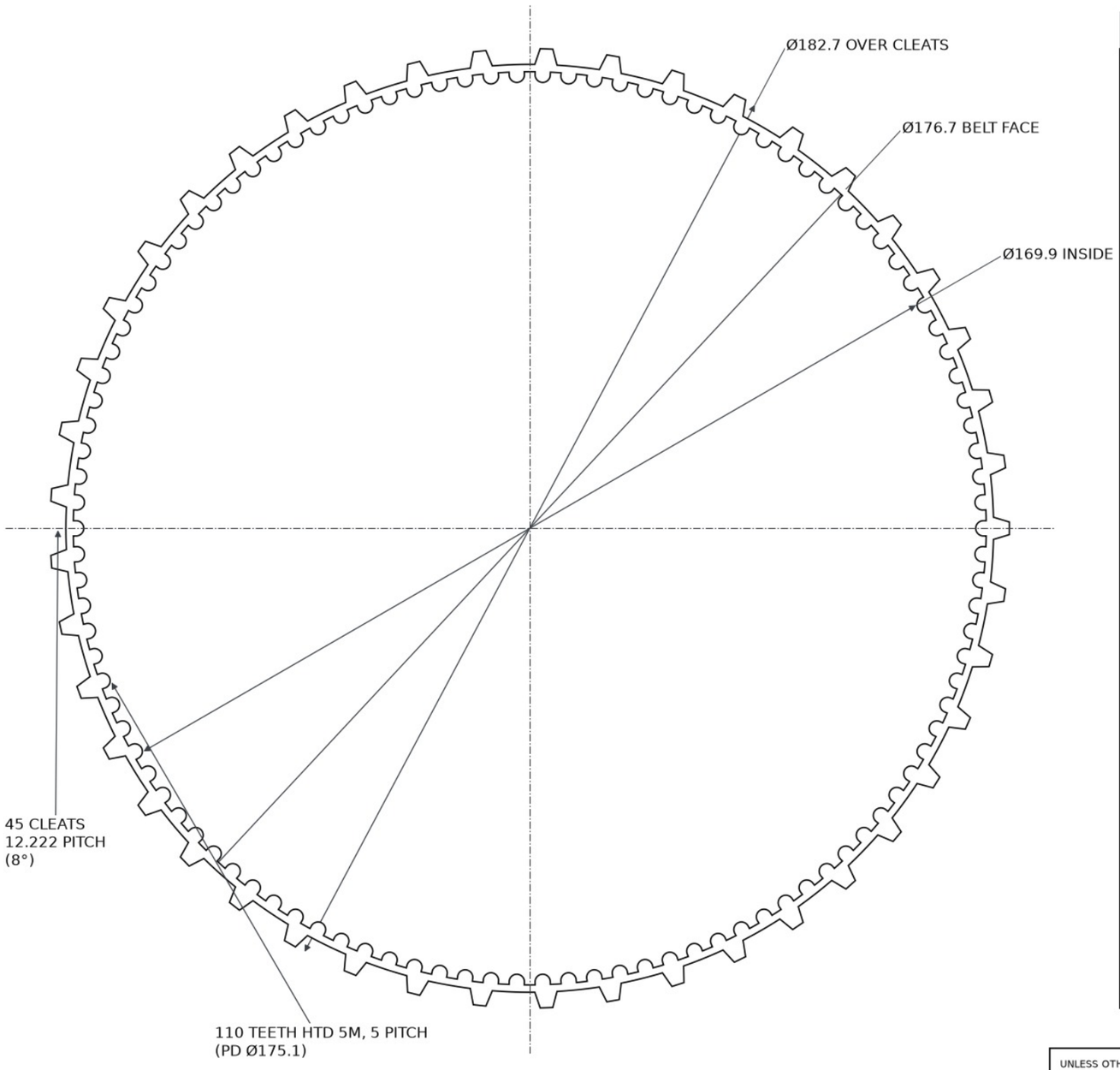
In short In house, no supplier lead time; cleat pitch matched to the ES1 step; 5 mm deck clearance. Depends on the TPU head being free.

Hinge stops with this track (section at the sector gear): the deck line against the cleat tips is the only thing that changes between the options.



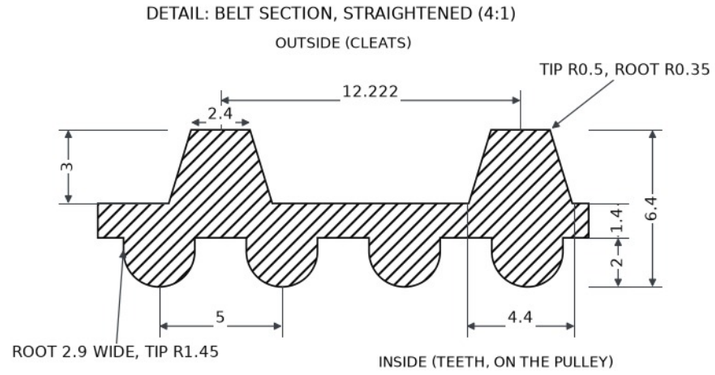


FRONT VIEW



EDGE VIEW

DETAIL: BELT SECTION, STRAIGHTENED (4:1)



NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

TRACK, 110T HTD 5M, 45 CLEATS

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO. TS-01

QTY. 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

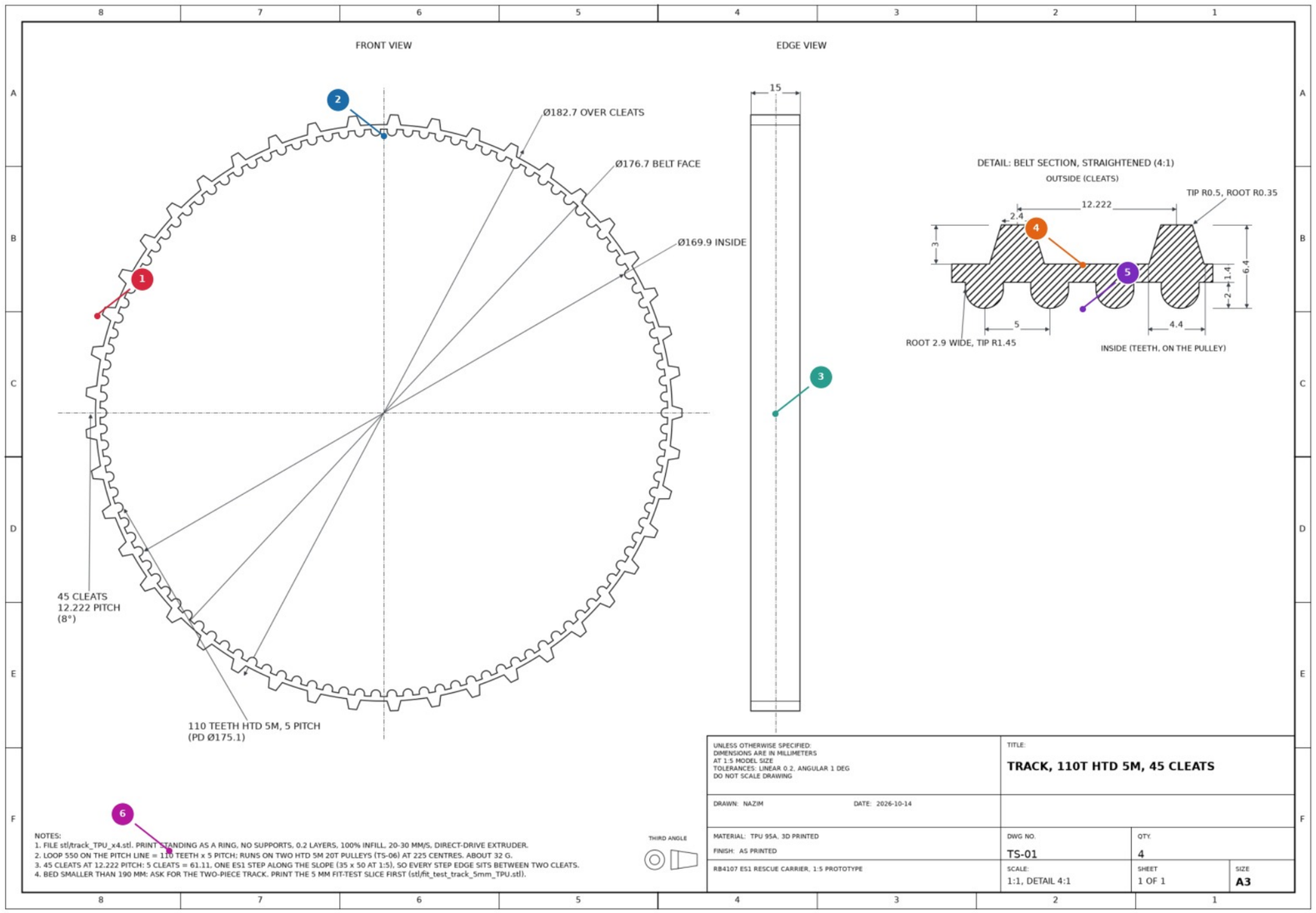
SCALE: 1:1, DETAIL 4:1

SHEET 1 OF 1

SIZE **A3**

How to read TS-01: the track

Markers below sit on a reduced copy of sheet TS-01 (previous page, full size). Numbers match the key on the right.



1 Cleats, outside

45 bumps 12.22 apart (8 degrees). Five of them span one ES1 step, so every step edge lands between two cleats and the track holds like a gear on a rack.

2 Teeth, inside

110 HTD 5M teeth, 5 apart. These mesh with the 20-tooth wheels; this is what stops the track slipping.

3 Edge view

The ring seen from the side: 15 wide, 182.7 tall over the cleats.

4 Cleat profile (4:1)

A cleat is 3 tall, 4.4 wide at its base and 2.4 at the tip, with rounded corners so it releases from the step.

5 Tooth profile (4:1)

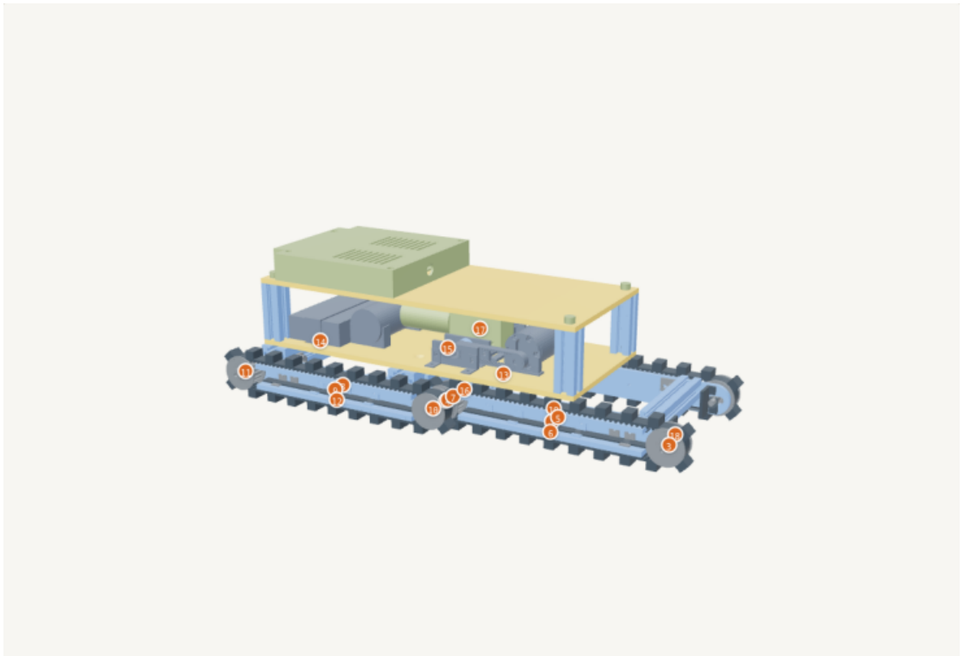
A tooth is 2 deep with a 1.45 radius tip and a 2.9 root; the 1.4 backing between teeth and cleats is the belt itself. Whole section 6.4 thick.

6 Notes

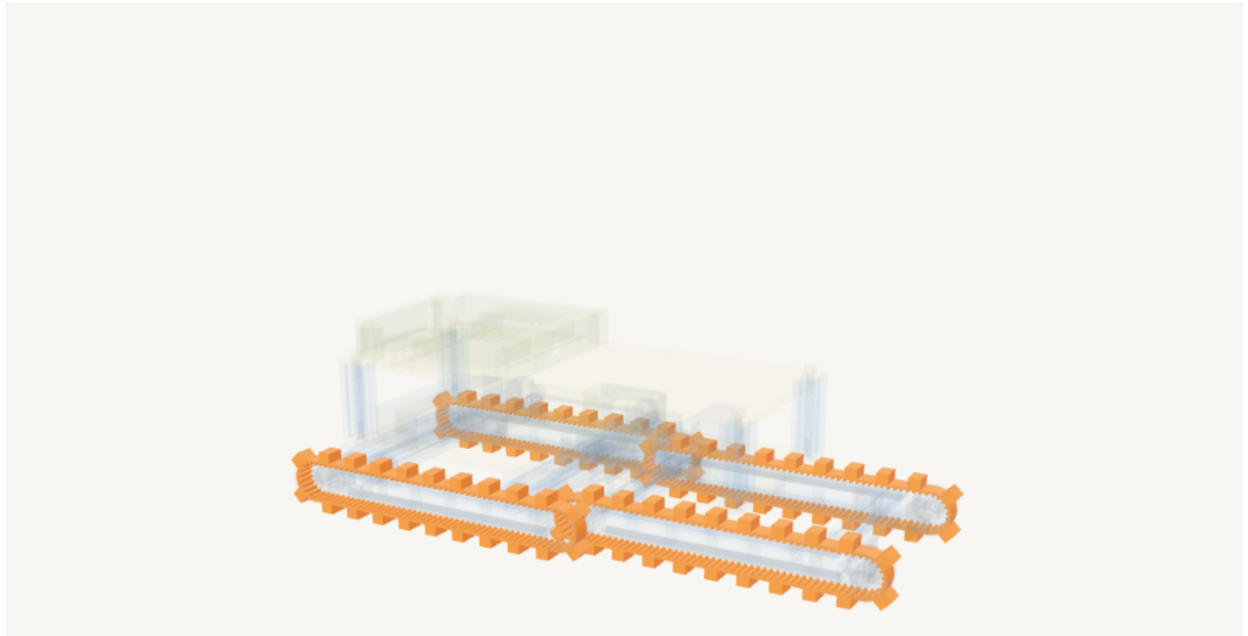
Print standing up as a ring, no supports, 100% infill, TPU 95A, on the Fab Lab UP300. Four tracks, about 32 g each.

19 Oct, chosen option: cleats cut to 5 mm (Nhien), tips 2.9 under the deck, so NO deck notch. Bonded by the belt maker, 1-2 weeks. TS-01B updated.

OPTION B: Nhien's bought belt with bonded cleats (Rev B BOM B10)



Assembled with this track (same model as the interactive page; Track menu: bought belt + cleats).



The four tracks of this option highlighted; everything else is common to both options.

What it is

Stock HTD 5M timing belt, 550 mm pitch length (110 teeth), 15 wide, e.g. SKF PHG 550-5M-15 (RS SG 058-6535), with 22 rubber cleats 10 x 7.9 x 15 bonded on the back every 5th tooth by a belt shop. Four belts, 88 cleats. Drawing TS-01B and its explainer follow.

Deck clearance

Deck: the cleat tips reach 25.0 mm above the hinge axis, exactly the deck underside, so they rub along the 135 mm the deck overhangs the front lanes. Notch P12 13 mm deep over each front lane from 25 mm behind the hinge to 110 mm ahead (the post corners stay), or cut the cleats to 5 mm for 3 mm clearance.

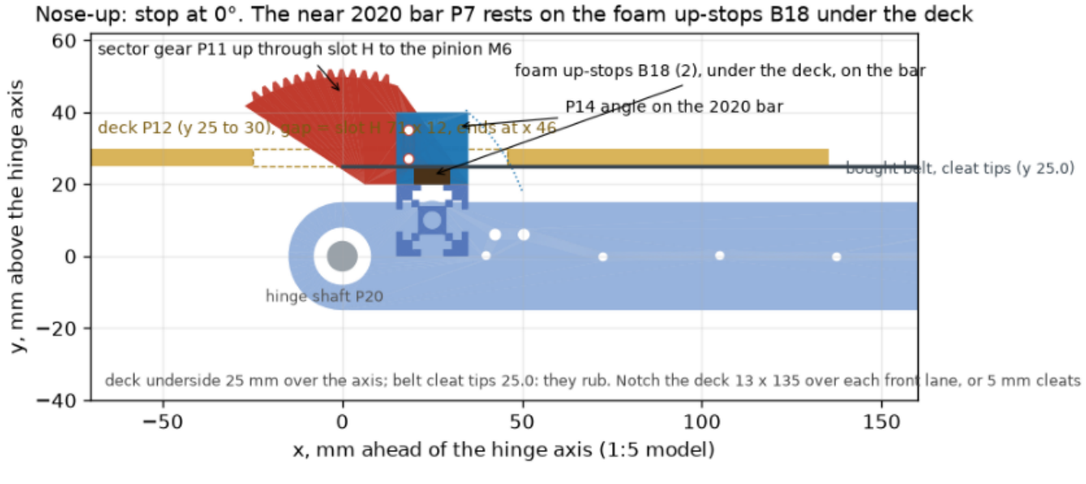
What to order

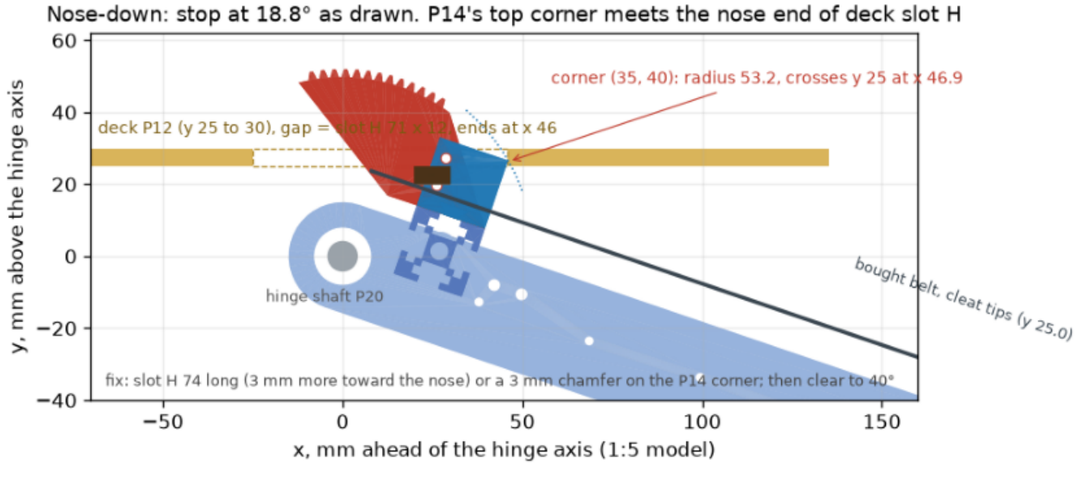
To order: 4 belts from RS SG (days); a belt shop (Fenner, Megadyne or Gates distributor) to cut and bond the 88 cleats, allow 1 to 2 weeks; the deck notch with Nhien's laser cut. Procurement list section 1B.

In short

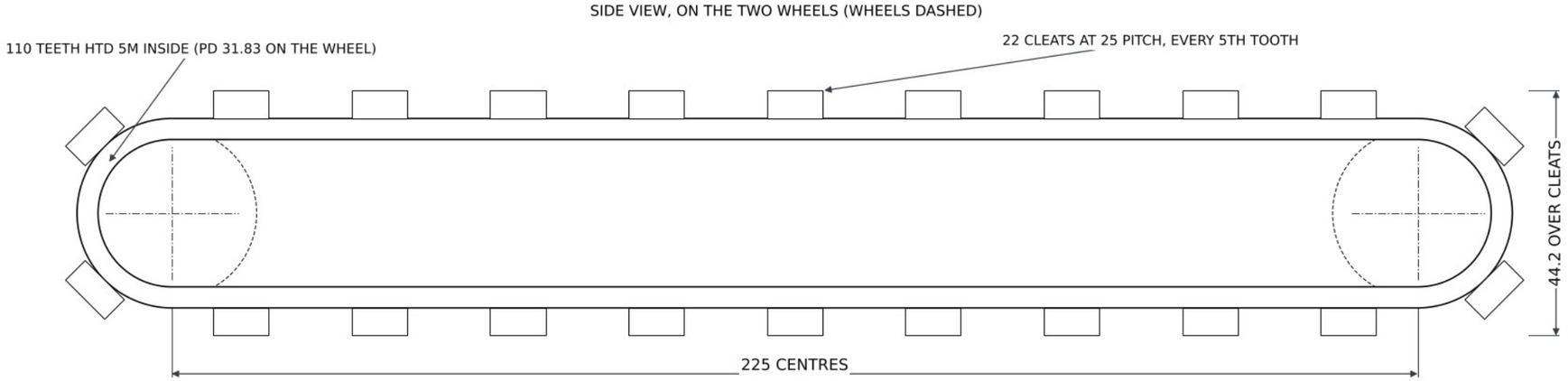
Bought belt, no printing; the cleats are a supplier job and the deck needs the notch. Same wheels, centres and joints as option A.

Hinge stops with this track (section at the sector gear): the deck line against the cleat tips is the only thing that changes between the options.

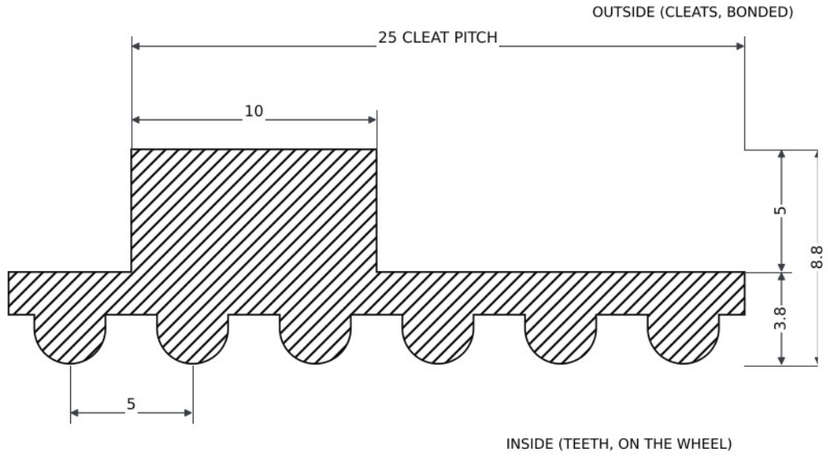




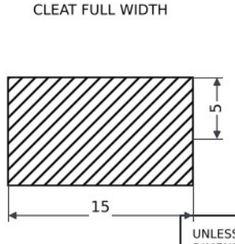
SIDE VIEW, ON THE TWO WHEELS (WHEELS DASHED)



DETAIL: SECTION ALONG THE BELT, STRAIGHTENED (4:1)



SECTION ACROSS THE BELT (2:1)



NOTES:
1. BELT: HTD 5M, 550 PITCH LENGTH (110 TEETH), 15 WIDE, E.G. SKF PHG 550-5M-15 (RS SG 058-6535). RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES.
2. CLEATS: 22 PER BELT, RUBBER 10 x 5 x 15, ONE EVERY 5TH TOOTH (25 PITCH), BONDED ON THE BELT BACK BY THE BELT SHOP (REV B BOM B10). 4 BELTS = 88 CLEATS.
3. CLEATS 5 HIGH (CUT DOWN FROM 7.9, NHIEN 19 OCT): TIPS 22.1 FROM THE AXLE, 2.9 UNDER THE DECK UNDERSIDE, SO DECK P12 NEEDS NO NOTCH.
4. ALTERNATIVE: TS-01 PRINTED TPU TRACK, 3 MM CLEATS, 5 MM UNDER THE DECK. SAME WHEELS, SAME CENTRES, SAME JOINTS.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

TRACK BELT, BOUGHT: HTD 550-5M-15 + 22 CLEATS

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: NEOPRENE HTD BELT, RUBBER CLEATS

FINISH: AS BOUGHT, CLEATS BONDED

DWG NO. TS-01B

QTY. 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

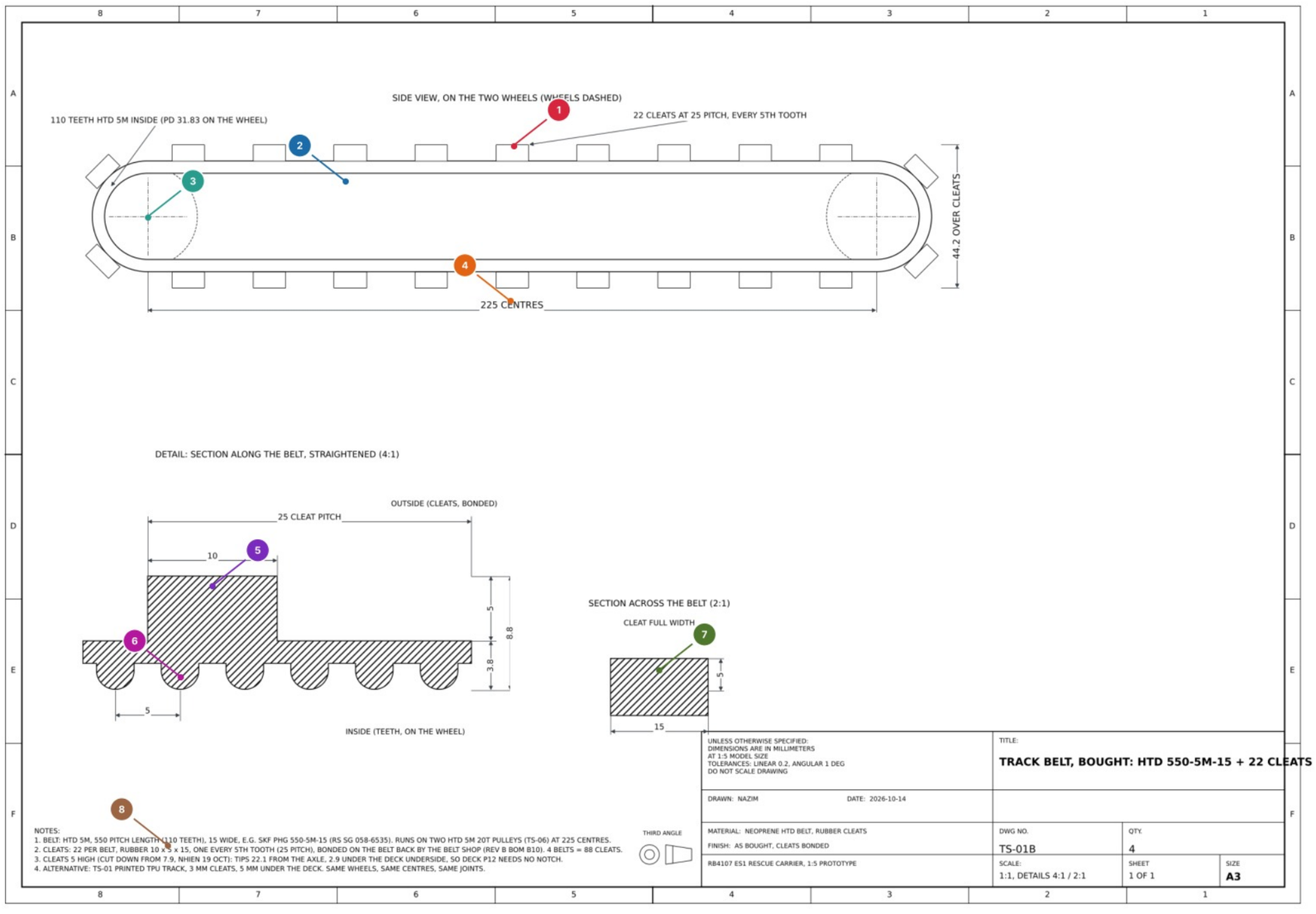
SCALE: 1:1, DETAILS 4:1 / 2:1

SHEET 1 OF 1

SIZE **A3**

How to read TS-01B: the bought track belt (Rev B B10)

Markers below sit on a reduced copy of sheet TS-01B (previous page, full size). Numbers match the key on the right.



1 Cleats, outside
22 rubber blocks 10 long, 5 high (cut down from 7.9) and the full 15 width, bonded on the belt back every 25 mm (every 5th tooth). They grip the step edges. Their tips are 22.1 mm from the axle, 2.9 under the deck, so the deck needs no notch.

2 Teeth, inside
110 HTD 5M teeth, 5 apart, moulded into the belt. They mesh with the 20-tooth wheels, so the belt cannot slip.

3 Hinge wheel (dashed)
The belt wraps the driven 20T wheel on the hinge shaft at the left and the idler at the right, 225 apart: 450 straight plus the two half wraps = 550, the belt's pitch length.

4 225 centres
Same wheel spacing as the printed track, so the plates, slots and idlers do not change between the two versions.

5 Cleat profile (4:1)
A cleat is a plain block 10 long and 5 high, bonded to the belt back; no taper. 8.8 total from tooth tip to cleat tip.

6 Tooth profile (4:1)
HTD 5M tooth, 2.06 deep, on a 1.74 backing: 3.8 belt thickness. This is the belt as bought.

7 Across the belt (2:1)
Belt and cleat are both 15 wide; the cleat spans the full width.

8 Notes
Buy SKF PHG 550-5M-15 (RS SG 058-6535) x4; a belt shop cuts and bonds 88 cleats, 5 high, so the belts fit under the deck with no notch.

PART B

Nhien's Rev B fabrication sheets for the parts the track runs on

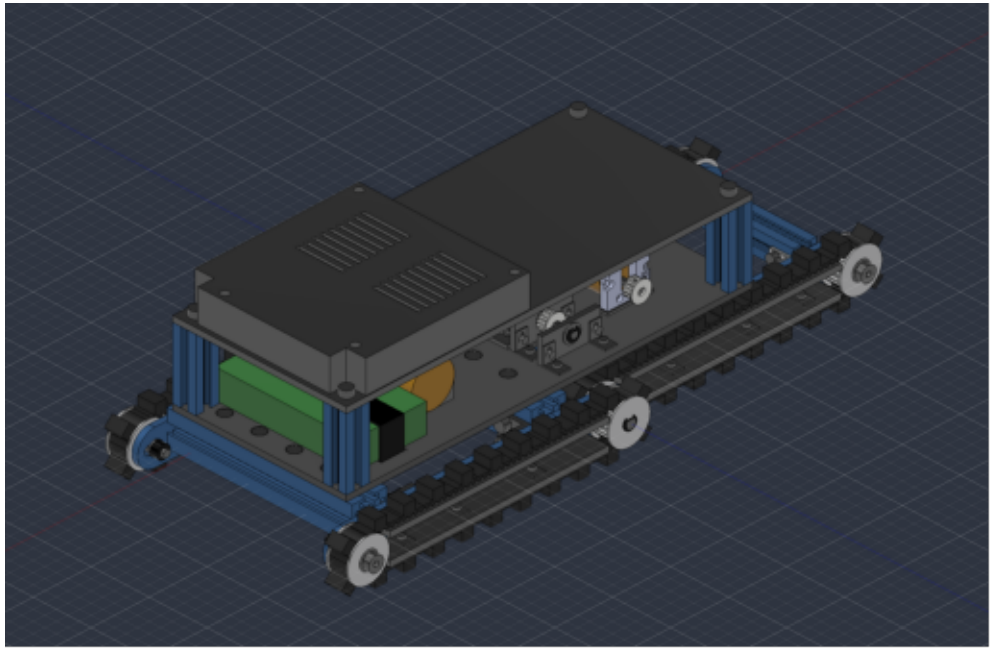
Taken unchanged from 'RB4107 Fabrication drawings (Rev B).pdf' (Nhien, 15 Oct 2026). These are the side plates, rails, bars, brackets, hinge supports, shafts, bought-part modifications and the sector gear that the track and wheels mount on, plus his bill of materials, which is the parts list this pack follows.

His sheet	Content
2	Assembly views
3	Joint / fastener schedule and assembly order
4	P1 side plate, front segment
5	P2 side plate, rear segment
18	P3 to P6 slide rails (MISUMI UPA plate, drilled)
19	P7 / P8 / P15 profile cut list
12	P18 corner bracket, 3030
13	P19 corner bracket, 2020
11	P17 hinge shaft bearing support
20	P20 / P21 shafts
21	M1 to M8 modifications to bought parts (track pulleys M1, M2)
9	P11 hinge sector gear
14	P14 sector mount angle
23, 24, 25	Bill of materials (also RB4107 BOM Rev B.xlsx)

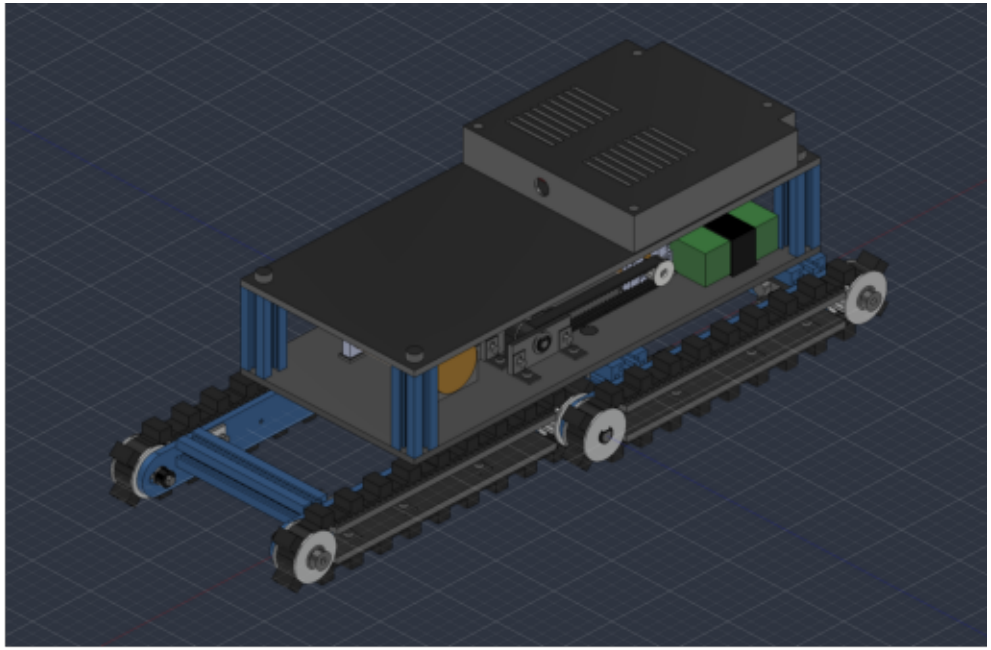
Superseded by the design review (my sheets take precedence): his P20 sheet shows the D-shaft with E-clip grooves and his M1 / M3 rows show D-bores; the build uses the plain round shaft, reamed bores and cross pins of TS-06 and TS-11. His rail sheet shows the P5, P10, P12, P13, P16, P22, P23, M7, M8 (186).

Not included here, drawn underneath and electronics sheets (P9, P17, P18, P19, P24, P25, M4, M6), which are Nhien's scope and unchanged by the track.

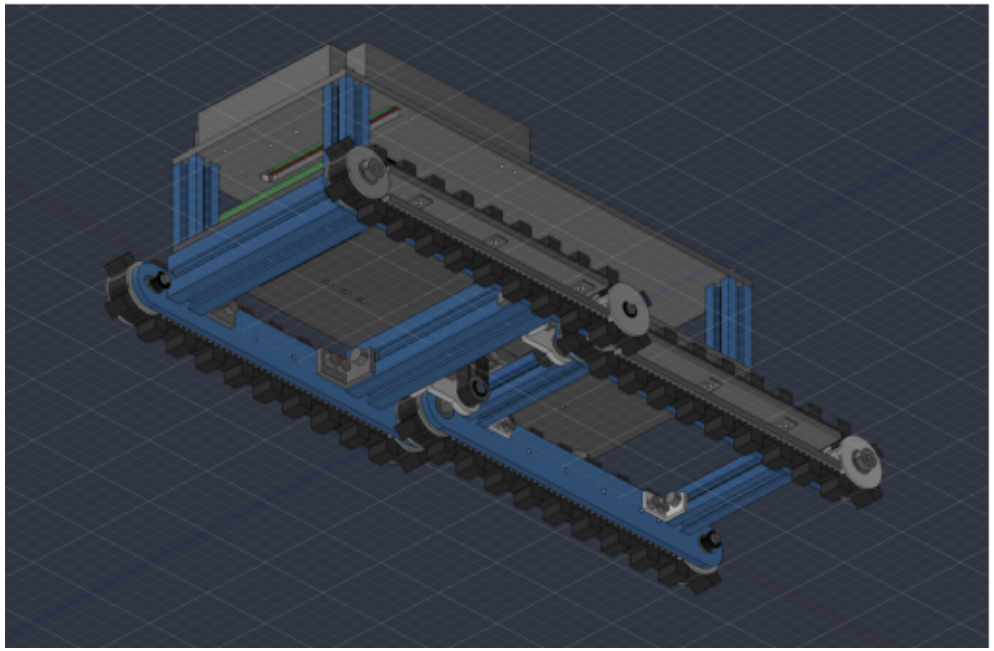
ASSEMBLY VIEWS (reference only)



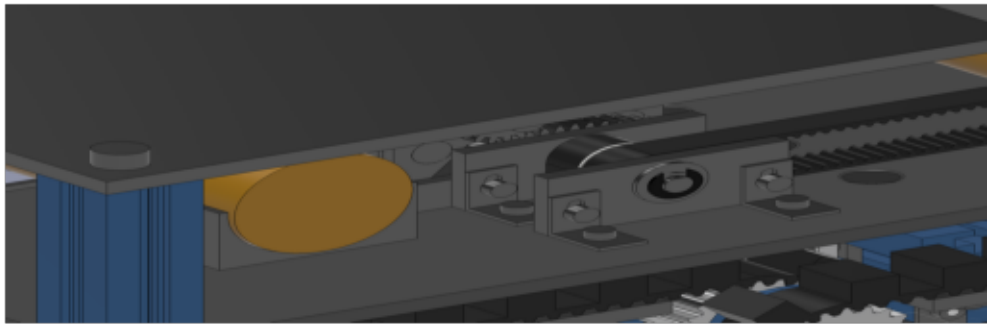
Overall, from the rear, B side



Overall, from the front, A side



From below: track sides, bars, bearings, idler slots



Hinge area

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	ASM Assembly views
MATERIAL	-
QTY	-
SCALE	NTS
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	2 of 25

JOINT / FASTENER SCHEDULE

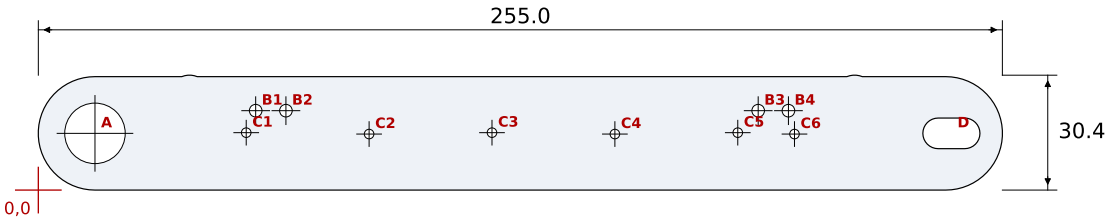
JOINT	FASTENERS
Side plate P2 - 3030 bar P8 (4 corners)	P18 bracket; M6x10 SHCS + slot-8 M6 T-nut; 2x M5x8 SHCS into plate
Side plate P1 - 2020 bar P7 (4 corners)	P19 bracket; M5x8 SHCS + slot-6 M5 T-nut; 2x M4x8 SHCS into plate
Deck P12 - 3030 bars P8 (rear + middle)	8x M5x12 csk ISO 10642 (from top) + slot-8 M5 T-nuts
Deck P12 - posts P15	4x M6x16 csk ISO 10642 from BELOW into tapped post ends
Tray P16 - posts P15	4x M6x16 SHCS ISO 4762 from top into tapped post ends
Hinge bearing support P17 - middle bar P8	2x M5x10 SHCS + slot-8 M5 T-nuts each (2 supports)
Shaft support plate P13 - deck	2x RS 427-991 (M7); M3x5 SHCS to plate (slot) + M3x5 to deck (tapped)
Sector gear P11 - angle P14 - bar P7	2x M3x8 csk into gear; 2x M4x8 csk + slot-6 M4 T-nuts into bar
Rail brackets (24) - side plates / rails	M3x6 SHCS into tapped plate; 3 mm x 5 PT screw into rail
Idlers (4) - side plates	ISO 7379 Ø8x25 M6 shoulder screw, 8x11x1 head washer, 8x11x0.5 shim, spacer 8x10x9, M6 washer + nyloc; 2x 688-2RS
Hinge drive shafts P20 (2)	3x 688-2RS each (rear plate, front plate, P17); E-clip DIN 6799-6 both ends; shim 8x11x1
Intermediate shafts P21 (2)	2x F686ZZ each (flanges out); E-clip DIN 6799-5 + shim 6x8x0.5 both ends
Drive motors (2)	Pololu #1995 bracket on P9 shim; M3x12 SHCS from below + nut; tail cradle P10 (M3 inserts)
Hinge motor 5840	M8 L-bracket; 4x M4x8 csk DIN 7991 into gearbox; 2x M5x8 + washer into deck
Controller	MDD10A on M3x6 nylon standoffs; ESP32 sled; BNO085 on M2.5x5; 5V reg. on M2x5; cover 4x M3x25
Battery	20 mm hook-and-loop strap through the two deck slots

TIGHTENING (A2/8.8 into aluminium): M3 1.0 N·m M4 2.0 N·m M5 4.0 N·m M6 7 N·m. Grub screws: Loctite 243.
ASSEMBLY ORDER
1. Press bearings into side plates, P17 supports and idlers (Loctite 641 on outer ring if loose).
2. Build the rear frame: P2 plates + P8 bars with P18 brackets (drop T-nuts into the bar slots first).
3. Build the front frame: P1 plates + P7 bars with P19 brackets; fit P14 + P11 sector gear.
4. Hinge: slide P20 shafts through rear pulley, rear plate, front pulley, front plate, P17, shim, pulley; E-clips; grubs.
5. Deck P12 onto the bars (M5 csk), posts P15 (M6 from below), tray P16 (M6 from top).
6. Drive: motors + P9 shims + cradles, P13 supports + P21 shafts, belts. Tension Stage-1 (motor slots) then Stage-2 (M7 slots).
7. Idlers on shoulder screws; tension each track by sliding the idler out, then tighten the nyloc nut.
8. Electronics on the tray, wiring through the slot (grommet), cover P22 last.

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	ASM Joint / fastener schedule
MATERIAL	-
QTY	-
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	3 of 25

P1 SIDE PLATE - FRONT SEGMENT

VIEW FROM INSIDE (bar side); hinge end left, nose end right (scale 0.5:1, thickness 5.0)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
A	14.98	15.00	Ø16.0 H7 THRU (688-2RS bearing seat)
B1	57.48	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
B2	65.48	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
B3	190.38	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
B4	198.38	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
C1	54.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C2	87.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C3	119.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C4	152.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C5	184.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C6	199.98	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
D	241.48	15.00	SLOT 8.1 wide x 15.1 long, along X, THRU

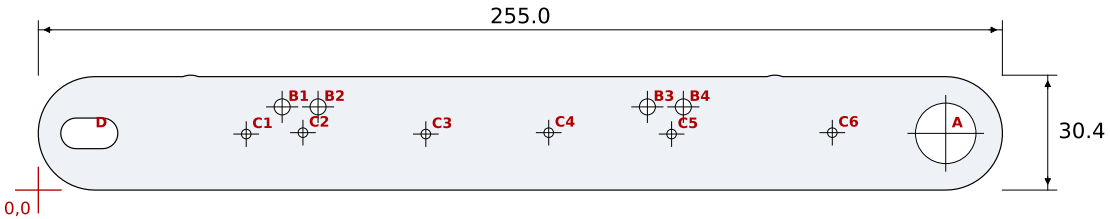
NOTES

1. Laser / waterjet cut from DXF, then drill and tap. Deburr, break edges 0.2-0.3.
2. Opposite-hand plate = mirror image (STEP 'opposite hand').
3. Ø16 H7 at the hinge end: press in a 688-2RS bearing with Loctite 641.
4. Slot at the nose end = idler tension slot for the ISO 7379 Ø8 shoulder screw.
5. M3 tapped holes: RS 427-991 rail brackets (check a sample bracket before tapping).
6. M4 tapped holes: 2020 inside corner brackets P19 (2 per bar end).
7. CAD: STEP/P1*.step DXF/P1*.dxf

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P1 Side plate - FRONT segment
MATERIAL	Al 6061-T6, 5 mm plate
QTY	2 (1 as drawn + 1 opposite hand)
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	4 of 25

P2 SIDE PLATE - REAR SEGMENT

VIEW FROM INSIDE (bar side); rear idler left, hinge end right (scale 0.5:1, thickness 5.0)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
A	239.98	15.00	Ø16.0 H7 THRU (688-2RS bearing seat)
B1	64.48	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
B2	73.98	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
B3	161.08	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
B4	170.58	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
C1	54.98	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C2	69.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C3	102.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C4	134.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C5	167.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C6	209.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
D	13.48	15.00	SLOT 8.1 wide x 15.1 long, along X, THRU

NOTES

1. Laser / waterjet cut from DXF, then drill and tap. Deburr, break edges 0.2-0.3.
2. Opposite-hand plate = mirror image (STEP 'opposite hand').
3. Ø16 H7 at the hinge end: press in a 688-2RS bearing with Loctite 641.
4. Slot at the rear end = idler tension slot for the ISO 7379 Ø8 shoulder screw.
5. M3 tapped holes: RS 427-991 rail brackets. M5 tapped holes: 3030 corner brackets P18.
6. CAD: STEP/P2*.step DXF/P2*.dxf

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P2 Side plate - REAR segment
MATERIAL	Al 6061-T6, 5 mm plate
QTY	2 (1 as drawn + 1 opposite hand)
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
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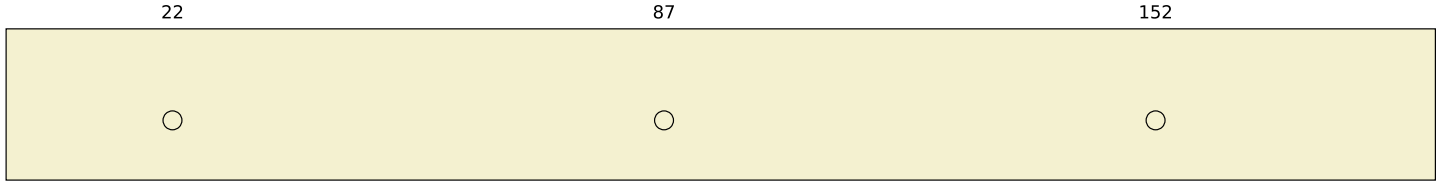
P3-P6 SLIDE RAILS = MISUMI UPA-189-20-5 UHMW-PE plate (bought, drill only)

P3

Slide rail - FRONT, UPPER

QTY 1 + 1 opp. hand

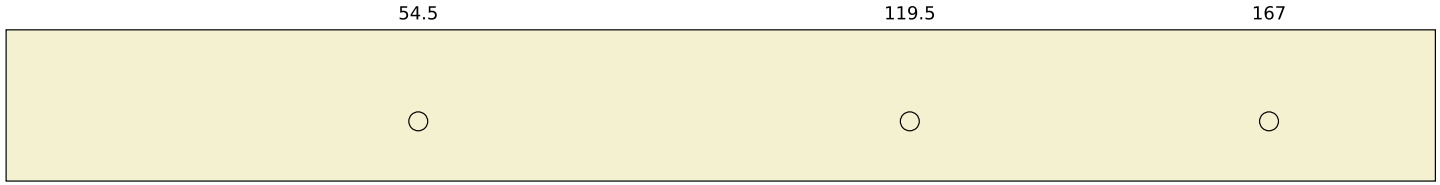


189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

P4

Slide rail - FRONT, LOWER

QTY 1 + 1 opp. hand

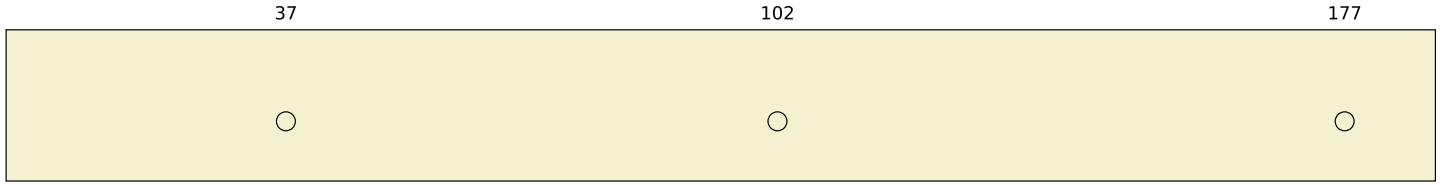


189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

P5

Slide rail - REAR, UPPER

QTY 1 + 1 opp. hand

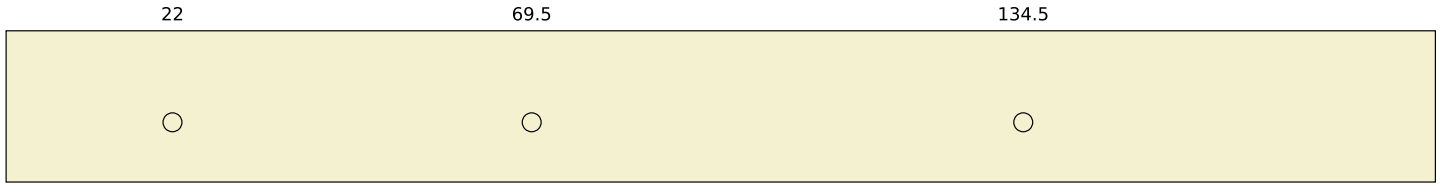


189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

P6

Slide rail - REAR, LOWER

QTY 1 + 1 opp. hand



189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

NOTES

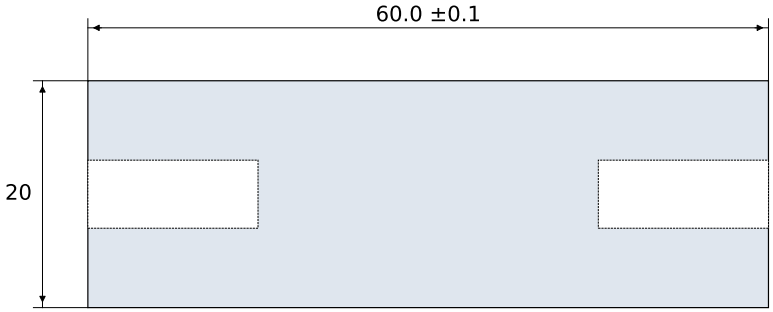
1. Order MISUMI UPA-189-20-5 (UHMW-PE plate, milky white, 189 x 20 x 5, cut to size) x 8. Only the 3 pilot holes are drilled.
   The strip is 5 mm wider than the 15 mm belt; the extra 5 mm overhangs on the OUTER side (does not touch the belt).
   Pilot holes must not break through the wear face.
2. 3 mm thread-forming screws for plastics (PT type) x 5, through the RS 427-991 bracket legs.

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P3-P6 Slide rails (MISUMI UPA-189-20-5)
MATERIAL	UHMW-PE 189 x 20 x 5
QTY	1 + 1 opp. each (8)
SCALE	1:1
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	18 of 25

P7 / P8 / P15 ALUMINIUM T-SLOT PROFILES - CUT LIST

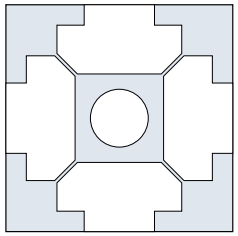
Nothing screws into the SIDE of a profile: side joints use T-nuts in the slots. Only P15 ends are tapped.

PART	DESCRIPTION	PROFILE	LENGTH	QTY	MACHINING
P7	Cross bar - front segment (hinge-side + nose)	2020 B-type, slot 6, bore Ø5.1	110.0 ±0.2	2	Cut square, deburr. No machining.
P8	Cross bar - rear + middle	3030 B-type, slot 8, bore Ø6.65 (80/20 30-3030 type)	160.0 ±0.2	2	Cut square, deburr. No machining.
P15	Corner post (deck to tray)	2020 B-type, slot 6, bore Ø5.1	60.0 ±0.1	4	Tap M6x1 15 deep BOTH ends (centre bore).

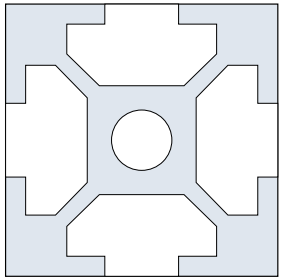


P15 corner post (1.5:1)

M6x1-6H x 15 deep, both ends, in the Ø5.1 centre bore



2020 section (1.5:1)



3030 section (1.2:1)

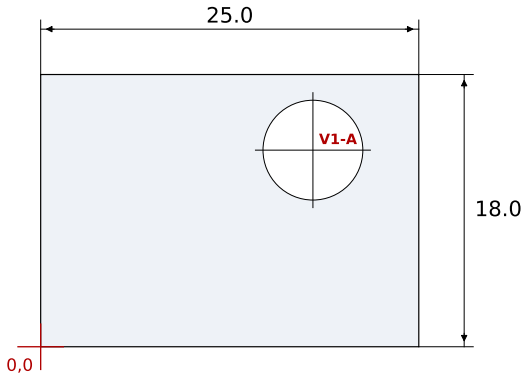
NOTES

1. Profiles: 6063-T5 / 6061 extrusion, B-type slots. Order cut-to-length from the profile supplier if possible.
2. Ends square to the axis within 0.1 so plates and brackets sit flat.
3. P8 ends are NOT tapped (Rev A tapped M5 - removed). They are fixed with P18 corner brackets + M6 T-nuts.
4. P7 ends are NOT tapped any more: fixed with P19 corner brackets + M5 T-nuts.
5. T-nuts to drop in BEFORE closing the frame: P8 rear: 4x M5 (deck, top slot) + 2x M6 (brackets, inner side slot).
   P8 middle: 4x M5 (deck, top) + 2x M6 (brackets, rear side) + 4x M5 (hinge bearing supports P17, front side).
   P7 hinge-side: 2x M4 (sector angle, top) + 2x M5 (brackets).   P7 nose: 2x M5 (brackets).
6. Use drop-in (roll-in) T-nuts so they can be added after assembly if one is forgotten.

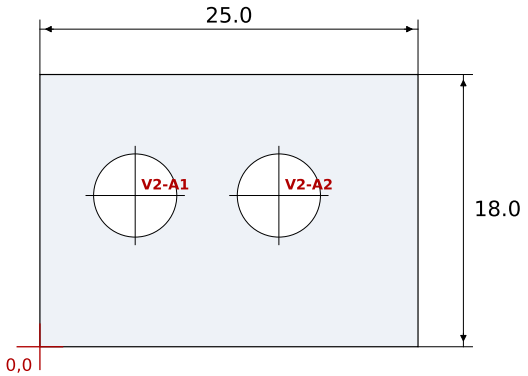
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P7/P8/P15 Profile cut list
MATERIAL	Al T-slot extrusion
QTY	2 / 2 / 4
SCALE	as shown
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	19 of 25

P18 INSIDE CORNER BRACKET - 3030 BARS

LEG ON BAR SIDE (T-nut) (scale 2:1)



LEG ON SIDE PLATE (scale 2:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A	18.00	13.00	Ø6.6 THRU (M6 clearance)
V2-A1	6.30	10.00	Ø5.5 THRU (M5 clearance)
V2-A2	15.80	10.00	Ø5.5 THRU (M5 clearance)

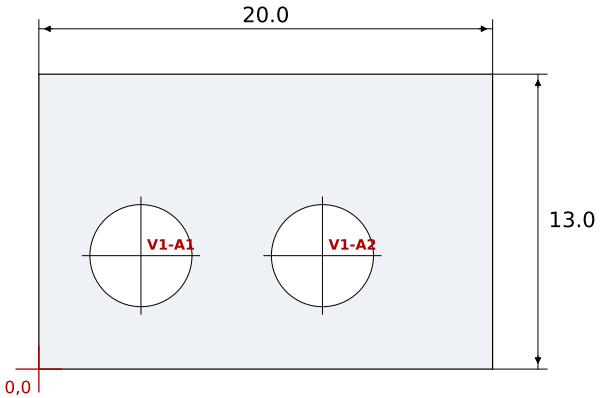
NOTES

1. Cut 18 long from 25x25x4 aluminium angle, drill only.
2. Bar leg: Ø6.6 for M6x10 + slot-8 M6 T-nut. Plate leg: 2x Ø5.5 for M5x8 into the tapped side plate.
3. CAD: STEP/P18*.step

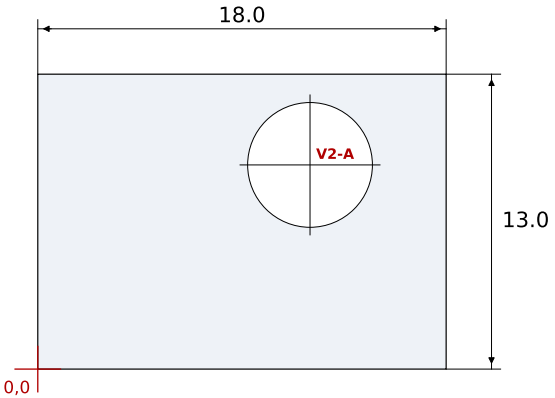
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P18 Inside corner bracket - 3030 bars
MATERIAL	Al angle 25x25x4 (6061/6063)
QTY	4
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	12 of 25

P19 INSIDE CORNER BRACKET - 2020 BARS

LEG ON SIDE PLATE (scale 3:1)



LEG ON BAR SIDE (T-nut) (scale 3:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A1	4.50	5.00	Ø4.5 THRU (M4 clearance)
V1-A2	12.50	5.00	Ø4.5 THRU (M4 clearance)
V2-A	12.00	9.00	Ø5.5 THRU (M5 clearance)

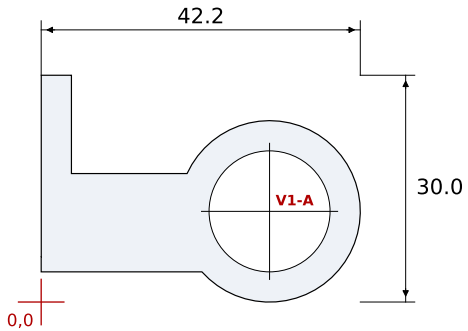
NOTES

1. Cut 13 long from 20x20x3 aluminium angle; trim the plate leg to 18. Drill only.
2. Plate leg: 2x Ø4.5 for M4x8. Bar leg: Ø5.5 for M5x8 + slot-6 M5 T-nut.
3. CAD: STEP/P19*.step

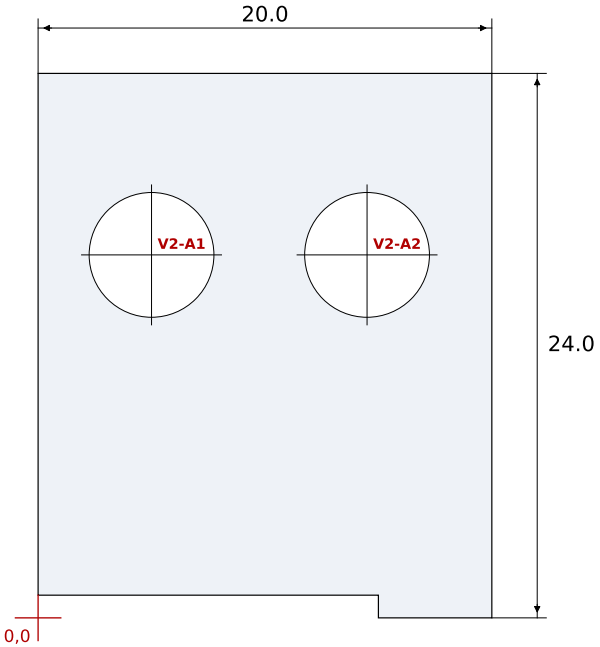
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P19 Inside corner bracket - 2020 bars
MATERIAL	Al angle 20x20x3 (6061/6063)
QTY	4
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	13 of 25

P17 HINGE SHAFT BEARING SUPPORT A (B = OPPOSITE HAND)

PLATE FACE (scale 1:1)



FOOT FACE (against 3030 bar side) (scale 3:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A	30.20	11.99	Ø16.0 H7 THRU (688-2RS bearing seat)
V2-A1	5.00	16.00	Ø5.5 THRU (M5 clearance)
V2-A2	14.50	16.00	Ø5.5 THRU (M5 clearance)

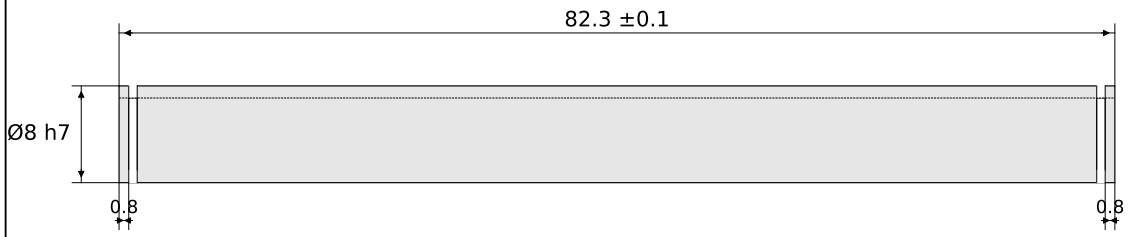
NOTES

1. One piece: 5 mm plate + 4 mm foot (machined from 25 x 45 bar, or 3D print in PETG-CF, 8 mm plate).
2. Ø16 H7 bearing seat for 688-2RS. Foot: 2x Ø5.5 for M5x10 into slot-8 T-nuts on the middle bar.
3. CAD: STEP/P17*.step DXF/P17*.dxf

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P17 Hinge shaft bearing support A (B = opposite hand)
MATERIAL	Al 6061-T6 (or PETG-CF print)
QTY	1 + 1 opposite hand
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	11 of 25

P20 / P21 SHAFTS (D-shaft stock, cut + groove)

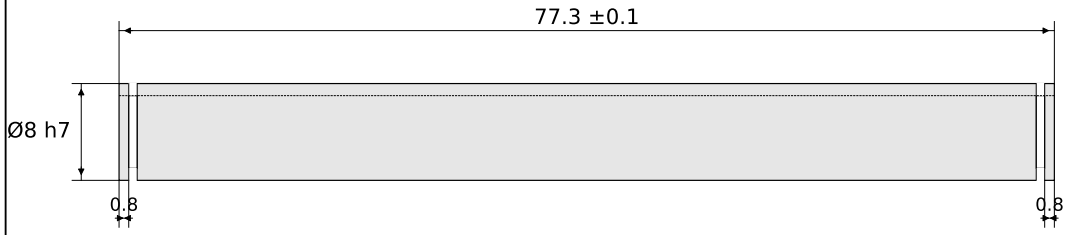
Start from ground D-shaft stock (Ø8 flat 7, Ø6 flat 5.5). Cut to length, chamfer 0.3x45°, turn the two E-clip grooves.



P20-A Hinge drive shaft A (QTY 1)

D-flat 7 across, FULL LENGTH (dashed)

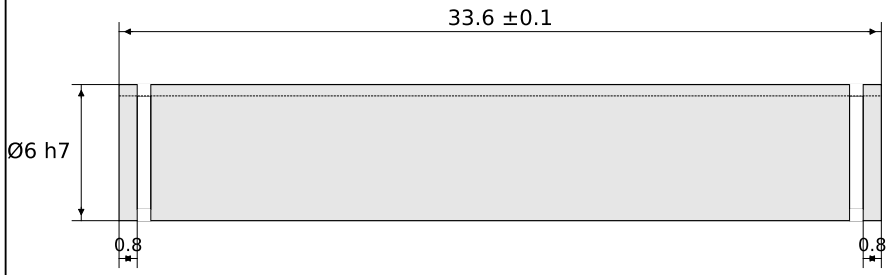
2x groove Ø6 x 0.7 wide for E-clip DIN 6799-6



P20-B Hinge drive shaft B (QTY 1)

D-flat 7 across, FULL LENGTH (dashed)

2x groove Ø6 x 0.7 wide for E-clip DIN 6799-6



P21 Intermediate shaft (QTY 2)

D-flat 5.5 across, FULL LENGTH (dashed)

2x groove Ø5 x 0.6 wide for E-clip DIN 6799-5

NOTES

1. Material: steel D-shaft (1045 / 303 stainless, ground h7). Hardness not required.
2. Grooves: P20 Ø6.0 +0/-0.06 x 0.7 (DIN 6799-6); P21 Ø5.0 +0/-0.06 x 0.6 (DIN 6799-5). Sharp groove corners.
3. Bearings (688-2RS / F686ZZ) slide over the D; fix inner rings with Loctite 641 at assembly.
4. Stack on P20 (from one end): rear drive pulley - rear-plate bearing - front drive pulley - front-plate bearing - support bearing (P17) - shim 8x11x1 - stage-2 pulley. E-clips at both ends trap the stack.
5. Front drive pulleys also get an M4x8 cup-point grub screw onto the flat (Loctite 243).

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P20/P21 Shafts
MATERIAL	Steel D-shaft Ø8 / Ø6
QTY	1+1 / 2
SCALE	1.6:1, 3:1
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	20 of 25

M1-M8 MODIFICATIONS TO BOUGHT / EXISTING PARTS

ID	PART	QTY	MODIFICATION
M1	Track drive pulley HTD 20T BF (rear x2, front x2)	4	Bore out pilot to Ø8 D (flat 7), or order D-bore 8. Keep both flanges (Ø36 x 1.5).
			FRONT pair only: radial M4 tapped hole at mid-width (9.5 from a flange face), onto the flat (M4x8 grub).
M2	Track idler pulley HTD 20T BF (rear x2, nose x2)	4	Bore Ø16 H7 through for 2x 688-2RS (press fit) + spacer sleeve 8x10x9 between them.
M3	Stage-2 pulley on hinge shaft (HTD 5M)	2	Bore: Ø8 D (flat 7).
M4	Intermediate pulleys, stage 1 + stage 2	4	Bore: Ø6 D (flat 5.5). M3 grub screw optional.
M5	Motor pulley, stage 1 (HTD 3M)	2	Radial M3 tapped hole mid-width onto the motor D-flat (M3x3 grub, max length 3).
M6	Hinge pinion m1 20T, 5 mm face	1	Bore: Ø8 D (flat 7) to suit the 5840 shaft. Radial M3 tapped hole (M3x4 grub).
M7	RS PRO 427-991 bracket (shaft supports)	8	Upright-leg hole elongated upward to a 3.4 x 6.4 slot (belt tension). Sheet M7.
M8	5840 L-bracket (motor kit)	1	4x Ø4.5 csk Ø8.4 x 90° on 40 x 28 (gearbox datasheet). Sheet M8.

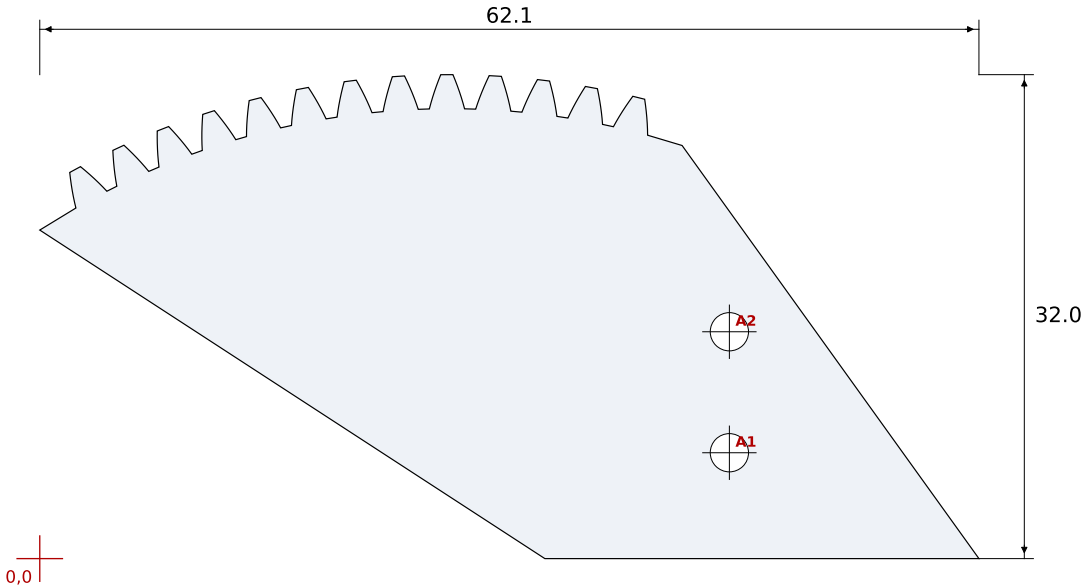
NOTES

1. Grub screws: ISO 4029 cup point, Loctite 243. Radial holes at mid-width of the hub/teeth, between teeth if possible.
2. If a sprocket / pulley is 3D printed, print the D-bore and use heat-set inserts for the grub screws.
3. Relief faces: only the hub boss may touch a bearing inner ring; nothing rotating may rub on a plate.
4. STEP files of each modified part are in STEP/M*.step for reference.
5. M1/M2 = standard HTD 5M 20T BF pulley for 15 mm belt: OD 30.69, PD 31.83, toothed 16, 2 flanges Ø36 x 1.5, total 19.
   M2 bore Ø16 is above the catalogue max (14) - leaves a 5.3 wall under the teeth; ask the supplier/lathe to bore it.

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	M1-M8 Modified bought parts
MATERIAL	various
QTY	see table
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	21 of 25

P11 HINGE SECTOR GEAR M1 R51 SEGMENT

VIEW FROM +Z (motor side). Origin of CAD = hinge axle   (scale 2:1, thickness 5.0)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
A1	45.60	7.00	M3x0.5-6H THRU (tap drill Ø2.5)
A2	45.60	15.00	M3x0.5-6H THRU (tap drill Ø2.5)

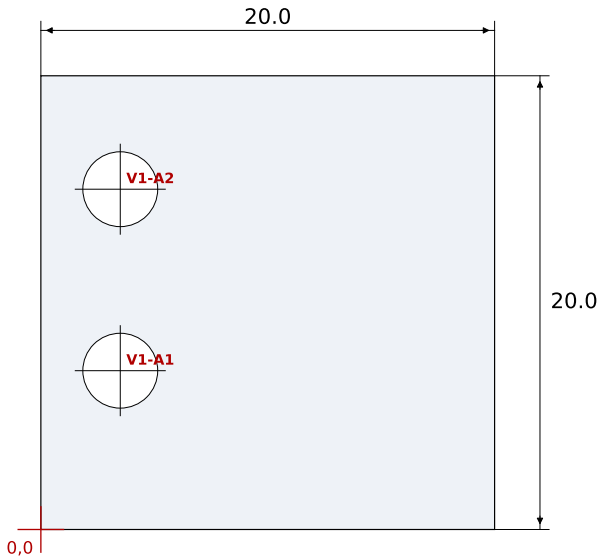
NOTES

1. Laser / wire cut from DXF (tooth flanks in the DXF are true involute, m1, 20° PA, pitch R51).
2. Rev B: old Ø4.5 holes removed. 2x M3 tapped holes for the P14 angle (M3x8 csk from the angle side).
3. Screw tips must end flush with the outer face (hinge bracket plate is 1 mm away).
4. CAD: STEP/P11*.step   DXF/P11*.dxf

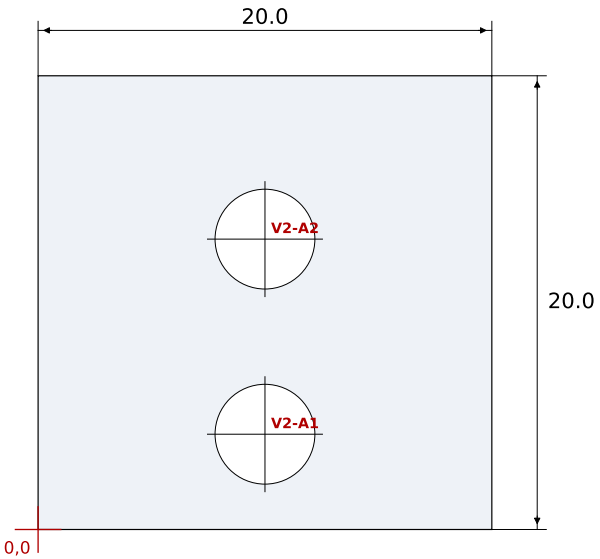
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P11 Hinge sector gear m1 R51 segment
MATERIAL	Steel S275 / mild, 5 mm
QTY	1
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	9 of 25

P14 SECTOR MOUNT ANGLE

VERTICAL LEG (against sector gear) (scale 3:1)



BASE LEG (on 2020 bar) (scale 3:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A1	3.50	7.00	M4x0.7-6H THRU (tap drill Ø3.3), CSK Ø6.5x90° THIS FACE
V1-A2	3.50	15.00	M4x0.7-6H THRU (tap drill Ø3.3), CSK Ø6.5x90° THIS FACE
V2-A1	10.00	4.20	Ø4.4 THRU (M4 clearance), CSK Ø4.4x90° THIS FACE
V2-A2	10.00	12.80	Ø4.4 THRU (M4 clearance), CSK Ø4.4x90° THIS FACE

NOTES

1. Cut 20 long from 20x20x3 angle. Drill + countersink.
2. Vertical leg: 2x M3 csk from the INSIDE of the angle (M3x8 into the tapped sector gear).
3. Base leg: 2x M4 csk on top, on the bar slot centre line (M4x8 + slot-6 T-nuts).
4. CAD: STEP/P14*.step

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P14 Sector mount angle
MATERIAL	Al angle 20x20x3
QTY	1
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
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BILL OF MATERIALS (1/3) - also in RB4107 BOM.xlsx

ID	DESCRIPTION	SPEC / STANDARD	QTY	SOURCE
== 1 Fabricated parts (make) ==				
P1	Side plate - front segment (1 + 1 opposite hand)	Al 6061-T6 5 mm, laser + drill/tap	2	make - drawings
P2	Side plate - rear segment (1 + 1 opposite hand)	Al 6061-T6 5 mm, laser + drill/tap	2	make - drawings
P3-P6	Slide rails / wear strips: MISUMI UPA-189-20-5, drill 3 pilot holes	UHMW-PE 189 x 20 x 5	8	make - drawings
P7	Cross bar, front segment (hinge-side + nose)	2020 profile x 110	2	make - drawings
P8	Cross bar, rear + middle	3030 profile x 160	2	make - drawings
P9	Motor shim plate, slotted	Al 2 mm, laser	2	make - drawings
P10	Drive-motor tail cradle	PETG print	2	make - drawings
P11	Hinge sector gear, m1 R51 segment	Steel 5 mm, laser/wire	1	make - drawings
P12	Deck plate	Al 6061-T6 5 mm, laser + drill/tap	1	make - drawings
P13	Intermediate shaft support plate (A1, A2, B1, B2)	Al 6061-T6 4 mm	4	make - drawings
P14	Sector mount angle	Al angle 20x20x3 x 20	1	make - drawings
P15	Corner post, M6 tapped both ends	2020 profile x 60	4	make - drawings
P16	Tray plate (controller deck)	Al 6061-T6 5 mm, laser + drill/tap	1	make - drawings
P17	Hinge shaft bearing support (A + B opposite hand)	Al 6061 or PETG-CF	2	make - drawings
P18	Inside corner bracket, 3030	Al angle 25x25x4 x 18	4	make - drawings
P19	Inside corner bracket, 2020	Al angle 20x20x3 x 13	4	make - drawings
P20	Hinge drive shaft A 82.3 / B 77.3	Steel D-shaft Ø8 (flat 7)	2	make - drawings
P21	Intermediate shaft 33.6	Steel D-shaft Ø6 (flat 5.5)	2	make - drawings
P22	Controller cover, one-piece, vented	PETG print	1	make - drawings
P23	ESP32 sled	PETG print	1	make - drawings
== 2 Bought parts to modify ==				
M1	Track drive pulley: D-bore 8 (+ M4 grub, front pair)	HTD5M 20T BF, 15 mm belt	4	buy + modify
M2	Track idler: Ø16 H7 bore for 2x 688-2RS	HTD5M 20T BF, 15 mm belt	4	buy + modify
M3	Stage-2 pulley on hinge shaft: D-bore 8	HTD 5M, 9 mm belt	2	buy + modify
M4	Intermediate pulleys (stage 1 + 2): D-bore 6	HTD 3M / 5M	4	buy + modify
M5	Motor pulley stage 1: radial M3 grub hole	HTD 3M, 6 mm D bore	2	buy + modify
M6	Hinge pinion: D-bore 8 + M3 grub hole	Spur gear m1 20T, 20° PA, 5 face	1	buy + modify
M7	RS PRO 427-991 bracket: leg hole -> slot (shaft supports)	see sheet M7	8	buy + modify
M8	5840 L-bracket: 4x Ø4.5 csk on 40 x 28	see sheet M8	1	buy + modify
== 3 Bought mechanical parts ==				
B1	Gearmotor 37D, 12 V, with encoder (drive A + B)	Pololu #2828	2	Pololu
B2	Machined bracket for 37D gearmotor	Pololu #1995	2	Pololu
B3	Worm gear motor 5840-31ZY 12 V + L-bracket kit (hinge)	5840-31ZY	1	AliExpress / RobotShop
B4	Ball bearing 688-2RS 8 x 16 x 5	688-2RS	14	RS / Misumi
B5	Flanged ball bearing F686ZZ 6 x 13 x 5 (flange 15 x 1.1)	F686ZZ	4	RS / Misumi
B6	Shoulder screw Ø8 x 25, M6 thread (idler stub axle)	ISO 7379 8x25	4	Misumi / RS
B7	Angle bracket 15 x 14.6 x 10, 1.2 mm steel (rails + shaft supports)	RS PRO 427-991	32	RS

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	BOM Bill of materials
MATERIAL	-
QTY	-
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
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BILL OF MATERIALS (2/3) - also in RB4107 BOM.xlsx

ID	DESCRIPTION	SPEC / STANDARD	QTY	SOURCE
B8	Stage-1 timing belt (motor -> intermediate)	HTD 3M, 9 wide, length per Rev A	2	per Rev A spec
B9	Stage-2 timing belt (intermediate -> hinge shaft)	HTD 5M, 9 wide, length per Rev A	2	per Rev A spec
B10	Track belt HTD 550-5M-15 (110 T) + 22 cleats every 5 teeth (25 mm), cleat 10 x	e.g. SKF PHG 550-5M-15, RS 058-6535	4	RS + cleats by belt sh
B11	Al 2020 B-type profile, slot 6, bore Ø5.1 (P7 + P15) - stock	2020, 1 m	1	Misumi / local
B12	Al 3030 B-type profile, slot 8 (80/20 30-3030 type) (P8) - stock	3030, 0.5 m	1	8020 / local
B13	Steel D-shaft Ø8, flat 7 (P20) - stock 200 mm	Ø8 h7 D-shaft	1	Misumi / local
B14	Steel D-shaft Ø6, flat 5.5 (P21) - stock 100 mm	Ø6 h7 D-shaft	1	Misumi / local
B15	Al angle 25x25x4 (P18) - 100 mm	6063	1	local
B16	Al angle 20x20x3 (P14, P19) - 100 mm	6063	1	local
B17	Battery strap 20 mm hook-and-loop with buckle, ~230 mm	Velcro	1	local
B18	Foam pad 10 x 10 x 8, adhesive-backed (hinge up-stops)	EVA / neoprene	2	local
B19	Cable gland PG7 (actuator / hinge leads through cover)	PG7, 3-6.5 mm cable	1	RS
B20	Grommet edge strip for tray wire slot, 0.4 m	1-2 mm panel	1	RS
== 4 Fasteners ==				
F1	Socket head cap screw M3 x 5	ISO 4762 A2	16	RS / Misumi / local
F2	Socket head cap screw M3 x 6 (rail brackets to plates)	ISO 4762 A2	24	RS / Misumi / local
F3	Thread-forming screw for plastics 3 x 5, pan head (rails)	PT / Delta-PT	24	RS / Misumi / local
F4	Countersunk screw M3 x 8	ISO 10642	2	RS / Misumi / local
F5	Countersunk screw M4 x 8	ISO 10642	2	RS / Misumi / local
F6	Countersunk screw M4 x 8 (bracket to 5840 gearbox)	DIN 7991	4	RS / Misumi / local
F7	Socket head cap screw M4 x 8	ISO 4762	8	RS / Misumi / local
F8	Socket head cap screw M5 x 8	ISO 4762	12	RS / Misumi / local
F9	Socket head cap screw M5 x 10	ISO 4762	4	RS / Misumi / local
F10	Countersunk screw M5 x 12	ISO 10642	8	RS / Misumi / local
F11	Socket head cap screw M6 x 10	ISO 4762	4	RS / Misumi / local
F12	Socket head cap screw M6 x 16	ISO 4762	4	RS / Misumi / local
F13	Countersunk screw M6 x 16	ISO 10642	4	RS / Misumi / local
F14	Grub screw M3 x 3, cup point	ISO 4029	2	RS / Misumi / local
F15	Grub screw M3 x 4, cup point	ISO 4029	1	RS / Misumi / local
F16	Grub screw M4 x 8, cup point	ISO 4029	2	RS / Misumi / local
F17	T-nut M4, slot 6 (2020), drop-in	B-type slot 6	2	RS / Misumi / local
F18	T-nut M5, slot 6 (2020), drop-in	B-type slot 6	4	RS / Misumi / local
F19	T-nut M5, slot 8 (3030), drop-in	B-type slot 8	12	RS / Misumi / local
F20	T-nut M6, slot 8 (3030), drop-in	B-type slot 8	4	RS / Misumi / local
F21	E-clip for Ø6 shaft (groove Ø5)	DIN 6799-5	4	RS / Misumi / local
F22	E-clip for Ø8 shaft (groove Ø6)	DIN 6799-6	4	RS / Misumi / local
F23	Shim washer 6 x 8 x 0.5	DIN 988	4	RS / Misumi / local
F24	Shim washer 8 x 11 x 1	DIN 988	6	RS / Misumi / local

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	BOM Bill of materials
MATERIAL	-
QTY	-
SCALE	-
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BILL OF MATERIALS (3/3) - also in RB4107 BOM.xlsx

ID	DESCRIPTION	SPEC / STANDARD	QTY	SOURCE
F24a	Shim washer 8 x 11 x 0.5	DIN 988	4	RS / Misumi / local
F24b	Bearing spacer sleeve 8 x 10 x 9 (Al tube, cut)	Al tube 10 x 1	4	RS / Misumi / local
F25	Washer M6	DIN 125 A	4	RS / Misumi / local
F26	Nyloc nut M6	DIN 985	4	RS / Misumi / local
F27	Nylon standoff M3 x 6, male-female (MDD10A)	PA66	8	RS / Misumi / local
F28	Nylon standoff M2.5 x 5, male-female (BNO085)	PA66	4	RS / Misumi / local
F29	Nylon standoff M2 x 5, male-female (5V regulator)	PA66	2	RS / Misumi / local
F30	Socket head cap screw M3 x 25 (controller cover) [not modelled]	ISO 4762	4	RS / local
F31	Nylon screw M3 x 6 (MDD10A to standoffs) [not modelled]	PA66	8	RS / local
F32	Nylon screw M2.5 x 5 (BNO085) + M2 x 4 (regulator) [not modelled]	PA66	6	RS / local
F33	Socket head cap screw M3 x 8 (ESP32 sled to tray) [not modelled]	ISO 4762	2	RS / local
F34	Socket head cap screw M3 x 12 + nut (Pololu bracket base, from below deck) [n	ISO 4762 / ISO 4032	6	RS / local
F35	Heat-set insert M3 x 5 + SHCS M3 x 10 (tail cradles) [not modelled]	brass insert	4	RS / local
F36	Socket head cap screw M5 x 8 + washer (5840 bracket foot) [not modelled]	ISO 4762 + DIN 125	2	RS / local
== 5 Electronics ==				
E1	Dual motor driver 10 A (drive A+B / hinge + actuator)	Cytron MDD10A	2	Cytron
E2	Microcontroller board	ESP32-DevKitC (WROOM-32)	1	Espressif / local
E3	9-axis IMU breakout	BNO085 (Adafruit 4754 size)	1	Adafruit / local
E4	5 V 5 A step-down regulator	Pololu D24V50F5	1	Pololu
E5	Battery 3S LiPo 3000 mAh 'shorty'	11.1 V	1	local
E6	Wiring, connectors, fuse 15 A, power switch	per wiring diagram	1	local
== 6 Consumables ==				
C1	Thread-locker, medium strength (grub screws)	Loctite 243	1	local
C2	Retaining compound (bearings on D-shafts / seats)	Loctite 641	1	local
C3	Cable ties 100 mm	-	20	local

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	BOM Bill of materials
MATERIAL	-
QTY	-
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
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